

Forecasting Volatility in Financial Markets: A Review

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1. Introduction

VOLATILITY FORECASTING IS AN important task in financial markets, and it has held the attention of academics and practitioners over the last two decades. At the time of writing, there are at least 93 published and working papers that study forecasting performance of various volatility models, and several times that number have been written on the subject of volatility modelling without the forecasting aspect. This extensive research reflects the importance of volatility in investment, security valuation, risk management, and monetary policy making.

Volatility is not the same as risk. When it is interpreted as uncertainty, it becomes a key input to many investment decisions and portfolio creations. Investors and portfolio managers have certain levels of risk which they can bear. A good forecast of the volatility of asset prices over the investment holding period is a good starting point for assessing investment risk.

Volatility is the most important variable in the pricing of derivative securities, whose trading volume has quadrupled in recent years. To price an option, we need to know the volatility of the underlying asset from now until the option expires. In fact, the market convention is to list option prices in terms of volatility units. Nowadays, one can buy derivatives that are written on volatility itself, in which case the definition and measurement of volatility will be clearly specified in the derivative contracts. In these new contracts, volatility now becomes the underlying “asset.” So a volatility forecast and a second prediction on the volatility of volatility over the defined period is needed to price such derivative contracts.

Financial risk management has taken a central role since the first Basle Accord was established in 1996. This effectively makes volatility forecasting a compulsory risk-management exercise for many financial institutions around the world. Banks and trading houses have to

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set aside reserve capital of at least three times that of value-at-risk (VaR), which is defined as the minimum expected loss with a 1-percent confidence level for a given time horizon (usually one or ten days). Sometimes, a 5-percent critical value is used. Such VaR estimates are readily available given volatility forecast, mean estimate, and a normal distribution assumption for the changes in total asset value. When the normal distribution assumption is disputed, which is very often the case, volatility is still needed in the simulation process used to produce the VaR figures.

Financial market volatility can have a wide repercussion on the economy as a whole. The incidents caused by the terrorists' attack on September 11, 2001, and the recent financial reporting scandals in the United States have caused great turmoil in financial markets on several continents and a negative impact on the world economy. This is clear evidence of the important link between financial market uncertainty and public confidence. For this reason, policy makers often rely on market estimates of volatility as a barometer for the vulnerability of financial markets and the economy. In the United States, the Federal Reserve explicitly takes into account the volatility of stocks, bonds, currencies, and commodities in establishing its monetary policy (Sylvia Nasar 1992). The Bank of England is also known to make frequent references to market sentiment and option implied densities of key financial variables in its monetary policy meetings.

Given the important role of volatility forecasting and that so much has been written on the subject, this paper aims to provide comprehensive coverage of the status of this research. Taking a utilitarian viewpoint, we believe that the success of a volatility model lies in its out-of-sample forecasting power. It is impossible, in practice, to perform tests on all volatility forecasting models on a large number of data sets and over many different periods. By carefully reviewing the methodologies and empirical findings in 93 papers, the contribution of this review is to provide a

bird's-eye view of the whole volatility forecasting literature and to provide some recommendations for the practice and future research. John Knight and Stephen Satchell (1998), which we draw upon frequently, is the first book to cover many issues and early empirical results related to volatility forecasting. Our focus here, however, is on the 93 papers and the collective findings in this pool of research. We have excluded in this review all papers that do not produce out-of-sample volatility forecasts and papers that forecast correlations, though the latter might be useful for forecasting portfolio risk.

The remaining sections are organized as follows. Section 2 provides some preliminaries such as the definition and measurement of volatility, and lists some confounding factors such as forecast horizons and data frequency. Section 3 introduces the two broad categories of methods widely used in making volatility forecasts, namely time series models and option ISD (implied standard deviation). Section 4 lists a number of forecast performance measures and raises various issues related to forecast evaluation. Sections 5 and 6 are the core sections of this paper; section 5 reviews research papers that forecast volatility based on historical price information only; section 6 reviews research papers that use option ISD alone or in addition to historical price information to forecast future volatility. Section 7 discusses our views about research and achievement in volatility forecasting and provides some directions for future research. Section 8 summarizes and concludes. The technical specifications of volatility models are listed in an appendix. A list containing a short summary of each of the 93 papers is provided at the end.

2. Some Preliminaries

2.1 Volatility, Standard Deviation, and Risk

Many investors and generations of finance students often have an incomplete appreciation

of the differences between volatility, standard deviation, and risk. It is worth elucidating some of the conceptual issues here. In finance, volatility is often used to refer to standard deviation, σ , or variance, σ^2 , computed from a set of observations as

$$\hat{\sigma}^2 = \frac{1}{N-1} \sum_{t=1}^N (R_t - \bar{R})^2, \quad (1)$$

where $\bar{R}$ is the mean return. The sample standard deviation statistic $\hat{\sigma}$ is a distribution free parameter representing the second moment characteristic of the sample. Only when σ is attached to a standard distribution, such as a normal or a t distribution, can the required probability density and cumulative probability density be derived analytically. Indeed, σ can be calculated from any irregular shape distribution, in which case the probability density will have to be derived empirically. In the continuous time setting, σ is a scale parameter that multiplies or reduces the size of the fluctuations generated by the standard wiener process. Depending on the dynamic of the underlying stochastic process and whether or not the parameters are time varying, very different shapes of returns distributions may result. So it is meaningless to use σ as a risk measure unless it is attached to a distribution or a pricing dynamic. When σ is used to measure uncertainty, the users usually have in mind, perhaps implicitly, a normal distribution for the returns distribution.

Standard deviation, σ , is the correct dispersion measure for the normal distribution and some other distributions, but not all. Other measures that have been suggested and found useful include the mean absolute return and the inter-quantile range. However, the link between volatility and risk is tenuous; in particular, risk is more often associated with small or negative returns, whereas most measures of dispersion make no such distinction. The Sharpe ratio, for example, defined as return in excess of risk

free rate divided by standard deviation, is frequently used as an investment performance measure. It incorrectly penalizes occasional high returns. The idea of “semi-variance,” an early suggestion by Harry Markowitz (1991), which only uses the squares of returns below the mean, has not been widely used, largely because it is not operationally easy to apply in portfolio construction.

2.2 Volatility Definition and Measurement

As mentioned previously, volatility is often calculated as the sample standard deviation, which is the square root of equation (1). Stephen Figlewski (1997) notes that since the statistical properties of sample mean make it a very inaccurate estimate of the true mean, especially for small samples, taking deviations around zero instead of the sample mean as in equation (1) typically increases volatility forecast accuracy. There are methods for estimating volatility that are designed to exploit or reduce the influence of extremes.² While equation (1) is an unbiased estimate of σ^2 , the square root of $\hat{\sigma}^2$ is a biased estimate of σ due to Jensen inequality.³ Zhuanxin Ding, Clive Granger, and Robert Engle (1993) suggest measuring volatility directly from absolute returns.⁴

To understand the continuous time analogue of (1), we assume for the ease of

² For example, the Maximum likelihood method proposed by Clifford Ball and Walter Torous (1984), the high-low method proposed by Michael Parkinson (1980) and Mark Garman and Michael Klass (1980).

³ See Jeff Fleming (1998, footnote 10.) John Cox and Mark Rubinstein (1985) explain how this bias can be corrected assuming a normal distribution for R_t . However, in most cases, the impact of this adjustment is small.

⁴ Marie Davidian and Raymond Carroll (1987) show absolute returns volatility specification is more robust against asymmetry and non-normality. There is some empirical evidence that deviations or absolute returns based models produce better volatility forecasts than models based on squared returns (Stephen Taylor 1986; Louis Ederinton and Wei Guan 2000a; and Michael McKenzie 1999) but the majority of time series volatility models are squared returns models.

exposition that the instantaneous returns are generated by the continuous time martingale,

$$dp_t = \sigma_t dW_{p,t} \quad (2)$$

where $dW_{p,t}$ denotes a standard wiener process. From (2) the conditional variance for the one-period returns, $r_{t+1} \equiv p_{t+1} - p_t$, is $\int_0^1 \sigma_{t+\tau}^2 d\tau$, which is also known as the *integrated volatility* over the period t to $t + 1$. This quantity is of central importance in the pricing of derivative securities under stochastic volatility (see John Hull and Alan White 1987). While p_t can be observed at time t , σ_t is an unobservable latent variable that scales the stochastic process $dW_{p,t}$ continuously through time.

Let m be the sampling frequency and there are m continuously compounded returns in one time unit such that

$$r_{m,t} \equiv p_t - r_{t-1/m}.$$

If the discretely sampled returns are serially uncorrelated and the sample path for σ_t is continuous, it follows from the theory of quadratic variation (Ioannis Karatzas and Stephen Shreve 1988) that

$$p \lim_{m \rightarrow \infty} \left(\int_0^1 \sigma_{t+\tau}^2 d\tau - \sum_{j=1, \dots, m} r_{m,t+j/m}^2 \right) = 0.$$

Hence, time t volatility is theoretically observable from the sample path of the return process so long as the sampling process is frequent enough. The term *realized volatility* has been used in William Fung and David Hsieh (1991), and Torben Andersen and Tim Bollerslev (1998), to mean the sum of intraday squared returns at short intervals such as fifteen- or five-minutes.⁵ Such a

volatility estimator has been shown to provide an accurate estimate of the latent process that defines volatility. Characteristics of financial market data used in these studies suggest that returns measured at an interval shorter than five minutes are plagued by spurious serial correlation caused by various market microstructure effects including nonsynchronous trading, discrete price observations, intraday periodic volatility pattern, and bid-ask bounce. Andersen and Bollerslev (1998) and George Christodoulakis and Satchell (1988) show how the inherent noise in the approximation of actual and unobservable volatility by square returns results in misleading forecast evaluation. These theoretical results turn out to have a major implication for volatility forecasting research. We shall return to this important issue in section 4.4.

2.3 Stylized Facts about Financial Market Volatility

There are several salient features about financial time series and financial market volatility that are now well documented. These include fat tail distributions of risky asset returns, volatility clustering, asymmetry and mean reversion, and comovements of volatilities across assets and financial markets. More recent research finds correlation among volatility is stronger than that among returns and both tend to increase during bear markets and financial crises. Since volatility of financial time series has complex structure, Francis Diebold et al. (1998) warn that forecast estimates will differ depending on the current level of volatility, volatility structure (e.g. the degree of persistence and mean reversion, etc.) and the forecast horizon. These will be made clearer in the discussions below.

If returns are *iid* (independent and identically distributed, or strict white noise), then variance of returns over a long horizon can be derived as a simple multiple of single period variance. But, this is clearly not the case for many financial time series because of the

⁵ In the foreign exchange markets, quotes for major exchange rates are available round the clock. In the case of stock markets, close-to-open square return is used in the volatility aggregation process during market close.

stylized facts listed above. While a point forecast of $\hat{\sigma}_{T-1,T|t-1}$ becomes very noisy as $T \rightarrow \infty$, a cumulative forecast, $\hat{\sigma}_{t,T|t-1}$, becomes more accurate because of errors cancellation and volatility mean reversion unless there is a fundamental change in the volatility level or structure.⁶

Some studies find volatility time series appear to have a unit root (Philip Perry 1982, and Adrian Pagan and G. William Schwert 1990). That is,

$$\sigma_t = \phi\sigma_{t-1} + \epsilon_t,$$

with ϕ indistinguishable from 1. Other papers find some volatility measures of daily and intra-day returns have a long memory property (see Granger, Ding, and Scott Spear 2000 for examples and references). The autocorrelations of variances, and particularly those of mean absolute deviations, stay positive and significantly above zero for lags up to a thousand or more. These findings are important because they imply that a shock in the volatility process will have a long-lasting impact.

Complication in relation to the choice of forecast horizon is partly due to volatility mean reversion. In general, volatility forecast accuracy improves as data sampling frequency increases relative to forecast horizon (Andersen, Bollerslev, and Steve Lange 1999). However, for volatility forecasts over a long horizon, Figlewski (1997) finds forecast error doubled in size when daily data, instead of monthly data, is used to forecast volatility over 24 months. In some cases, e.g. when the forecast horizon exceeds ten years, a volatility estimate calculated using

weekly or monthly data is better because volatility mean reversion is difficult to adjust using high frequency data. In general, model based forecasts lose supremacy when the forecast horizon increases with respect to the data frequency. For forecast horizons that are longer than six months, a simple historical method using low frequency data over a period at least as long as the forecast horizon works best (Andrew Alford and James Boatsman 1995; and Figlewski 1997).

As far as sampling frequency is concerned, Feike Drost and Theo Nijman (1993) prove, theoretically and for a special case (i.e. the GARCH(1,1) process, which will be introduced in section 3.1.2 later), that volatility structure should be preserved through intertemporal aggregation. This means that whether one models volatility at the hourly, daily, or monthly intervals, the volatility structure should be the same. But it is well known that this is not the case in practice; volatility persistence, which is highly significant in daily data, weakens as the frequency of data decreases.⁷ This further complicates any attempt to generalize volatility patterns and forecasting results.

3. Models Used in Volatility Forecasting

In this section, we first describe various popular time series volatility models that use the historical information set to formulate volatility forecasts and a second approach that derives market estimates of future volatility from traded option prices. Nonparametric methods for volatility forecasting have been suggested. But, as nonparametric methods were reported to perform

⁶ $\hat{\sigma}_{t,T|t-1}$ denotes a volatility forecast formulated at time $t-1$ for volatility over the period from t to T . In pricing options, the required volatility parameter is the expected volatility over the life of the option. The pricing model relies on a riskless hedge to be followed through until the option reaches maturity. Therefore the required volatility input, or the implied volatility derived, is a cumulative volatility forecast over the option maturity and not a point forecast of volatility at option maturity. The interest in forecasting $\hat{\sigma}_{t,T|t-1}$ goes beyond the riskless hedge argument, however.

⁷ See Diebold (1988), Richard Baillie and Bollerslev (1989), and Poon and Stephen Taylor (1992) for examples. Note that Daniel Nelson (1992) points out separately that as the sampling frequency becomes shorter, volatility modelled using a discrete time model approaches its diffusion limit and persistence is to be expected provided that the underlying return is a diffusion or a near diffusion process with no jumps.

poorly (Pagan and Schwert 1990; and Kenneth West and Dongchul Cho 1995), they will not be discussed here. Also excluded from discussion here are volatility models that are based on neural networks (Michael Hu and Christ Tsoukalas 1999; genetic programming, e.g. Zumbach, Pictet, and Masutti 2001; time change and duration, e.g. Cho and Frees 1988, and Engle and Russell 1998).

3.1 *Times Series Volatility Forecasting Models*

Stephen Brown (1990), Engle (1993), and Abdurrahman Aydemir (1998) contain lists of time series models for estimating and modelling volatility. Kroner (1996) explains how volatility forecasts can be created and used. In this section, we narrow our discussion to models that are used in the 93 papers reviewed here. The specifications of these volatility models are provided in appendix A.

All models described in this section capture volatility persistence or clustering. Others take into account volatility asymmetry also. It is quite easy to construct a supply and demand model for financial assets, with supply a constant and demand partly driven by an external instrument that enters non-linearity, that will produce a model for financial returns that is heteroskedastic. Such a model is to some extent “theory based” but is not necessarily realistic. The pure time series models discussed in this section are not based on theoretical foundations but are selected to capture the main features of volatility found with actual returns. If successful in this, it is reasonable to expect that they will have some forecasting ability.

3.1.1 *Predictions Based on Past Standard Deviations*

This group of models starts on the basis that $\sigma_{t-\tau}$ for all $\tau > 0$ is known or can be estimated at time $t - 1$. The simplest historical price model is the *Random Walk* model, where σ_{t-1} is used as a forecast for σ_t .

Extending this idea, we have the *Historical Average* method, the simple *Moving Average* method, the *Exponential Smoothing* method and the *Exponentially Weighted Moving Average* method. The *Historical Average* method makes use of all historical standard deviations while the *Moving Average* method discards the older estimates. Similarly, the *Exponential Smoothing* method uses all historical estimates, and the *Exponentially Weighted Moving Average* (EWMA) method uses only the more recent ones. But unlike the previous two, the two exponential methods place greater weights on the more recent volatility estimates. All together, the four methods reflect a tradeoff between increasing the number of observations and sampling nearer to time t .

The *Riskmetrics*TM model uses the EWMA method. The *Smooth Transition Exponential Smoothing* model, proposed by James Taylor (2001), is a more flexible version of exponential smoothing where the weight depends on the size, and sometimes the sign as well, of the previous return. Next we have the *Simple Regression* method that expresses volatility as a function of its past values and an error term. The *Simple Regression* method is principally autoregressive. If past volatility errors are also included, one gets the ARMA model for volatility. Introducing a differencing order $I(d)$, we get ARIMA when $d = 1$ and ARFIMA when $d < 1$. Finally, we have the *Threshold Autoregressive* model, where the thresholds separate volatility into states with independent *simple regression* models and noise processes for volatility in each state.

Apart from *Random Walk* and *Historical Average*, successful applications of models described in this section normally involve searching for the optimal lag length or weighting scheme in an estimation period for out-of-sample forecasting. Such optimization generally involves minimizing in-sample volatility forecast errors. A more sophisticated forecasting procedure would

involve constant updating of parameter estimates when new information is observed and absorbed into the estimation period.

3.1.2 *ARCH Class Conditional Volatility Models*

A more sophisticated group of time series models is the ARCH family, which is extensively surveyed in Anil Bera and Matthew Higgins (1993), Bollerslev, Ray Chou, and Kenneth Kroner (1992), Bollerslev, Engle, and Nelson (1994), and Diebold and Jose Lopez (1995). In contrast to models described in section 3.1.1, ARCH class models do not make use of sample standard deviations, but formulate conditional variance, h_t , of returns via maximum likelihood procedure. Moreover, because of the way ARCH class models are constructed, h_t is known at time $t - 1$. So the one-step ahead forecast is readily available. Forecasts that are more than one step ahead can be formulated based on an iterative procedure.

The first example of ARCH model is ARCH(q) (Engle 1982) where h_t is a function of q past squared returns. In GARCH(p, q) (Bollerslev 1986, and Taylor 1986), additional dependencies are permitted on p lags of past h_t . Empirical findings suggest that GARCH is a more parsimonious model than ARCH, and GARCH(1,1) is the most popular structure for many financial time series. It turns out that *Riskmetrics*TM EWMA is a non-stationary version of GARCH(1,1) where the persistence parameters sum to 1 and there is no finite fourth moment. Such a model is often called an integrated model, which should not be confused with integrated volatility described in section 2.2. While unconvincing theoretically as a volatility generating process, an integrated model for volatility can nevertheless be estimated and has been shown to be powerful for prediction over a short horizon, as it is not conditioned on a mean level of volatility, and as a result it adjusts to changes in unconditional volatility quickly.

The EGARCH (Exponential GARCH) model (Nelson 1991) specifies conditional variance in logarithmic form, which means that there is no need to impose estimation constraint in order to avoid negative variance. With appropriate conditioning of the parameters, this specification captures the stylized fact that a negative shock leads to a higher conditional variance in the subsequent period than a positive shock would. Other models that allow for nonsymmetrical dependencies are the TGARCH (Threshold GARCH) which is similar to the GJR-GARCH (Lawrence Glosten, Ravi Jagannathan, and David Runkle 1993), QGARCH (Quadratic GARCH) and various other nonlinear GARCH reviewed in Philip Franses and Dick van Dijk (2000).

Both ARCH and GARCH models have been implemented with a James Hamilton (1989) type regime switching framework, where volatility persistence can take different values depending on whether it is in high or low volatility regimes. The most generalized form of regime switching model is the RS-GARCH(1,1) model used in Stephen Gray (1996) and Franc Klaassen (2002).

As mentioned before, volatility persistence is a feature that many time series models are designed to capture. A GARCH model features an exponential decay in the autocorrelation of conditional variances. However, it has been noted that squared and absolute returns of financial assets typically have serial correlations that are slow to decay, similar to those of an $I(d)$ process. A shock in the volatility series seems to have very "long memory" and impact on future volatility over a long horizon. The Integrated GARCH (IGARCH) model of Engle and Bollerslev (1986) captures this effect but a shock in this model impacts upon future volatility over an infinite horizon, and the unconditional variance does not exist for this model. This gives rise to FIGARCH(p, d, q) in Richard Baillie, Bollerslev, and Hans Mikkelsen (1996) and FIEGARCH(p, d, q) in Bollerslev and

Mikkelsen (1996) with $d \geq 0$. Provided that $d < 0.5$, the fractional integrated model is covariance stationary. However, as Soosung Hwang and Satchell (1998) and Granger (2001) point out, positive $I(d)$ process has a positive drift term or a time trend in volatility level which is not observed in practice. This is a major weakness of the fractionally integrated model for it to be adopted as a theoretically sound model for volatility.

It is important to note that there are many data generating processes, other than an $I(d)$ process, that also exhibit long memory in covariances. The short-memory stationary series with occasional breaks in mean in Granger and Namwon Hyung (2000) is an example. Diebold and Atsushi Inoue (2001) show stochastic regime switching can be easily confused with long memory if only a small amount of regime switching occurs. Gilles Zumbach (2002), on the other hand, captures long memory using IGARCH(2) (i.e. the sum of two IGARCH) and an LM model which aggregates high frequency squared returns with a set of power law weights.

3.1.3 Stochastic Volatility Models

In the stochastic volatility (SV) modelling framework, volatility is subject to a source of innovations that may or may not be related to those that drive returns. Modelling volatility as a stochastic variable immediately leads to fat tail distributions for returns. Autoregressive term in the volatility process introduces persistence, and correlation between the two innovative terms in the volatility process and the return process produces volatility asymmetry (Hull and White 1987, 1988). Long memory SV models have also been proposed by allowing volatility to have a fractional integrated order (see Andrew Harvey 1998).

For an excellent survey of SV work see Eric Ghysels, Harvey, and Eric Renault (1996), but the subject is rapidly changing. The volatility noise term makes the SV model a lot more flexible, but as a result the

SV model has no closed form, and hence cannot be estimated directly by maximum likelihood. The quasi-maximum likelihood estimation (QMLE) approach of Harvey, Esther Ruiz, and Neil Shephard (1994) is inefficient if volatility proxies are non-Gaussian (Andersen and Bent Sorensen 1997). The alternatives are the generalized method of moments (GMM) approach through simulations (Durrell Duffie and Kenneth Singleton 1993), or analytical solutions (Singleton 2001), and the likelihood approach through numerical integration (Moshe Fridman and Lawrence Harris 1988) or Monte Carlo integration using either importance sampling (Jon Danielsson 1994; Michael Pitt and Shephard 1997; J. Durbin and S. J. Koopman 2000) or Markov Chain (e.g. Eric Jacquier, Nicholas Polson, and Peter Rossi 1994; Sangjoon Kim, Shephard, and Siddhartha Chib 1998).

3.2 Options-Based Volatility Forecasts

A European style call (put) option is a right, but not an obligation, to purchase (sell) an asset at a strike price on option maturity date, T . An American style option is a European option that can be exercised prior to T . The Black-Scholes model for pricing European equity options (Fischer Black and Myron Scholes 1973) assumes stock price has the following dynamics

$$dS = \mu S dt + \sigma S dz, \quad (3)$$

and for the growth rate on stock

$$\frac{dS}{S} = \mu dt + \sigma dz. \quad (4)$$

From Ito lemma, the logarithmic of stock price has the following dynamics

$$d \ln S = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dz, \quad (5)$$

which means that stock price has a lognormal distribution or the logarithm of stock price has a normal distribution. Using a riskless

hedge argument, Black-Scholes proved that under certain assumptions, options prices could be derived using a risk neutral valuation relationship where all derivative assets generate only risk-free returns. Under this risk-neutral setting, investor risk preference and the required rate of returns on stock, μ in (4), are irrelevant as far as the pricing of derivatives is concerned. The Black-Scholes assumptions include constant volatility, σ , short sell with full use of proceeds, no transaction costs or taxes, divisible securities, no dividend before option maturity, no arbitrage, continuous trading, and a constant risk-free interest rate, r .

Empirical findings suggest that option pricing is not sensitive to the assumption of a constant interest rate. There are now good approximating solutions for pricing American-style options which can be exercised early and options that encounter dividend payments before option maturity. The impact of stochastic volatility on option pricing is much more profound, an issue we shall return to shortly. Apart from the constant volatility assumption, the violation of any of the remaining assumptions will result in the option price being traded within a band instead of at the theoretical price.

The Black-Scholes European option pricing formula states that option price at time t is a function of S_t (the price of the underlying asset), X (the strike price), r (the risk-free interest rate), T (time to option maturity) and σ (volatility of the underlying asset over the period from t to T). Given that S_t , X , r , and T are observable, once the market has produced a price (either a quote or a transaction price) for the option, we could use a backward induction technique to derive σ that the market used as an input. Such a volatility estimate is called option implied volatility. Since the reference period is from t to T in the future, option implied volatility is often interpreted as a market's expectation of volatility over the option's maturity, i.e. the period from t to T .

Given that each asset can have only one σ , it is a well-known puzzle that options of the

same time to maturity but differing in strikes appeared to produce different implied volatility estimates for the same underlying asset. Volatility smile, smirk, and sneer are names given to nonlinear shapes of implied volatility plots (against strike price). There are at least two theoretical explanations (viz. distributional assumption and stochastic volatility) for this puzzle. Other explanations that are based on market microstructure and measurement errors (e.g. liquidity, bid-ask spread and tick size) and investor risk preference (e.g. model risk, lottery premium and portfolio insurance) have also been proposed.

3.2.1 *Distributional Assumption*

To understand how Black-Scholes distributional assumption produces volatility smile, we need to make use of the positive relationship between volatility and option price, and the put-call parity⁸

$$c_t + Xe^{r(T-t)} = p_t + S_t \quad (6)$$

which established the positive relationship between call and put option prices. Since implied volatility is positively related to option price, equation (6) suggests there is also a positive relationship between implied volatilities derived from call and put options that have the same strike price and the same time to maturity.

As mentioned before Black-Scholes requires stock price in (5) to follow a lognormal distribution or the logarithmic stock returns to have a normal distribution. There is now widely documented empirical evidence that risky financial asset returns have leptokurtic tails. In the case where strike price is very high, the call option is deep-out-of-the-money⁹ and the probability for this option to

⁸ Much of the discussion here is from Hull (2000).

⁹ In option terminology, an option is out of the money when it is not profitable to exercise the option. For call option, this happens when $S < X$, and in the case of put, the condition is $S > X$. The reverse is true for in-the-money option. A call or a put is said to be at the money (ATM) when $S = X$. Near-the-money option is an option that is not exactly ATM, but close to being ATM. Sometimes, discounted values of S and X are used in the conditions.

be exercised is very low. Nevertheless, a leptokurtic right tail will give this option a higher probability, than that from a normal distribution, for the terminal asset price to exceed the strike price and the call option to finish in the money. This higher probability leads to a higher call price and a higher Black-Scholes implied volatility at high strike.

Next, we look at the case when strike price is low. First note that option value has two components; intrinsic value and time value. Intrinsic value reflects how deep the option is in the money. Time value reflects the amount of uncertainty before the option expires; hence it is most influenced by volatility. Deep-in-the-money call option has high intrinsic value and little time value, and a small amount of bid-ask spread or transaction tick size is sufficient to perturb the implied volatility estimation. We could use the argument in the previous paragraph but apply it to out-of-the-money (OTM) put option at low strike price. OTM put option has a close to nil intrinsic value and the put option price is due mainly to time value. Again because of the thicker tail on the left, we expect the probability that OTM put option finishes in the money to be higher than that for a normal distribution. Hence the put option price (and hence the call option price through put-call parity) should be greater than that predicted by Black-Scholes. If we use Black-Scholes to invert volatility estimates from these option prices, the Black-Scholes implied will be higher than actual volatility. This results in volatility smile where implied volatility is much higher at very low and very high strikes.

The above arguments apply readily to the currency market where exchange rate returns exhibit thick tail distributions that are approximately symmetrical. In the stock market, volatility skew (i.e., low implied at high strike but high implied at low strike) is more common than volatility smile after the October 1987 stock market crash. Since the distribution is skewed to the far left, the right

tail can be thinner than the normal distribution. In this case implied volatility at high strike will be lower than that expected from a volatility smile.

3.2.2 Effect of Stochastic Volatility

The thick tail and nonsymmetrical distribution referred to in the previous section could be a result of volatility being stochastic. First, we rewrite (3) as

$$dS_t = \mu_s S_t dt + \sigma_t S_t dz_S, \quad (7)$$

and now σ_t has its own dynamics

$$d\sigma_t^2 = (\mu_v - \beta\sigma_t^2) dt + \sigma_v\sigma_t^2 dz_v, \quad (8)$$

where β is the speed of the volatility process mean reverting to the long-run average (μ_v/β) , σ_v is the *volatility of volatility*, and ρ , not shown above, is the correlation between dz_S and dz_v .

When $\rho = 0$, the price process and the volatility process are not correlated; σ_v alone is enough to produce kurtosis and Black-Scholes volatility smile. When $\rho < 0$, large negative return corresponds to high volatility stretching the left tail further into the left. On the other hand, when return is very high, volatility is low, “squashing” the right tail nearer to the centre. This will give rise to low implied volatility at high strikes and volatility skew. The reverse is true when $\rho > 0$.

Given that volatility is not a directly tradable asset, the hedging mechanism used in Black-Scholes may not apply and the risk neutral valuation principle has to be modified since volatility may command a risk premium. Different approaches to this problem have been adopted. Hull and White (1987) assume volatility risk is not priced. Wiggins (1987) derives various specifications of volatility risk premium according to different assumptions for risk preference. Steven Heston (1993) provides a specification where volatility risk premium is proportional to variance and extracts this volatility risk premium from option prices in the same manner

as implied volatility is extracted. Despite the variety of approaches adopted, consensus emerges on the degree of Black-Scholes pricing bias as a result of stochastic volatility. In the case where volatility is stochastic and $\rho = 0$, Black-Scholes overprices near-the-money (NTM) or at-the-money (ATM) options and the degree of overpricing increases with maturity. On the other hand, Black-Scholes underprices both in- and out-of-the-money options. In terms of implied volatility, ATM implied volatility would be lower than actual volatility while implied volatility of far-from-the-money options (i.e. either very high or very low strikes) will be higher than actual volatility. The pattern of pricing bias will be much harder to predict if ρ is not zero, there is a premium for bearing volatility risk, and if either or both values vary through time.

Some of the early work on option implied volatility focused on finding an optimal weighting scheme to aggregate implied volatility of options across strikes. (See David Bates 1996 for a comprehensive survey of these weighting schemes.) Since the plot of implied volatility against strikes can take many shapes, it is not likely that a single weighting scheme will remove all pricing errors consistently. For this reason and together with the liquidity argument presented below, ATM option implied volatility is often used for volatility forecast but not implied volatilities at other strikes.

3.2.3 *Market Microstructure and Measurement Errors*

Early studies of option implied volatility suffered many estimation problems¹⁰ such as the improper use of the Black-Scholes model for American-style options, the omission of dividend payments, the option price and the underlying asset prices not being recorded at the same time, or the use of

stale prices. Since transactions may take place at bid or ask prices, transaction prices of option and the underlying assets are subject to bid-ask bounce making the implied volatility estimation unstable. Finally, in the case of S&P 100 OEX option, the privilege of a wildcard option is often omitted.¹¹ In more recent studies, much of these measurement errors have been taken into account. Many studies use futures and options futures because these markets are more active than the cash markets and hence the smaller risk of prices being stale.

Conditions in the Black-Scholes model include no arbitrage, transaction cost of zero and continuous trading. The lack of such a trading environment will result in options being traded within a band around the theoretical price. This means that implied volatility estimates extracted from market option prices will also lie within a band even without the complications described in sections 3.2.1 and 3.2.2. Figlewski (1997) shows that implied volatility estimates can differ by several percentage points due to bid-ask spread and discrete tick size alone. To smooth out errors caused by bid-ask bounce, Harvey and Whaley (1992) use a nonlinear regression of ATM option prices observed in a ten-minute interval before the market close on model prices.

Indication of a non-ideal trading environment is usually reflected in poor trading volume. This means implied volatility of options written on different underlying assets will have different forecasting power. For most option contracts, ATM option has the largest trading volume. This supports the popularity of ATM implied volatility referred to in section 3.2.2.

3.2.4 *Investor Risk Preference*

In the Black-Scholes's world, investor risk preference is irrelevant in pricing options.

¹⁰ Stewart Mayhew (1995) gives a detailed discussion on such complications involved in estimating implied volatility from option prices, and Hentschel (2001) provides a discussion of the confidence intervals for implied volatility estimates.

¹¹ This wildcard option arises because the stock market closes later than the option market. An option trader is given the choice to decide, before the stock market closes, whether or not to trade on an option whose price is fixed at an earlier closing time.

Given that some of the Black-Scholes assumptions have been shown to be invalid, there is now a model risk. Figlewski and T. Clifton Green (1999) simulate option writers' positions in the S&P 500, DM/\$, US LIBOR, and T-Bond markets using actual cash data over a 25-year period. The most striking result from the simulations is that delta hedged short maturity options, with no transaction costs and a perfect knowledge of realised volatility, finished with losses on average in all four markets. This is clear evidence of Black-Scholes model risk. If option writers are aware of this model risk and mark up option prices accordingly, the Black-Scholes implied volatility would be greater than the true volatility.

In some situations, investor risk preference may override the risk neutral valuation relationship. Figlewski (1997), for example, compares the purchase of an OTM option to buying a lottery ticket. Investors are willing to pay a price that is higher than the fair price because they like the potential payoff and the option premium is so low that mispricing becomes negligible. On the other hand, we also have fund managers who are willing to buy comparatively expensive put options for fear of the collapse of their portfolio value. Both types of behavior could cause the market price of an option to be higher than the Black-Scholes price, translating into a higher Black-Scholes implied volatility. The arbitrage argument does not apply here because there is unique risk preference (or aversion) associated with some groups of individuals. Guenter Franke, Richard Stapleton, and Martin Subrahmanyam (1998) provide a theoretical framework in which such option trading behavior may be analyzed.

3.2.5 *Option Implied Volatility Measure and Forecast*

From the discussion above, we may deduce that the construction of VIX by the Chicago Board of Options Exchange is an example of good practice. VIX, short for volatility index, is an implied volatility com-

posite compiled from eight options written on the S&P 100. It is constructed such that it is at-the-money (by combining just-in and just-out-of-the-money options) and has a constant 28 calendar days to expiry (by combining the first nearby and second nearby options around the targeted 28 calendar days to maturity). Eight option prices are used, including four calls and four puts, to reduce any pricing bias and measurement errors caused by staleness in the recorded index level. Since options written on the S&P 100 are American style, a cash-dividend adjusted binomial model was used to capture the effect of early exercise. The mid bid-ask option price is used instead of traded price because transaction prices are subject to bid-ask bounce. (See Robert Whaley 1993; and Fleming, Barbara Ostdiek, and Whaley 1995 for further details.) Due to the calendar day adjustment, VIX is about 1.2 times (i.e. $\sqrt{365/252}$) greater than historical volatility computed using trading day data.

Once an implied volatility estimate is obtained, it is usually scaled by $\sqrt{n}$ to get an n -day ahead volatility forecast. In some cases, a regression model may be used to adjust for historical bias (e.g. Louis Ederington and Wei Guan 2000), or the implied volatility may be parameterized within a GARCH/ARFIMA model with or without its own persistence adjustment (e.g. Ted Day and Craig Lewis 1992; Bevan Blair, Poon, and Taylor 2001; Hwang and Satchell 1998).

As mentioned earlier, option implied volatility is perceived as a market's expectation of future volatility and hence it is a market based volatility forecast. Arguably it should be superior to a time series volatility forecast. On the other hand, we explained before that option model based forecast requires a number of assumptions to hold for the option theory to produce a useful volatility estimate. Moreover, option implied also suffers from many market driven pricing irregularities detailed above. Nevertheless, as we will learn in section 6, option implied volatility appears to have superior forecasting

capability, outperforming many historical price volatility models and matching the performance of forecasts generated from time series models that use a large amount of high frequency data.

4. Forecast Evaluation

Comparing forecasting performance of competing models is one of the most important aspects of any forecasting exercise. In contrast to the efforts made in the construction of volatility models and forecasts, little attention has been paid to forecast evaluation in the volatility forecasting literature.

4.1 Measuring Forecast Errors

Ideally, an evaluation exercise should measure the relative or absolute usefulness of a volatility forecast to investors. However, to do that one needs to know the decision process that will include these forecasts and the costs or benefits that result from using these forecasts. Other utility-based criterion, such as that used in West, Edison, and Cho (1993), requires some assumptions about the shape and property of the utility function. In practice these costs, benefits and utility function are not known and it is usual to simply use measures suggested by statisticians.

Popular evaluation measures used in the literature include *Mean Error* (ME), *Mean Square Error* (MSE), *Root Mean Square Error* (RMSE), *Mean Absolute Error* (MAE), and *Mean Absolute Percent Error* (MAPE). Other less commonly used measures include *Mean Logarithm of Absolute Errors* (MLAE), Theil-U statistic and LINEX. Except for the last two performance measures, all the other performance measures are self-explanatory. Assume that the subject of interest is X_t , $\hat{X}_t$ is the forecast of X_t , and that there are N forecasts. The Theil-U measure is:

$$\text{Theil-U} = \frac{\sum_{i=1}^N (\hat{X}_i - X_i)^2}{\sum_{i=1}^N (\hat{X}_i^{BM} - X_i)^2} \quad (9)$$

where $\hat{X}_i^{BM}$ is the Benchmark forecast, used here to remove the effect of any scalar transformation applied to X .

In the LINEX loss function below the positive errors are weighted differently from the negative errors:

$$\text{LINEX} = \frac{1}{N} \sum_{i=1}^N \left[\exp \left\{ -a(\hat{X}_i - X_i) \right\} + a(\hat{X}_i - X_i) - 1 \right]. \quad (10)$$

The choice of the parameter a is subjective. If $a > 0$, the function is approximately linear for over-prediction and exponential for under-prediction. Granger (1999) describes a variety of nonsymmetric cost, or loss, functions of which the LINEX is an example. Given that most investors would treat gains and losses differently, use of such functions may be advisable, but their use is not common in the literature.

4.2 Comparing Forecast Errors of Different Models

In the special case where the error distribution of one forecasting model dominates that of another forecasting model, the comparison is straightforward (Granger 1999). In practice, this is rarely the case, and most comparisons are based on the average figure of some statistical measures described in section 4.1. For statistical inference, West (1996), West and Cho (1995), and West and M. McCracken (1998) show how standard errors for ME, MSE, MAE, and RMSE may be derived taking into account serial correlation in the forecast errors and uncertainty inherent in model parameters estimates that were used to produce the forecasts. In general, West's (1996) asymptotic theory works for recursive scheme only, where newly observed data is used to expand the estimation period. However, a rolling fixed estimation-period method, where the oldest data is dropped whenever new data is added,

might be more appropriate if there is nonstationarity or time variation in model parameters estimates.

Diebold and Roberto Mariano (1995) propose three tests for “equal accuracy” between two forecasting models. The tests relate prediction error to some very general loss function and analyze loss differential derived from errors produced by two competing models. The three tests include an asymptotic test that corrects for series correlation and two exact finite sample tests based on the sign test and the Wilcoxon’s signed-rank test. Simulation results show that the three tests are robust against non-Gaussian, nonzero mean, serially, and contemporaneously correlated forecast errors. The two sign-based tests in particular continue to work well among small samples.

Instead of striving to make some statistical inference, model performance could be judged on some measures of economic significance. Examples of such an approach include portfolio improvement based on volatility forecasts (Fleming, Chris Kirby, and Ostdiek 2000, 2002). Some papers test forecast accuracy by measuring the impact on option pricing errors (G. Andrew Karolyi 1993). In this case, if there is any pricing error in the option model, the mistake in volatility forecast will be cancelled out when the option implied is reintroduced into the pricing formula. So it is not surprising that evaluation that involves comparing option pricing errors often prefers the implied volatility method to all other time series methods.

What has not yet been done in the literature is to separate the forecasting period into “normal” and “exceptional” periods. It is conceivable that different forecasting methods are suited for different trading environments.

entails regressing the “actual”, X_i , on the forecasts, $\hat{X}_i$, as shown below

$$X_i = \alpha + \beta \hat{X}_i + v_i. \quad (11)$$

Conditioning upon the forecast, the prediction is unbiased only if $\alpha = 0$ and $\beta = 1$. The standard errors of the parameter estimates are often computed based on Hansen and Hodrick (1980) since the error term, v_i , is heteroskedastic and serially correlated when overlapping forecasts are evaluated. In cases where there are more than one forecasting models, additional forecasts are added to the right-hand side of (11) to check for incremental explanatory power. Such forecast encompassing testing dates back to Henri Theil (1966). Yock Chong and David Hendry (1986), and Ray Fair and Robert Shiller (1989, 1990), provide further theoretical exposition of such method for testing forecast efficiency. The first forecast is said to subsume information contained in other forecasts if these additional forecasts do not significantly increase the adjusted regression R^2 . Alternatively, an orthogonality test may be conducted by regressing the residuals from (11) on other forecasts. If these forecasts are orthogonal, i.e. do not contain additional information, then the regression coefficients will not be different from zero.

While it is useful to have an unbiased forecast, it is important to distinguish between biasness and predictive power. A biased forecast can have predictive power if the bias can be corrected. An unbiased forecast is useless if forecast errors are always big. For X_i to be considered as a good forecast, $\text{Var}(v_i)$ should be small and R^2 for the regression should tend to 100 percent. Blair, Poon, and Taylor (2001) use the proportion of explained variability, P , to measure explanatory power

4.3 Regression Based Forecast Efficiency and Orthogonality Test

The regression-based method for examining the informational content of forecasts

$$P = 1 - \frac{\sum (X_i - \hat{X}_i)^2}{\sum (X_i - \mu_X)^2}. \quad (12)$$

The ratio in the right-hand side of (12) compares the sum of squared prediction errors

(assuming $\alpha = 0$ and $\beta = 1$ in (11)) with the sum of squared variations of X_t . P compares the amount of variations in the forecast errors with that in actual volatility. If prediction errors are small, P is closer to 1. Given that a regression model that produces (12) is more restrictive than (11), P is likely to be smaller than conventional R^2 . P can even be negative since the ratio in the right hand side of (12) can be greater than 1. A negative P means that the forecast errors have a greater amount of variations than the actual volatility, which is not a desirable characteristic for a well-behaved forecasting model.

4.4 Using Squared Return to Proxy Actual Volatility

Given that volatility is a latent variable, the actual volatility X is often estimated from a sample using equation (1), which is not entirely satisfactory when the sample size is small. Before high frequency data becomes widely available, many researchers have resorted to using daily squared return, calculated from market closing prices, to proxy daily volatility. As shown in Lopez (2001), while ϵ_t^2 is an unbiased estimator of σ_t^2 , it is very imprecise due to its asymmetric distribution. Let

$$Y_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t \quad (13)$$

and $z_t \sim (0,1)$. Then

$$E[\epsilon_t^2 | \Phi_{t-1}] = \sigma_t^2 E[z_t^2 | \Phi_{t-1}] = \sigma_t^2$$

since $z_t^2 \sim \chi_{(1)}^2$. However, since the median of a $\chi_{(1)}^2$ distribution is 0.455, $\epsilon_t^2 < \frac{1}{2} \sigma_t^2$ more than 50 percent of the time. In fact

$$\begin{aligned} Pr\left(\epsilon_t^2 \in \left[\frac{1}{2}\sigma_t^2, \frac{3}{2}\sigma_t^2\right]\right) &= Pr\left(z_t^2 \in \left[\frac{1}{2}, \frac{3}{2}\right]\right) \\ &= 0.2588, \end{aligned}$$

which means that ϵ_t^2 is 50 percent greater or smaller than σ_t^2 nearly 75 percent of the time.

Under the null hypothesis that returns in (13) are generated by a GARCH(1,1) process,

Andersen and Bollerslev (1998) show that the population R^2 for the regression

$$\epsilon_t^2 = \alpha + \beta \hat{\sigma}_t^2 + \nu_t$$

is equal to κ^{-1} where κ is the kurtosis of the standardized innovations and κ is finite. For conditional Gaussian error, the R^2 from a correctly specified GARCH(1,1) model is bounded from above by $\frac{1}{3}$. Christodoulakis and Satchell (1998) extend the results to include compound normals and the Gram-Charlier class of distributions and show that the misestimation of forecast performance is likely to be worsened by non-normality known to present in financial data.

Hence, the use of ϵ_t^2 as volatility proxy will lead to low R^2 and undermine the inference regarding forecast accuracy. Extra caution is called for when interpreting empirical findings in studies that adopt such a noisy volatility estimator. Blair, Poon, and Taylor (2001) report an increase of R^2 by three to four times for the one-day ahead forecast when intra-day five-minute square returns instead of daily square returns are used to proxy the actual volatility.

4.5 Further Issues in Forecast Evaluation

In all forecast evaluations, it is important to distinguish in-sample and out-of-sample forecasts. In-sample forecast, which is based on parameters estimated using all data in the sample, implicitly assumes parameter estimates are stable through time. In practice, time variation of parameter estimates is a critical issue in forecasting. A good forecasting model should be one that can withstand the robustness of an out-of-sample test, a test design that is closer to reality. In our analyses of empirical findings in sections 5 and 6, we focus our attention only on studies that implement out-of-sample forecasts.

An issue not addressed previously is whether volatility X in (9), (10), (11), and (12) should be standard deviation or variance. The complication generated by this choice could be further compounded with the choice of performance measure (e.g.

MAE or MSE). The square of a variance error is the 4th power of the same error measured from standard deviation. This can complicate the task of forecast evaluation, given the difficulty in estimating fourth moments with common distributions let alone the thick-tailed ones in finance. The confidence interval of the mean error statistic can be very wide when forecast errors are measured from variances and worse if they are squared. This leads to difficulty in finding significant differences between alternative forecasting methods. For this reason, one may even consider using a logarithmic transformation (as in Pagan and Schwert 1990) to reduce the impact of outliers.

Marie Davidian and Raymond Carroll (1987) make similar observations in their study of variance function estimation for heteroskedastic regression. Using high order theory, they show that the use of square returns for modelling variance is appropriate only for approximately normally distributed data, and becomes non-robust when there is a small departure from normality. Estimation of the variance function that is based on logarithmic transformation or absolute returns is more robust against asymmetry and non-normality. More recently, Andersen Bollerslev, Diebold, and Labys (2001) and Andersen, Bollerslev, Diebold, and Ebens (2001) find realized volatility estimated from high frequency currency and stock returns are approximately lognormal. These findings are generally consistent with X being logarithmic volatility.

Bollerslev and Ghysels (1996) further suggest a heteroskedasticity-adjusted version of MSE called HMSE where

$$\text{HMSE} = \frac{1}{N} \sum_{t=1}^N \left[\frac{X_{T+t}}{\hat{X}_{T+t}} - 1 \right]^2.$$

In this case, the forecast error is effectively scaled by actual volatility. This type of performance measure is not appropriate if the absolute magnitude of the forecast error is a major concern.

5. Volatility Forecasting Based On Time Series Models

In this section, we review major findings in 44 papers that construct volatility forecasts based on historical information only. We will make some references to implied volatility forecasts when we discuss forecasting performance of SV and long memory volatility models. Main findings regarding implied volatility forecasts will be discussed in section 6.

5.1 Pre-ARCH Era and Non-ARCH Debate

Taylor (1987) is one of the earliest to test time series volatility forecasting models before ARCH/GARCH permeated the volatility literature. Taylor (1987) studies the use of high, low, and closing prices to forecast one to twenty days DM/\$ futures volatility and finds a weighted average composite forecast to perform best. Wiggins (1992) also gives support to extreme value volatility estimators.

In the pre-ARCH era, there were many other findings covering a wide range of issues. Dimson and Marsh (1990) find *ex ante* time-varying optimized weighting schemes do not always work well in out-of-sample forecasts. Sill (1993) finds S&P 500 volatility is higher during recession and that commercial T-Bill spread helps to predict stock market volatility. Andrew Alford and James Boatman (1995) find, from a sample of 6,879 stocks, that adjusting historical volatility towards volatility estimates of comparable firms in the same industry and size provides a better five-year ahead volatility forecast. Alford and Boatman (1995), Figlewski (1997), and Figlewski and Green (1999) all stress the importance of having a long enough estimation period to make good volatility forecasts over a long horizon.

5.2 The Explosion of ARCH/GARCH Forecasting Contests

Vedat Akigray (1989) is one of the earliest to test the predictive power of GARCH and

is commonly cited in many later GARCH studies, though an earlier investigation appeared in Taylor (1986). In the following decade, there were no less than twenty papers testing GARCH's predictive power against other time series methods and against option implied volatility forecasts. The majority of these forecast volatility of major stock indices and exchange rates.

The ARCH/GARCH models and their variants have many supporters. Akgiray finds GARCH consistently outperforms EWMA and HIS (i.e. historical volatility derived from standard deviation of past returns over a fixed interval) in all subperiods and under all evaluation measures. Pagan and Schwert (1990) find EGARCH is best especially in contrast to nonparametric methods. Despite a low R^2 , Cumby, Figlewski, and Hasbrouck (1993) conclude that EGARCH is better than naïve historical methods. Figlewski (1997) finds GARCH superiority confined to stock market and for forecasting volatility over a short horizon only. Cao and Tsay (1992) find TAR provides the best forecast for large stocks and EGARCH gives the best forecast for small stocks, and they suspect that the latter might be due to a leverage effect. Bali (2000) documents the usefulness of GARCH models, the nonlinear ones in particular, in forecasting one-week ahead volatility of U.S. T-Bill yields.

Other studies find no clear-cut result. These include Keun Yeong Lee (1991), West and Cho (1995), Chris Brooks (1998), and David McMillan, Alan Speight, and Dwain Gwilym (2000). Some models work best under different error statistics (e.g. MAE, MSE), different sampling schemes (e.g. rolling fixed sample estimation, or recursive expanding sample estimation), different time periods and for different assets. Timothy Brailsford and Robert Faff (1996) comment that the GJR-GARCH model has a marginal lead while Franses and Van Dijk (1996) claim the GJR forecast cannot be recommended. Many of these inconclusive studies share one or more of the following

characteristics: (i) they test a large number of very similar models all designed to capture volatility persistence; (ii) they use a large number of error statistics, each of which has a very different loss function; (iii) they forecast and calculate error statistics for variance and not standard deviation, which makes the difference between forecasts of different models even smaller, yet the standard error is large as the fourth moment is unstable; and (iv) they use squared daily, weekly, or monthly returns to proxy daily, weekly, or monthly "actual volatility," which results in extremely noisy volatility estimates; the noise makes the small differences between forecasts of similar models indistinguishable.

Unlike the ARCH class model, the "simpler" methods, including the EWMA method, do not separate volatility persistence from volatility shocks and most of them do not incorporate volatility mean reversion. The GJR model allows the volatility persistence to change relatively quickly when return switches sign from positive to negative and vice versa. If unconditional volatility of all parametric volatility models is the same, then GJR will have the largest probability of an underforecast.¹² The "simpler" methods tend to provide larger volatility forecasts most of the time because there is no constraint on stationarity or convergence to the unconditional variance, and may result in larger forecast errors. This possibly explains why GJR was the worst performing model in Franses and Van Dijk (1996) because they use MedSE (median squared error) as their sole evaluation criteria. In Brailsford and Faff (1996), the GJR(1,1) model outperforms the other models when MAE, RMSE, and MAPE are used.

¹² This characteristic is clearly evidenced in table 2 of Brailsford and Faff (1996). The GJR(1,1) model under-forecasts 76 (out of 90) times. The RW model has an equal chance of under- and over-forecasts, whereas all the other methods overforecast more than 50 (out of 90) times.

There are some merits to using “simpler” methods, and especially models that include long distributed lags. As ARCH class models assume variance stationarity, the forecasting performance suffers when there are changes in volatility level. Parameters estimation becomes unstable when data period is short or when there is a change in volatility level. This has led to GARCH convergence problem in several studies (e.g. Tse and Tung 1992, and Walsh and Tsou 1998). Stephen Taylor (1986), Tse (1991), Tse and Tung (1992), Boudoukh, Richardson, and Whitelaw (1997), Walsh and Tsou (1998), Ederington and Guan (1999), Ferreira (1999), and James Taylor (2001) all favor some forms of exponential smoothing method to GARCH for forecasting volatility of a wide range of assets across equities, exchange rates and interest rates.

In general, models that allow for volatility asymmetry came out well in the forecasting contest because of the strong negative relationship between volatility and shock. Charles Cao and Ruey Tsay (1992), Ronald Heynen and Harry Kat (1994), Lee (1991), and Adrian Pagan and G. William Schwert (1990) favor the EGARCH model for volatility of stock indices and exchange rates, whereas Brailsford and Faff (1996) and Taylor (2001) find GJR-GARCH to outperform GARCH in stock indices. Turan Bali (2000) finds a range of nonlinear models works well for interest rate volatility.

5.3 *The Arrival of SV Forecasts*

The SV model has an additional innovative term in the volatility dynamics and, hence, is more flexible than ARCH class models and was found to fit financial market returns better and have residuals closer to standard normal. It is also closer to theoretical models in finance and especially those in derivatives pricing. However, largely due to the computation difficulty, volatility forecast based on the SV model was not studied until the mid 1990's, a decade later than ARCH/GARCH development. In a PhD

thesis, Heynen (1995) finds SV forecast is best for a number of stock indices across several continents. At the time of writing, there are only six other SV studies and their view about SV forecasting performance is by no means unanimous.

Heynen and Kat (1994) forecast volatility for seven stock indices and five exchange rates and find SV provides the best forecast for indices but produces forecast errors that are ten times larger than EGARCH's and GARCH's for exchange rates. Jun Yu (2002) ranks SV top for forecasting New Zealand stock market volatility, but the margin is very small, partly because the evaluation is based on variance and not standard deviation. Lopez (2001) finds no difference between SV and other time series forecasts using conventional error statistics. All three papers have the 1987 crash in the in-sample period, and the impact of the 1987 crash on their result is unclear.

Three other studies—Hagen Bluhm and Yu (2000); Chris Dunis, Jason Laws, and Stephane Chauvin (2000); and Eugenie Hol and Koopman (2002)—compare SV and other time series forecasts with option implied volatility forecast. Dunis et al. (2000) find combined forecast is the best for six exchange rates so long as the SV forecast is excluded. Bluhm and Yu (2000) rank SV equal to GARCH. Both Bluhm and Yu (2000) and Hol and Koopman (2002) conclude that implied is better than SV for forecasting stock index volatility.

5.4 *Recent Development in Long Memory Volatility Models*

Volatility forecasts based on models that exploit the long memory (LM) characteristics of volatility appear rather late in the literature. These include Andersen, Bollerslev, Diebold and Labys (2002), Jon Vilasuso (2002), Zumbach (2002) and three other papers that compare LM forecasts with option implied volatility, viz. Kai Li (2002), Martens and Jason Zein (2002), and Shiu-Yan Pong et al. (2002). We pointed out in

section 3.1.2 that other short memory models (e.g. extreme values, breaks, mixture of distribution, and regime switching) are also capable of producing long memory in second moments, and each of them entails a different data generating process. At the time of writing, there is no direct contest between these and the LM models.

An earlier LM paper by Hwang and Satchell (1998) uses LM models to forecast Black-Scholes implied volatility of equity option. This paper contains some useful insights about properties of LM models, but since we are focusing on forecasting volatility of the underlying asset rather than implied volatility, the results of Hwang and Satchell (1998) will not be discussed here.

Examples of LM models include the FIGARCH in Baillie, Bollerslev, and Mikkelsen (1996) and FIEGARCH in Bollerslev and Mikkelsen (1996). In Andersen, Bollerslev, Diebold, and Labys (2002) a vector autoregressive model with long distributed lags was built on realized volatility of three exchange rates, which they called the VAR-RV model. In Zumbach (2002) the weights apply to time series of realized volatility following a power law, which he called the LM-ARCH model.

As noted before in section 3.1.2, all fractional integrated models of volatility have a non-zero drift in the volatility process. In practice the estimation of fractional integrated models requires an arbitrary truncation of the infinite lags and as a result, the mean will be biased. Zumbach's (2002) LM-ARCH will not have this problem because of the fixed number of lags and the way in which the weights are calculated. Hwang and Satchell's (1998) scaled-truncated log-ARFIMA model is mean adjusted to control for the bias that is due to this truncation and the log transformation.

Among the historical price models, Vilasuso (2002) finds FIGARCH produces significantly better one- and ten-day ahead volatility forecasts for five major exchange rates. Zumbach (2002) produces only

one-day ahead forecasts and finds no difference among model performance. Andersen, Bollerslev, Diebold, and Labys (2002) find the realized volatility constructed VAR model, i.e. VAR-RV, produces the best one- and ten-day ahead volatility forecasts. It is difficult to attribute this superior performance to the LM model alone because the VAR structure allows a cross series linkage that is absent in all other univariate models, and we also know that the more accurate realized volatility estimates would result in improved forecasting performance, everything else equal.

The other three papers that compare forecasts from LM models with implied volatility forecasts generally find implied volatility forecast produces the highest explanatory power. Martiens and Zein (2002) find log-ARFIMA forecast beats implied in S&P 500 futures but not in ¥US\$ and crude oil futures. Li (2002) finds implied produces better short-horizon forecast, whereas the ARFIMA provides better forecast for a six-month horizon. However, when regression coefficients are constrained to be $\alpha = 0$ and $\beta = 1$, the regression R^2 becomes negative at long horizons. From our discussion in section 4.3, this suggests that volatility at the six-month horizon might be better forecast using the unconditional variance instead of model-based forecasts.

As all LM papers in this group were written very recently and after the publication of Andersen and Bollerslev (1998), the realized volatilities are constructed from intra-day high frequency data. When comparison is made with implied volatility forecast, however, the implied volatility is usually extracted from daily closing option prices. Despite the lower data frequency, option implied volatility appears to outperform forecasts from LM models built on high-frequency data.

5.5 *Regime Switching Models*

It has long been argued that the financial market reacts to large and small shocks differently, and the rate of mean reversion is faster for large shocks. Benjamin Friedman

and David Laibson (1989), Charles Jones, Owen Lamont, and Robin Lumsdaine (1998) and Ederington and Lee (2001) all provide explanations and empirical support for the conjecture that volatility adjustment in high and low volatility states follows a twin-speed process; slower adjustment and more persistent volatility in a low volatility state and faster adjustment and less volatility persistence in a high volatility state.

One approach for modelling changing volatility persistence is to use a Hamilton (1989) type regime switching (RS) model, which like GARCH model is strictly stationary and covariance stationary. The TAR model used in Cao and Tsay (1992) is similar to a SV model with regime switching, and Cao and Tsay (1992) prefers TAR to EGARCH and GARCH. The earlier RS applications, such as Pagan and Schwert (1990) and Hamilton and Susmel (1994) tend to be more rigid, where conditional variance is state dependent but not time dependent. Until recently, only ARCH class conditional variance is permitted. Recent extensions by Gray (1996) and Klaassen (2002) allow GARCH type heteroskedasticity in each state and the probability of switching between states to be time dependent.

Hamilton and Rauli Susmel (1994) find regime switching ARCH with leverage effect produces better volatility forecast than asymmetry version of GARCH. Hamilton and Gang Lin (1996) use a bivariate RS model and find stock market returns are more volatile during recession periods. Gray (1996) fits an RSGARCH (1,1) model to U.S. one-month T-Bill rates, where the rate of mean level reversion is permitted to differ under different regimes, and find substantial improvement in forecasting performance. Klaassen (2002) also applies RSGARCH (1,1) to the foreign exchange market and finds a superior, though less dramatic, performance.

It is worth noting that interest rates are different from the other assets in that interest rates exhibit a “level” effect, i.e., volatility depends on the level of the interest rate. It is

plausible that it is this level effect that Gray (1996) is picking up that results in superior forecasting performance. This level effect also appears in some European short rates (Ferreira 1999). There is no such level effect in exchange rates and so it is not surprising that Klaassen (2002) does not find similar dramatic improvement.¹³

5.6 *Extreme Values and Outliers*

There are at least two stylized facts about volatility in the financial markets that were not captured by ARCH models: (i) The standardized residuals from ARCH models still display large kurtosis (see Thomas McCurdy and Ieuan Morgan 1987; Anders Milhoj 1987; Hsieh 1989; and Baillie and Bollerslev 1989). Conditional heteroskedasticity alone could not account for all the tail thickness. This is true even when the Student-t distribution is used to construct the likelihood function (see Bollerslev 1987; and Hsieh 1989); (ii) The ARCH effect is significantly reduced or disappears once large shocks are controlled for (Reena Aggarwal, Carla Inclan, and Ricardo Leal 1999). Franses and Hendrik Ghijsels (1999) find forecasting performance of the GARCH model is substantially improved in four out of five stock markets studied when the additive outliers are removed.

Diebold and Peter Pauly (1987), and Christopher Lamoureux and William Lastrapes (1990) show that the high volatility persistence in the GARCH model could be due to structural changes in the variance process. A shift in unconditional variance will result in volatility persistence in GARCH that assumes covariance stationarity. Philip Kearns and Pagan (1993) investigate whether volatility persistence was an artifact of extremes or outliers by symmetrically trimming the scores of the largest observations, but find volatility persistence remained in Australian stock index returns. On the other hand, Aggarwal, Inclan, and Leal (1999) use the Inclan and George Tiao

¹³ We thank a referee for this important insight.

(1994) method to identify and adjust for volatility level changes and find a big difference after these level shifts are controlled for. The difference between the two approaches is tenuous, however. As the time between volatility level changes gets smaller, the second approach converges to the first.

At the time of writing, there is no consensus about the treatment of extreme values and outliers; whether they should simply be removed or trimmed or their impact on volatility be separately handled. The financial market literature is also rather loose in its terminology regarding outliers and extremes, which can lead to opaque discussions. To a statistician, there are two extremes in each sample, the minimum and the maximum, although this can extend to the first few ordinal statistics at each end. It follows that the number of extremes does not increase with sample size, n . The number of terms in the 5-percent tail of the distribution does increase with sample size, being $0.05n$, so that tails and extremes are not identical concepts; extremes lie in the tails but tails include data that are not extremes. Of course, for small samples, the two sets become almost identical, but they are quite different for large samples. Tails are part of the distribution but for large samples true extremes can be considered to be drawn from an extreme-value distribution. In the far tail the two could be the same. "Outliers" could be drawn from a quite different distribution when the market goes into a different mode, in which case the observed distribution is a mixture of the two. An obvious problem is that there are rather few observations from the outlier distributions and so estimation is difficult.

The volume-volatility literature has documented a strong link between contemporaneous trading volume and conditional volatility. It is plausible that residuals that are scaled by trading volume might be approximately Gaussian and the thick tails removed. No doubt, the Christmas wish list of volatility forecasters must be for a method of forecasting crashes, even if it is for a short

period ahead. It may be possible to find such a method using options or very high frequency data, but a great deal of further exploration is required.

5.7 *Getting the Right Conditional Variance and Forecast with the "Wrong" Models*

Many of the time series volatility models including the GARCH models can be thought of as approximating a deeper time-varying volatility construction, possibly involving several important economic explanatory variables. Since time series models involve only lagged returns but have considered many forms of specification, it seems likely that they will provide an adequate, possibly even a very good approximation to actuality for long periods but not at all times. This means that they will forecast well on some occasions, but less well on others, depending on fluctuations in the underlying driving variables.

Daniel Nelson (1992) proves that if the true process is a diffusion or near-diffusion model with no jumps, then even when misspecified, appropriately defined sequences of ARCH terms with a large number of lagged residuals may still serve as consistent estimators for the volatility of the true underlying diffusion, in the sense that the difference between the true instantaneous volatility and the ARCH estimates converges to zero in probability as the length of the sampling frequency diminishes. Nelson (1992) shows that such ARCH models may misspecify both the conditional mean and the dynamic of the conditional variance; in fact the misspecification may be so severe that the models make no sense as data generating processes; they could still produce consistent one-step-ahead conditional variance estimates and short-term forecasts.

Nelson and Dean Foster (1995) provide further conditions for such misspecified ARCH models to produce consistent forecasts over medium and long terms. They show that forecasts of the process and its volatility generated by these misspecified

models will converge in probability to the forecast generated by the true diffusion or near diffusion process provided that all unobservable state variables are consistently estimated and that the conditional mean and conditional covariances of all state variables are correctly specified. An example of a true diffusion process given by Nelson and Foster (1995) is the stochastic volatility model described in section 3.2.2.

These important theoretical results confirm our empirical observations that under normal circumstances, i.e. no big jumps in prices, there may be little practical difference in choosing between volatility models provided that the sampling frequency is small and that whichever model one has chosen, it contains long enough lagged residuals. This might be an explanation for the success of high-frequency and long-memory volatility models (e.g. Blair, Poon, and Taylor 2001; and Andersen et al. 2002).

6. Volatility Forecasting Based On Option ISD

In contrast to time series volatility forecasting models described in section 6, the use of option ISD (Implied Standard Deviation) as a volatility forecast involves some extra complexities. A test on the forecasting power of option ISD is a joint test of option market efficiency and a correct option pricing model. Since trading frictions differ across assets, some options are easier to replicate and hedge than others. It is therefore reasonable to expect different levels of efficiency and different forecasting power for options written on different assets. We will focus on this aspect of the forecasting contest in section 6.1 and compare implied and time series volatility forecasts within each asset class.

While each historical price constitutes an observation in the sample used in calculating volatility forecast, each option price constitutes a volatility forecast over the option maturity, and there can be many option prices

at any one time. As mentioned in section 3.2, there are also problems of volatility smile and volatility skew. Options of different strike prices produce different Black-Scholes implied volatility estimates. Section 6.2 discusses the information content of ISD across strikes and the effectiveness of different weighting schemes used to produce an implied volatility composite for volatility forecasting.

The issue of a correct option pricing model is more fundamental in finance. Option pricing has a long history and various extensions have been made since Black-Scholes to cope with dividend payments, early exercise, and stochastic volatility. However, none of the option pricing models (except Heston 1993) that appeared in the volatility forecasting literature allows for a premium for bearing volatility risk. In the presence of a volatility risk premium, we expect the option price to be higher, which means implied volatility derived using an option pricing model that assumes zero volatility risk premium (such as the Black-Scholes model) will also be higher, and hence automatically be more biased. Section 6.3 examines the issue of biasness of ISD forecasts and evaluates the extent to which implied biasness is due to the omission of volatility risk premium.

6.1 Predictability Across Different Assets

As noted in section 3.2.3, early studies that test forecasting power of option ISD are fraught with various deficiencies. Despite these complexities, option ISD has been found empirically to contain a significant amount of information about future volatility and it often beats volatility forecasts produced by sophisticated time series models. Such a superior performance appears to be common across assets.

6.1.1 Individual Stocks

Henry Latane and Richard Rendleman (1976) were the first to discover the forecasting capability of option ISD. They find actual volatilities of 24 stocks calculated from

in-sample periods and extended partially into the future are more closely related to implied than historical volatility. Donald Chiras and Steven Manaster (1978) and Stan Beckers (1981) find prediction from implied can explain a large amount of the cross-sectional variations of individual stock volatilities. Chiras and Manaster (1978) document an R^2 of 34–70 percent for a large sample of stock options traded on CBOE whereas Beckers (1981) reports R^2 of 13–50 percent for a sample that varies from 62 to 116 U.S. stocks over the sample period. Gordon Gemmill (1986) produces R^2 of 12–40 percent for a sample of thirteen U.K. stocks. Richard Schmalensee and Robert Trippi (1978) find implied volatility rises when stock price falls and that implied volatilities of different stocks tend to move together. From a time series perspective, Lamoureux and Lastrapes (1993) and George Vasilellis and Nigel Meade (1996) find implied could also predict time series variations of equity volatility better than forecasts produced from time series models.

The forecast horizons of this group of studies are usually quite long, ranging from three months to three years. Studies that examine incremental information content of time series forecasts find volatility historical average provides significant incremental information in both cross-sectional (Beckers 1981; Chiras and Manaster 1978; Gemmill 1986) and time series settings (Lamoureux and Lastrapes 1993) and that combining GARCH and implied produces the best forecast (Vasilellis and Meade 1996). These findings have been interpreted as an evidence of stock option market inefficiency since option implied does not subsume all information. In general, stock option implied exhibits instability and suffers most from measurement errors and bid-ask spread because of the low liquidity.

6.1.2 *Stock Market Index*

There are 22 studies that use index option ISD to forecast stock index volatility; seven of these forecast volatility of S&P 100, ten

forecast volatility of S&P 500, and the remaining five forecast index volatility of smaller stock markets. The S&P 100 and S&P 500 forecasting results make an interesting contrast as almost all studies that forecast S&P 500 volatility use S&P 500 futures options, which are more liquid and less prone to measurement errors than the OEX stock index option written on S&P 100. We will return to the issue of measurement errors when we discuss biasness in section 6.3.

All but one study (viz. Linda Canina and Stephen Figlewski 1993) conclude that implied contains useful information about future volatility. Blair, Poon, and Taylor (2001) and Allen Potesman (2000) record the highest R^2 for S&P 100 and S&P 500 respectively. About 50 percent of index volatility is predictable up to a four-week horizon when actual volatility is estimated more accurately using very high frequency intra-day returns.

Similar, but less marked, forecasting performance emerged from the smaller stock markets, which include the German, Australian, Canadian, and Swedish markets. For a small market such as the Swedish market, Per Frennberg and Bjorn Hanssén (1996) find seasonality to be prominent and that implied forecast cannot beat simple historical models such as the autoregressive model and random walk. Very erratic and unstable forecasting results were reported in Brace and Hodgson (1991) for the Australian market. Craig Doidge and Jason Wei (1998) find the Canadian Toronto index is best forecast with GARCH and implied combined, whereas Bluhm and Yu (2000) find VDAX, the German version of VIX, produces the best forecast for the German stock index volatility.

A range of forecast horizons were tested among this group of studies though the most popular choice is one month. There is evidence that the S&P implied contains more information since the 1987 crash (see Christensen and Prabhala 1998 for S&P 100; and Ederington and Guan 2002 for S&P 500). Some described this as the “awakening” of the S&P option markets.

About half of the papers in this group test if there is incremental information contained in time series forecasts. Day and Lewis (1992), Ederington and Guan (1999, 2002), and Martin and Zein (2002) find ARCH class models and volatility historical average add a few percentage points to the R^2 , whereas Blair, Poon, and Taylor (2001); Christensen and Prabhala (1998); Fleming (1998); Fleming, Ost diek, and Whaley (1995); Hol and Koopman (2002); and Szamany, Ors, Kim, and Davidson (2002) all find option implied dominates time series forecasts.

6.1.3 Exchange Rate

The strong forecasting power of implied is again confirmed in the currency markets. Sixteen papers study currency options for a number of major currencies, the most popular of which are DM/US\$ and ¥/US\$. Most studies find implied to contain information about future volatility for short horizons up to three months. Li (2002) and Scott and Tucker (1989) find implied forecasts well for up to a six- to nine-months horizon. Both studies register the highest R^2 in the region of 40–50 percent.

A number of studies in this group find implied beats time series forecasts including volatility historical average (see Hung-Gay Fung, Chin-Jen Lie, and Abel Moreno 1990; and Wei and Frankel 1991) and ARCH class models (see Dajiang Guo 1996a,b; Phillip Jorion 1995, 1996; Martens and Zein 2002; Pong et al. 2002; Andrew Szakmary et al. 2002; and Xu and Taylor 1995). Some studies find combined forecast is the best choice (see Dunis, Law, and Chauvin 2000; and Taylor and Xu 1997).

Two studies find high frequency intra-day data can produce more accurate time series forecast than implied. Fung and Hsieh (1991) find one-day ahead time series forecast from a long-lag autoregressive model fitted to fifteen-minute returns is better than implied. Li (2002) finds the ARFIMA model outperformed implied in long horizon forecasts while implied dominates over

shorter horizons. Implied forecasts were found to produce higher R^2 than other long memory models, such as the Log-ARFIMA model in Martens and Zein (2002), and Pong et al. (2002). All these long memory forecasting models are more recent and are built on volatility compiled from high frequency intra-day returns, while the implied volatility remains to be constructed from less frequent daily option prices.

6.1.4 Other Assets

The forecasting power of implied from interest rate options was tested in Malcolm Edey and Graham Elliot (1992), Fung and Hsieh (1991), and Kaushik Amin and Victor Ng (1997). Interest rate option models are very different from other option pricing models because of the need to price all interest rate derivatives consistently at the same time in order to rule out arbitrage opportunities. Trading in interest rate instruments is highly liquid as trading friction and execution cost are negligible. Practitioners are more concerned about the term structure fit than the time series fit, as millions of pounds of arbitrage profits could change hands instantly if there is any inconsistency in contemporaneous prices.

Earlier studies such as Edey and Elliot (1992), and Fung and Hsieh (1991) use the Black model (a modified version of Black-Scholes), which prices each interest rate option without cross referencing to prices of other interest rate derivatives. The single factor Heath-Jarrow-Morton model used in Amin and Ng (1997), and fitted to short rate only, works in the same way, although these authors have added different constraints to the short rate dynamics, as the main focus of their paper is to compare different variants of short rate dynamics. Despite the added complications, all three studies find significant forecasting power in implied volatility of interest rate (futures) options. Amin and Ng (1997) in particular report R^2 of 21 percent for twenty-day ahead volatility forecasts, and volatility historical average adds only a few percentage points to the R^2 .

Implied volatilities from options written on non-financial assets were examined in Day and Lewis (1993, crude oil); Kroner, Kneafsey, and Claessens (1995, agriculture and metals); Martens and Zein (2002, crude oil); and Szakmary et al. (2002; 35 futures options contracts across nine markets including S&P 500, interest rates, currency, energy, metals, agriculture, and livestock futures). All four studies find implied dominates time series forecasts, although Kroner, Kneafsey, and Claessens (1995) find combining GARCH and implied produces the best forecast.

6.2 *ATM (At-the-Money) or Weighted Implied?*

Since options of different strikes have been known to produce different implied volatilities, a decision has to be made as to which of these implied volatilities should be used, or which weighting scheme should be adopted, to produce a forecast that is most superior. The most common strategy is to choose the implied derived from ATM option based on the argument that ATM option is the most liquid and hence ATM implied is least prone to measurement errors. ATM implied is also theoretically the most sound. Feinstein (1989a) shows that for the stochastic volatility process described in Hull and White (1987), implied volatility from ATM and near expiration option provides the closest approximation to the average volatility over the life of the option provided that volatility risk premium is either zero or a constant. This means that if volatility is stochastic, ATM implied is least prone to bias as well compared with implied at other strikes.

If ATM implied is not available, then NTM (nearest-to-the-money) option is used instead. Sometimes, to reduce measurement errors and the effect of bid-ask bounce, an average is taken from a group of NTM implied volatilities. VIX described in section 3.2.5, for example, is an average of eight implied volatilities derived from four calls and four puts and a weighting scheme aiming to produce a composite implied that is of a con-

stant maturity of 28 calendar days and approximately ATM. Other weighting schemes that also give greater weight to ATM implied are vega (i.e. the partial derivative of option price w.r.t. volatility) weighted or trading volume weighted, weighted least square (WLS) and some multiplicative versions of these three. The WLS method, which first appeared in Whaley (1982), aims to minimize the sum of squared errors between the market and the theoretical prices of a group of options. Since ATM option has the highest trading volume and ATM option price is the most sensitive to volatility input, all three weighting schemes (and the combinations thereof) have the effect of placing the greatest weight on ATM implied. Other weighting schemes such as equally weighted, and weight based on the elasticity of option price to volatility, that do not emphasize ATM implied, are less popular.

The forecasting power of individual and composite implied volatilities has been tested in Ederington and Guan (2000); Fung, Lie, and Moreno (1990); Gemmill (1986); Kroner, Kevin Kneafsey, and Stijn Claessens (1995); Scott and Tucker (1989); and Vasilellis and Meade (1996). The general consensus is that among the weighted implied volatilities, those that have a VIX-style composite weight seem to be the best, followed by schemes that favor ATM option such as the WLS and the vega weighted implied. The worst performing ones are equally weighted and elasticity weighted implied using options across all strikes. Different findings emerged as to whether an individual implied forecasts better than a composite implied. Becker (1981), Feinstein (1989b), Fung, Lie, and Moreno (1990), and Gemmill (1986) find evidence to support individual implied although they all prefer a different implied (viz. ATM, Just-OTM, OTM, and ITM respectively for the four studies). Kroner, Kneafsey, and Claessens find composite implied forecasts better than ATM implied. On the other hand, Scott and Tucker (1989) conclude that when emphasis

is placed on ATM implied, which weighting scheme one chooses does not really matter.

As mentioned in section 6.1.1, implied volatility, especially that of stock option, can be quite unstable across time. Beckers (1981) finds taking a five-day average improves forecasting power of stock option implied. Shaikh Hamid (1998) finds such an intertemporal averaging is also useful for stock index option during very turbulent periods. On a slightly different note, Xu and Taylor (1995) find implied estimated from sophisticated volatility term structure model produces similar forecasting performance as implied from the shortest maturity option.

A series of studies by Ederington and Guan have reported some interesting findings. Ederington and Guan (1999) report that information content of implied volatility of S&P 500 futures options exhibits a frown shape across strikes with options that are NTM and have moderately high strike (i.e. OTM calls and ITM puts) possess the largest information content with R^2 equal to 17 percent for calls and 36 percent for puts. In a follow-on paper, Ederington and Guan (2000) find that using regression coefficients that are produced from in-sample regression of forecast against realized volatility is very effective in correcting implied forecasting bias. They also find that after such a bias correction, there is little to be gained from averaging implied across strikes. This means that ATM implied together with a bias correction scheme could be the simplest, and yet the best, way forward.

Findings in Ederington and Guan (1999, 2000) raise a very profound issue in finance. So far the volatility forecasting literature has been relying heavily on ATM implied. Implied from options that are far away from ATM were found to be bad forecasts and are persistently biased. This reflects a crucial fact that we have not yet found an option pricing model that is capable of pricing far-from-the-money option accurately and consistently. Although this is a tremendously important issue in finance, it is a research

question different from the one we are addressing here, and we shall leave it as a challenge for future research.

6.3 Implied Biasness

Usually, forecast unbiasedness is not an overriding issue in any forecasting exercise. Forecast bias can be estimated and corrected if the degree of bias remains stable through time. However, biasness in implied volatility can have a more serious undertone since it means options might be over- or under-priced, which can only be a result of an incorrect option pricing model or an inefficient option market. Both deficiencies have important implications in finance.

As mentioned in section 4.3, testing for biasness is usually carried out using regression equation (11), where $\hat{X}_i = \hat{X}_t$ is the implied forecast of period t volatility. For a forecast to be unbiased, one would require $\alpha = 0$ and $\beta = 1$. Implied forecast is upwardly biased if $\alpha > 0$ and $\beta = 1$, or $\alpha = 0$ and $\beta > 1$. In the case where $\alpha > 0$ and $\beta < 1$, which is the most common scenario, implied under-forecasts low volatility and over-forecasts high volatility.

It has been argued that implied bias will persist only if it is difficult to perform arbitrage trades that may remove the mispricing. This is more likely in the case of stock index options and less likely for futures options. Stocks and stock options are traded in different markets. Since trading of a basket of stocks is cumbersome, arbitrage trades in relation to a mispriced stock index option may have to be done indirectly via index futures. On the other hand, futures and futures options are traded alongside each other. Trading in these two contracts are highly liquid. Despite these differences in trading friction, implied biasness is reported in both the S&P 100 OEX market (Canina and Figlewski 1993; Christensen and Prabhala 1998; Fleming, Ostdiek, and Whaley 1995; and Fleming 1998) and the S&P 500 futures options market (Feinstein 1989b; and Ederington and Guan 1999, 2002).

Biasness is equally widespread among implied volatilities of currency options (see Guo 1996b; Jorion 1995; Li 2002; Scott and Tucker 1989; and Wei and Frankel 1991). The only exception is Jorion (1996) who cannot reject the null hypothesis that the one-day ahead forecasts from implied are unbiased. The five studies that reported biasness use implied to forecast exchange rate volatility over a much longer horizon, from one to nine months.

Unbiasness of implied forecast was not rejected in the Swedish market (Frennberg and Hansson 1996) though we already know from section 6.1.2 that the forecasting power of implied is not very strong anyway in this small stock market. Unbiasness of implied forecast was rejected for U.K. stock options (Gemmell 1986), U.S. stock options (Lamoureux and Lastrapes 1993), options and futures options across a range of assets in Australia (Edey and Elliot 1992), and for 35 futures options contracts traded over nine markets ranging from interest rate to livestock futures (Szakmary et al. 2002). Similarly, Amin and Ng (1997) find the hypothesis that $\alpha = 0$ and $\beta = 1$ cannot be rejected for the Eurodollar futures options market.

Where unbiasedness was rejected, the bias in all but two cases is due to $\alpha > 0$ and $\beta < 1$. These two exceptions are Fleming (1998), who reports $\alpha = 0$ and $\beta < 1$ for S&P 100 OEX options, and Day and Lewis (1993) who find $\alpha > 0$ and $\beta = 1$ for distant term oil futures options contracts.

Christensen and Prabhala (1998) argue that implied is biased because of error-in-variable caused by measurement errors described in section 3.2.3. Using last period implied and last period historical volatility as instrumental variables to correct for these measurement errors, Christensen and Prabhala (1998) find unbiasedness cannot be rejected for implied of S&P 100 OEX options. Ederington and Guan (1999, 2002) find bias in S&P 500 futures options implied also disappeared when similar instrument variables were used.

It has been suggested to us that implied biasness could not have been caused by model misspecification or measurement errors because this has relatively small effects for ATM options, which is used in most of the studies that report implied biasness. In addition, the clientele effect cannot explain the bias either because it only affects OTM options. Research now turns to volatility risk premium as an explanation.¹⁴

Poteshman (2000) finds half of the bias in S&P 500 futures options implied was removed when actual volatility was estimated with a more efficient volatility estimator based on intra-day five-minute returns. The other half of the bias was almost completely removed when a more sophisticated and less restrictive option pricing model, i.e. the Heston (1993) model, was used. The Heston model allows volatility to be stochastic similar to the Hull-White model used in Guo (1996b) and Lamoureux and Lastrapes (1993), who both report implied biasness. But unlike the Hull-White model, the Heston model also allows the market price of risk to be non-zero.

Further research on option volatility risk premium is currently underway in Luca Benzoni (2001) and Mikhail Chernov (2001). Chernov (2001) finds, similar to Poteshman (2000), when implied volatility is discounted by a volatility risk premium and when the errors-in-variables problems in measures of historical and realized volatility are removed, the unbiasedness of VIX cannot be rejected over the sample period from 1986 to 2000. The volatility risk premium debate on the questions continues if we are able to predict the magnitude and variations of the volatility premium and if implied from an option pricing model that permits a non-zero market price of risk will continue to outperform time series models when all forecasts (including forecasts of volatility risk premium) are made in an ex ante manner.

¹⁴ We thank a referee for this important insight.

7. *Where Next?*

The volatility forecasting literature is still very active. Many more new results are expected in the near future. There are several areas where future research could seek to improve upon. First is the issue about forecast evaluation and combining forecasts of different models. It will be useful if statistical tests were conducted to test if the forecast errors from Model A are significantly smaller, in some sense, than those from Model B, and so on for all pairs. Even if Model A is found to be better than all the other models, the conclusion is NOT one should henceforth forecast volatility with Model A and ignore the other models, as it is very likely that a combination of all the forecasts will be superior. To find the weights one can either run a regression of empirical volatility (the quantity being forecast) on the individual forecasts, or as approximation just use equal weights. Testing the effectiveness of a composite forecast is just as important as testing the superiority of the individual models, but this has not been done more often and across different data sets.

A mere plot of any measure of volatility against time will show the familiar “volatility clustering” which indicates some degree of forecastability. The biggest challenge lies in predicting changes in volatility. If implied volatility is agreed to be the best performing forecast, on average, this is in agreement with general forecast theory, which emphasizes the use of a wider information set than just the past of the process being forecast. Implied volatility uses option prices and so potentially the information set is richer. What needs further consideration is whether all of its information is being extracted and if it could still be widened to further improve forecast accuracy, especially of long horizon forecast. To achieve this we need to understand better the cause of volatility (both historical and implied). Such understanding will help improve time series methods, which are the only viable methods

when options, or market based forecast, are not available.

Closely related to the above is to understand the source of volatility persistence, and the volume-volatility research appears to be promising in providing a framework in which volatility persistence may be closely scrutinized. The Mixture of Distribution Hypothesis (MDH) proposed by Peter Clark (1973), the link between volume-volatility and market trading mechanism in George Tauchen and Mark Pitts (1983), and the empirical findings of the volume-volatility relationship surveyed in Jonathan Karpoff (1987) are useful starting points. Given that Lamoureux and Lastrapes (1990) find volume to be strongly significant when it is inserted into the ARCH variance process, while returns shocks become insignificant, and that Ronald Gallant, Rossi, and Tauchen (1993) find lagged volume substantially attenuates the “leverage” effect, the volume-volatility research may lead to a new and better way for modelling returns distributions. To this end, Andersen (1996) puts forward a generalized framework for the MDH where the joint dynamics of returns and volume are estimated, and reports a significant reduction in the estimated volatility persistence. Such a model may be useful for analyzing the economic factors behind the observed volatility clustering in returns, but this line of research has not yet been pursued vigorously.

There are many old issues that have been around for a long time. These include consistent forecasts of interest rate volatilities that satisfy the no-arbitrage relationships simultaneously across all interest rate instruments; more tests on the use of absolute returns models in comparison with squared returns models in forecasting volatility; and a multivariate approach to volatility forecasting where cross correlation and volatility spillover may be accommodated.

There are many new adventures that are currently underway as well.¹⁵ These include

¹⁵ We thank a referee for these suggestions.

the realized volatility approach noticeably driven by Andersen, Bollerslev, Diebold, and various co-authors, the estimation and forecast of volatility risk premium described in section 6.3, the use of spot and option price data simultaneously (e.g., Chernov and Ghysels 2000), and the use of Bayesian and other methods to estimate stochastic volatility models (e.g. Jones 2001), etc.

It is difficult to envisage in which direction volatility forecasting research will flourish in the next five years. If, within the next five years, we can cut the forecast error by half and remove the option pricing bias in ex ante forecast, this will be a very good achievement indeed. Producing by then forecasts of large events will also be worthwhile.

8. *Summary and Conclusion*

This survey has concentrated on two questions: is volatility forecastable? If it is, which method will provide the best forecasts? To consider these questions, a number of basic methodological viewpoints need to be discussed, mostly about the evaluation of forecasts. What exactly is being forecast? Does the time interval (the observation interval) matter? Are the results similar for different speculative markets? How does one measure predictive performance?

Volatility forecasts are classified in this section as belonging in one of the following four categories:

- HISVOL; for historical volatility models, which includes random walk, historical averages of squared returns, or absolute returns. Also included in this category are time series models based on historical volatility using moving averages, exponential weights, autoregressive models, or even fractionally integrated autoregressive absolute returns, for example. Note that HISVOL models can be highly sophisticated. The multivariate VAR realized volatility model in Andersen et al. (2002) is classified here

as a "HISVOL" model. All models in this group model volatility directly omitting the goodness of fit of the returns distribution or any other variables such as option prices.

- GARCH; any members of the ARCH, GARCH, EGARCH, and so forth families are included.
- ISD; for option implied standard deviation, based on the Black-Scholes model and various generalizations.
- SV; for stochastic volatility model forecasts.

The survey of papers includes 93 studies, but 27 of them did not involve comparisons between methods from at least two of these groups, and so were not helpful for comparison purposes.

The following table involves just pair-wise comparisons. Of the 66 studies that were relevant, some compared just one pair of forecasting techniques, others compared several. For those involving both HISVOL and GARCH models, 22 found HISVOL better at forecasting than GARCH (56 percent of the total), and seventeen found GARCH superior to HISVOL (44 percent). The full table is:

	Number of Studies	Studies Percentage
HISVOL > GARCH	22	56%
GARCH > HISVOL	17	44%
HISVOL > ISD	8	24%
ISD > HISVOL	26	76%
GARCH > ISD	1	6
ISD > GARCH	17	94
SV > HISVOL	3	
SV > GARCH	3	
GARCH > SV	1	
ISD > SV	1	

The combination of forecasts has a mixed picture. Two studies find it to be helpful but another does not.

The overall ranking suggests that ISD provides the best forecasting with HISVOL

and GARCH roughly equal, although possibly HISVOL does somewhat better in the comparisons. The success of the implied volatility should not be surprising, as these forecasts use a larger, and more relevant, information set than the alternative methods as they use option prices. They are also less practical, not being available for all assets.

Among the 93 papers, seventeen studies compared alternative versions of GARCH. It is clear that GARCH dominates ARCH. In general, models that incorporate volatility asymmetry such as EGARCH and GJR-GARCH perform better than GARCH. But certain specialized specifications, such as fractionally integrated GARCH (FIGARCH) and regime switching GARCH (RSGARCH) do better in some studies. However, it seems clear that one form of study that is included is conducted just to support a viewpoint that a particular method is useful. It might not have been submitted for publication if the required result had not been reached. This is one of the obvious weaknesses of a comparison such as this; the papers being reported are being prepared for different reasons, use different data sets, many kinds of assets, various intervals between readings, and a variety of evaluation techniques. Rarely discussed is if one method is significantly better than another. Thus, although a suggestion can be made that a particular method of forecasting volatility is the best, no statement is available about the cost-benefit from using it rather than something simpler, or how far ahead the benefits will occur.

Financial market volatility is clearly forecastable. The debate is on how far ahead one could accurately forecast and to what extent could volatility changes be predicted. This conclusion does not violate market efficiency since accurate volatility forecast is not in conflict with underlying asset and option prices being correct. The option implied volatility being a market based volatility forecast has been shown to contain most information about future volatility. The supremacy among historical time series models depends on the type of asset being

modelled. But, as a rule of thumb, historical volatility methods work equally well compared with more sophisticated ARCH class and SV models. Better reward could be gained by making sure that actual volatility is measured accurately. These are broad-brush conclusions omitting the fine details which we outline in this document. Because of the complex issues involved and the importance of volatility measure, volatility forecasting will continue to remain as a specialist subject and be studied vigorously.

Appendix A: Historical Price Volatility Models

A.1 Prediction Models Built on Sample Standard Deviations

Volatility, σ_t , in this section is the sample standard deviation of period t returns, and $\hat{\sigma}_t$ is the forecast of σ_t . If t is a month, then σ_t is often calculated as the sample standard deviation of all daily returns in the month. For a long time, σ_t is proxied by daily squared return if t is a day. More recently and with the availability of high frequency data, daily σ_t is derived from the cumulation of intra-day returns.

Random Walk (RW)

$$\hat{\sigma}_t = \sigma_{t-1} \quad (14)$$

Historical Average (HA)

$$\hat{\sigma}_t = (\sigma_{t-1} + \sigma_{t-2} + \dots + \sigma_1)/(t-1) \quad (15)$$

Moving Average (MA)

$$\hat{\sigma}_t = (\sigma_{t-1} + \sigma_{t-2} + \dots + \sigma_{t-\tau})/\tau \quad (16)$$

Exponential Smoothing (ES)

$$\hat{\sigma}_t = (1 - \beta)\sigma_{t-1} + \beta\hat{\sigma}_{t-1} \text{ and } 0 \leq \beta \leq 1 \quad (17)$$

Exponentially Weighted Moving Average (EWMA)

$$\hat{\sigma}_t = \sum_{i=1}^{\tau} \beta^i \sigma_{t-i} / \sum_{i=1}^{\tau} \beta^i \quad (18)$$

EWMA is a truncated version of ES with a finite τ .

Smooth Transition Exponential Smoothing (STES)

$$\hat{\sigma}_t = \alpha_{t-1} \epsilon_{t-1}^2 + (1 - \alpha_{t-1}) \hat{\sigma}_{t-1}^2 \quad (19)$$

$$\alpha_{t-1} = \frac{1}{1 + \exp(\beta + \gamma V_{t-1})}$$

where V_{t-1} is the transition variable; $V_{t-1} = \varepsilon_{t-1}$ for STES-E, $V_{t-1} = |\varepsilon_{t-1}|$ for STES-AE and V_{t-1} is a function of both ε_{t-1} and $|\varepsilon_{t-1}|$ for STES-EAE.

Simple Regression (SR)

$$\hat{\sigma}_t = \gamma_{1,t-1}\sigma_{t-1} + \gamma_{2,t-1}\sigma_{t-2} + \dots \quad (20)$$

Threshold autoregressive (TAR)

$$\hat{\sigma}_t = \phi_0^{(i)} + \phi_1^{(i)}\sigma_{t-1} + \dots + \phi_p^{(i)}\sigma_{t-p}, \quad (21)$$

$$i = 1, 2, \dots, k$$

A.2 ARCH Class Conditional Volatility Models

For all models described in this section, returns, r_t , has the following process

$$r_t = \mu + \varepsilon_t$$

$$\varepsilon_t = \sqrt{h_t}z_t$$

and h_t follows one of the following ARCH class models.

ARCH (q)

$$h_t = \omega + \sum_{k=1}^q \alpha_k \varepsilon_{t-k}^2 \quad (22)$$

where $\omega > 0$ and $\alpha_k \geq 0$.

GARCH (p, q)

$$h_t = \omega + \sum_{k=1}^q \alpha_k \varepsilon_{t-k}^2 + \sum_{j=1}^p \beta_j h_{t-j} \quad (23)$$

where $\omega > 0$. (See Nelson and Cao 1992 for constraints on α_k and β_j .) For finite variance, $\sum \alpha_k + \sum \beta_j < 1$.

EGARCH (p, q)

$$\ln h_t = \alpha_0 + \sum_{j=1}^q \beta_j \ln h_{t-j} + \sum_{k=1}^p \alpha_k \left[\theta \psi_{t-k} + \gamma \left(|\psi_{t-k}| - \left(\frac{2}{\pi} \right)^{\frac{1}{2}} \right) \right] \quad (24)$$

$$\psi_t = \mu_t / \sqrt{h_t}$$

Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
1. Akigray (1989)	CRSP VW & EW indices	Jan. 63–Dec. 86 (Precrash) Split into 4 subperiods of 6 years each.	D	GARCH(1,1) ARCH(2) EWMA HIS (ranked)
2. Alford and Boatman (1995)	6879 stocks listed in NYSE/ASE & NASDAQ	12/66–6/87	W, M	“Shrinkage” forecast (HIS adjusted towards comparable firms) HIS Median HIS vol of “comparable” firm (ranked)
3. Amin and Ng (1997)	3M Eurodollar futures & futures options	1/1/88–1/11/92	D	Implied ^{American All Call+Put} (WLS, 5 variants of the HJM model) HIS (ranked)
4. Andersen and Bollerslev (1998)	DM/\$, ¥/\$	In: 1/10/87–30/9/92 Out: 1/10/92–30/9/93	D (5 min)	GARCH(1,1)

TGARCH (1,1) or GJR-GARCH (1,1)

$$h_t = \omega + \alpha \epsilon_{t-1}^2 + \delta D_{t-1} \epsilon_{t-1}^2 + \beta h_{t-1} \quad (25)$$

$$D_{t-1} = \begin{cases} 1 & \text{if } \epsilon_{t-1} < 0 \\ 0 & \text{if } \epsilon_{t-1} \geq 0 \end{cases}$$

QGARCH (1,1)

$$h_t = \omega + \alpha (\epsilon_{t-1} - \gamma)^2 + \beta h_{t-1} \quad (26)$$

STGARCH (*Smooth Transition GARCH*)

$$h_t = \omega + [1 - F(\epsilon_{t-1})] \alpha \epsilon_{t-1}^2 + F(\epsilon_{t-1}) \delta \epsilon_{t-1}^2 + \beta h_{t-1} \quad (27)$$

where

$$F(\epsilon_{t-1}) = \frac{1}{1 + \exp(-\theta \epsilon_{t-1})}$$

for logistic STGARCH,

$$F(\epsilon_{t-1}) = 1 + \exp(-\theta \epsilon_{t-1}^2)$$

for exponential STGARCH,

GARCH(1,1) *regime switching*

$$h_{t,S_{t-1}} = \omega_{S_{t-1}} + \alpha_{S_{t-1}} \epsilon_{t-1}^2 + \beta_{S_{t-1}} h_{t-1,S_{t-1}}$$

where S_t indicates the state of regime at time t .

CGARCH(1,1) (Component GARCH)

$$h_t = \omega_t + \alpha_t (\epsilon_{t-1}^2 - \omega_{t-1}) + \beta_1 (h_{t-1} - \omega_{t-1}) \quad (28)$$

$$\omega_t = \omega + \rho \omega_{t-1} + \xi (\epsilon_{t-1}^2 - h_{t-1})$$

where ω_t represents a time-varying trend or permanent component in volatility which is driven by volatility prediction error ($\epsilon_{t-1}^2 - h_{t-1}$) and is integrated if $\rho = 1$.

A.3 Stochastic Volatility Model

Stochastic Volatility (SV)

$$r_t = \mu + \epsilon_t$$

$$\epsilon_t = z_t \exp(0.5h_t) \quad (29)$$

$$h_t = \omega + \beta h_{t-1} + v_t$$

v_t may or may not be independent of z_t .

Forecasting Horizon	Evaluation & R-square	Comments
20 days ahead estimated from rolling 4 years data. Daily returns used to construct “actual vol”; adjusted for serial correlation.	ME, RMSE, MAE, MAPE	GARCH is least biased and produced best forecast especially in periods of high volatility and when changes in volatility persist. Heteroskedasticity is less strong in low frequency data and monthly returns are approximately normal.
5 years starting from 6 months after firm’s fiscal year	MedE, MedAE	To predict 5-year monthly volatility one should use 5 years worth of weekly or monthly data. Adjusting historical forecast using industry and size produced best forecast.
20 days ahead (1 day ahead forecast produced from in-sample with lag implied in GARCH/GJR not discussed here.)	R^2 is 21% for implied and 24% for combined. $H_0: \alpha_{\text{implied}} = 0, \beta_{\text{implied}} = 1$ cannot be rejected with robust SE.	Interest rate models that incorporate volatility term structure (e.g. Vasicek) perform best. Interaction term capturing rate level and volatility contribute additional forecasting power.
1 day ahead, use 5-min returns to construct “actual vol”	R^2 is 5 to 10% for daily squared returns, 50% for 5-min square returns.	R^2 increases monotonically with sample frequency.

(Continued)

Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
5. Andersen, Bollerslev, Diebold and Labys (2002)	¥/US\$, DM/US\$ Reuters FXFX quotes	1/12/86–30/6/99 In: 1/12/86–1/12/96, 10 years Out: 2/12/96–30/6/99, 2.5 years	Tick (30min)	VAR-RV, AR-RV, FIEGARCH-RV GARCH-D, RM-D, FIEGARCH-D VAR-ABS (ranked)
6. Andersen, Bollerslev and Lange (1999)	DM/US\$ Reuters quotes	1/12/86–30/11/96 In: 1/10/87–30/9/92	5 min	GARCH(1,1) at 5-min, 10-min, 1-hr, 8-hr, 1-day, 5-day, 20-day interval.
7. Bali (2000)	3-, 6-, 12-month T-Bill rates	8/1/54–25/12/98	W	NGARCH GJR, TGARCH AGARCH, QGARCH TSGARCH GARCH VGARCH Constant vol (CKLS) (ranked, forecast both level and volatility)
8. Beckers (1981)	62 to 116 Stocks options	28/4/75–21/10/77	D	FBSD Implied _{ATM call, 5 days ave} Implied _{vega call, 5 days ave} RW _{last quarter} (ranked, both implieds are 5-day average because of large variations in daily stock implied.)
	50 stock option	4 dates: 18/10/76, 24/1/77, 18/4/77, 18/7/77	D (from Tick)	TISD _{vega} Implied _{ATM call, 1 days ave} (ranked)
9. Bera and Higgins (1997)	Daily SP500, Weekly \$/£, Monthly US Ind Prod	SP 1/1/88–28/5/93 \$/\$ 12/12/85–28/2/91 IndProd 1/60–3/93	D W M	GARCH Bilinear model (ranked)
10. Blair, Poon & Taylor (2001)	S&P 100 (VIX)	2/1/87–31/12/99 Out: 4/1/93–31/2/99	Tick	Implied _{VIX} GJR HIS ₁₀₀ (ranked)

Forecasting Horizon	Evaluation & R-square	Comments
1 and 10 days ahead. "Actual vol" derived from 30-min returns.	1-day ahead R^2 ranges between 27–40% (1-day ahead) and 20–33% (10-day ahead).	RV is realised volatility, D is daily return, and ABS is daily absolute return. VAR allows all series to share the same fractional integrated order and cross series linkages. Forecast improvement is largely due to the use of high frequency data (and realised volatility) instead of the model(s).
1, 5 and 20 days ahead, use 5-min returns to construct "actual vol"	RMSE, MAE, HRMSE, HMAE, LL	HRMSE and HMAE are heteroskedasticity adjusted error statistics; LL is the logarithmic loss function. High frequency returns and high frequency GARCH(1,1) models improve forecast accuracy. But, for sampling frequencies shorter than 1 hour, the theoretical results and forecast improvement break down.
1 week ahead. Use weekly interest rate absolute change to proxy "actual vol".	R^2 increases from 2% to 60% by allowing for asymmetries, level effect and changing volatility.	CKLS: Chan, Karolyi, Longstaff and Sanders (1992).
Over option's maturity (3 months), 10 non-overlapping cycles. Use sample SD of daily returns over option maturity to proxy "actual vol".	MPE, MAPE. Cross sectional R^2 ranges between 34–70% across models and expiry cycles. FBSD appears to be least biased with $\alpha = 0$, $\beta = 1$. $\alpha > 0$, $\beta < 1$ for the other two implieds.	FBSD: Fisher Black's option pricing service takes into account stock vol tend to move together, mean revert, leverage effect and implied can predict future. ATM, based on vega WLS, outperforms vega weighted implied, and is not sensitive to <i>ad hoc</i> dividend adjustment. Incremental information from all measures suggests option market inefficiency. Most forecasts are upwardly biased as actual vol was on a decreasing trend.
Ditto	Cross sectional R^2 ranges between 27–72% across models and expiry cycles.	TISD: Single intra-day transaction data that has the highest vega. The superiority of TISD over implied of closing option prices suggest significant non-simultaneity and bid-ask spread problems.
One step ahead. Reserve 90% of data for estimation	Cox MLE RMSE (LE: logarithmic error)	Consider if heteroskedasticity is due to bilinear in level. Forecasting results show strong preference for GARCH.
1, 5, 10 and 20 days ahead estimated using a rolling sample of 1,000 days. Daily actual volatility is calculated from 5-min returns.	1-day ahead R^2 is 45% for VIX, and 50% for combined. VIX is downward biased in out-of-sample period.	Using squared returns reduces R^2 to 36% for both VIX and combined. Implied volatility has its own persistence structure. GJR has no incremental information though integrated HIS vol can almost match IV forecasting power.

(Continued)

Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
11. Bluhm and Yu (2000)	German DAX stock index and VDAX the DAX volatility index	In: 1/1/88–28/6/96 Out: 1/7/66–30/6/99	D	Implied _{VDAX} GARCH(-M), SV EWMA, EGARCH, GJR, HIS (approx ranked)
12. Boudoukh, Richardson and Whitelaw (1997)	3-month US T-bill	1983–92	D	EWMA, MDE GARCH(1,1), HIS (ranked)
13. Brace and Hodgson (1991)	Futures option on Australian Stock Index (Marking to market is needed for this options)	1986–87	D	HIS _{5, 20, 65 days} Implied _{NTM call, 20–75 days} (ranked)
14. Brailsford and Faff (1996)	Australian Statex-Actuaries Accumulation Index for top 56	1/1/74–30/6/93 In: Jan. 74–Dec. 85 Out: Jan. 86–Dec. 93 (include 87's crash period)	D	GJR, Regr, HIS, GARCH, MA, EWMA, RW, ES (rank sensitive to error statistics)
15. Brooks (1998)	DJ Composite	17/11/78–30/12/88 Out: 17/10/86–30/12/88	D	RW, HIS, MA, ES, EWMA, AR, GARCH, EGARCH, GJR, Neural network (all about the same)
16. Canina and Figlewski (1993)	S&P 100 (OEX)	15/3/83–28/3/87 (Pre-crash)	D	HIS _{60 calendar days} Implied _{Binomial Call} (ranked) Implied in 4 maturity gp, each subdivided into 8 intrinsic gp.
17. Cao and Tsay (1992)	Excess returns for S&P, VW EW indices	1928–89	M	TAR EGARCH(1,0) ARMA(1,1) GARCH(1,1) (ranked)
18. Chiras and Manaster (1978)	All stock options from CBOE	23 months from June 73–April 75	M	Implied (weighted by price elasticity) HIS _{20 monthss} (ranked)

Forecasting Horizon	Evaluation & R-square	Comments
45 calendar days, 1, 10 and 180 trading days. "Actual" is the sum of daily squared returns.	MAPE, LINEX	Ranking varies a lot depend on forecast horizons and performance measures.
1 day ahead based on 150-day rolling period estimation. Realized volatility is the daily squared changes averaged across $t + 1$ to $t + 5$.	MSE & regression. MDE has the highest R^2 while EWMA has the smallest MSE.	MDE is multivariate density estimation where volatility weights depend on interest rate level and term spread. EWMA and MDE have comparable performance and are better than HIS and GARCH.
20 days ahead. Use daily returns to calculate standard deviations.	Adj R^2 are 20% (HIS), 17% (HIS+ implied). All $\alpha > 0$ & sig. Some uni. regr. coeff. are sig. negative (for both HIS & implied).	Large fluctuations of R^2 from month to month. Results could be due to the difficulty in valuing futures style options.
1 month ahead. Models estimated from a rolling 12-year window.	ME, MAE, RMSE, MAPE, and a collection of asymmetric loss functions.	Though the ranks are sensitive, some models dominate others; MA12 > MA5 and Regr > MA > EWMA > ES. GJR came out quite well but is the only model that always underpredict.
1 day ahead squared returns using rolling 2,000 observations for estimation.	MSE, MAE of variance, % over-predict. R^2 is around 4% increases to 24% for pre-crash data.	Similar performance across models especially when 87's crash is excluded. Sophisticated models such as GARCH and neural net did not dominate. Volume did not help in forecasting volatility.
7 to 127 calendar days matching option maturity, overlapping forecasts with Hansen std error. Use sample SD of daily returns to proxy "actual vol".	Combined R^2 is 17% with little contribution from implied. All $\alpha_{\text{implied}} > 0$, implied $\beta < 1$ with robust SE.	Implied has no correlation with future volatility and does not incorporate info contained in recently observed volatility. Results appear to be peculiar for pre-crash period. Time horizon of "actual vol" changes day to day. Different level of implied aggregation produces similar results.
1 to 30 months. Estimation period ranges from 684 to 743 months.	MSE, MAE Daily returns used to construct "actual vol".	TAR provides best forecasts for large stocks. EGARCH gives best long-horizon forecasts for small stocks (may be due to Leverage effect). Difference in MAE can be as large as 38%.
20 month ahead. Use SD of 20 monthly returns to proxy "actual vol".	Cross sectional R^2 of implied ranges 13-50% across 23 months. HIS adds 0-15% to R^2 .	Implied outperformed HIS especially in the last 14 months. Find implied increases and better behave after dividend adjustments and evidence of mispricing possibly due to the use European pricing model on American style options.

(Continued)

Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
19. Christensen and Prabhala (1998)	S&P 100 (OEX) Monthly expiry cycle	Nov. 83–May 95	M	Implied _{BS} ATM 1-month Call HIS _{18 days} (ranked)
20. Christoffersen and Diebold (2000)	4 stk indices 4 ex rates US 10 year T-Bond	1/1/73–1/5/97	D	No model. (No rank; Evaluate volatility forecastability (or persistence) by checking interval forecasts.)
21. Cumby, Figlewski and Hasbrouck (1993)	¥/\$, stocks(¥, \$), bonds (¥, \$)	7/77 –9/90	W	EGARCH HIS (ranked)
22. Day and Lewis (1992)	S&P 100 OEX option Reconstructed S&P 100	Out: 11/11/83–31/12/89 In: 2/1/76–11/11/83	W	Implied _{BS} Call (shortest but > 7 days, volume WLS) HIS _{1 week} GARCH EGARCH (ranked)
23. Day and Lewis (1993)	Crude oil futures options Crude oil futures	14/11/86–18/3/91 8/4/83–18/3/91 coincide with Kuwait invasion by Iraq in second half of sample.	D	Implied _{Binomial} ATM Call HIS _{forecast horizon} GARCH-M EGARCH-AR(1) (ranked)
24. Dimson and Marsh (1990)	UK FT All Share	1955–89	Q	ES, Regression RW, HA, MA (ranked)
25. Doidge and Wei (1998)	Toronto 35 stock index & European options	In: 2/8/88–31/12/91 Out: 1/92–7/95	D	Combine3 Combine2 GARCH EGARCH HIS _{100 days} Combine1 Implied _{BS} Call + Put (All maturities > 7 days, volume WLS) (ranked)

Forecasting Horizon	Evaluation & R-square	Comments
Non-overlapping 24 calendar (or 18 trading) days. Use SD of daily returns to proxy “actual vol”.	R^2 of log var are 39% (implied), 32% (HIS) and 41% (combined). $\alpha < 0$ (because of log), $\beta < 1$ with robust SE. Implied is more biased before the crash.	Not adj for dividend and early exercise. Implied dominates HIS. HIS has no additional information in subperiod analysis. Proved that results in Canina & Figlewski (1993) is due to pre-crash characteristics and high degree of data overlap relative to time series length. Implied is unbiased after controlling for measurement errors using implied _{t-1} and HIS _{t-1} .
1 to 20 days	Run tests and Markov transition matrix eigenvalues (which is basically 1st-order serial coefficient of the hit sequence in the run test).	Equity & FX: forecastability decrease rapidly from 1 to 10 days. Bond: may extend as long as 15 to 20 days. Estimate bond returns from bond yields by assuming coupon equal to yield.
1 week ahead, estimation period ranges from 299 to 689 weeks.	R^2 varies from 0.3% to 10.6%.	EGARCH is better than naïve in forecasting volatility though R-square is low. Forecasting correlation is less successful.
1 week ahead estimated from a rolling sample of 410 observations. Use sample variance of daily returns to proxy weekly “actual vol”.	R^2 of variance regr are 2.6% (implied) & 3.8% (encomp). All forecasts add marginal info. $H_0: \alpha_{\text{implied}} = 0$, $\beta_{\text{implied}} = 1$ cannot be rejected with robust SE.	Omit early exercise. Effect of 87’s crash is unclear. When weekly squared returns were used to proxy “actual vol”, R^2 increase and was max for HIS contrary to expectation (9% compared with 3.7% for implied).
Option maturity of 4 nearby contracts, (average 13.9, 32.5, 50.4 & 68 trading days to maturity). Estimated from rolling 500 observations.	ME, RMSE, MAE. R^2 of variance regr are 72% (short mat) and 49% (long maturity). With robust SE $\alpha > 0$ for short and $\beta = 0$ for long, $\beta = 1$ for all maturity.	Implied performed extremely well. Performance of HIS and GARCH are similar. EGARCH much inferior. Bias adjusted and combined forecasts do not perform as well as unadjusted implied. GARCH has no incremental information. Result likely to be driven by Kuwait invasion by Iraq.
Next quarter. Use daily returns to construct “actual vol”.	MSE, RMSE, MAE, RMAE	Recommend exponential smoothing and regression model using fixed weights. Find ex ante time-varying optimization of weights does not work well ex post.
1 month ahead from rolling sample estimation. No mention on how “actual vol” was derived.	MAE, MAPE, RMSE	Combine1 equal weight for GARCH and implied forecasts. Combine2 weighs GARCH and implied based on their recent forecast accuracy. Combine3 puts implied in GARCH conditional variance. Combine3 was estimated using full sample due to convergence problem; so not really out-of-sample forecast.

(Continued)

Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
26. Dunis, Laws and Chauvin (2000)	DM/¥, £/DM, £/\$, \$/CHF, \$/DM, \$/¥	In: 2/1/91–27/2/98 Out: 2/3/98–31/12/98	D	GARCH(1,1) AR(10)-Sq returns AR(10)-Abs returns SV(1) in log form HIS _{21 or 63 trading days} 1- & 3-M forward Implied _{ATM quotes} Combine Combine (except SV) (rank changes across currencies and forecast horizons)
27. Ederington and Guan (1999) “Frown”	S&P 500 futures options	1 Jan. 88–30 April 98	D	Implied _{BK 16 Calls 16 Puts} HIS _{40 days} (ranked)
28. Ederington and Guan (2000) “Forecasting Volatility”	5 DJ Stocks S&P 500 3m Euro\$ rate 10y T-Bond yield DM/\$	2/7/62–30/12/94 2/7/62–29/12/95 1/1/73–20/6/97 2/1/62–13/6/97 1/1/71–30/6/97	D	GW _{MAD} GW _{STD} GARCH, EGARCH AGARCH HIS _{MAD,n} , HIS _{STD,n} (ranked, error statistics are close; GW _{MAD} leads consistently though with only small margin.)
29. Ederington and Guan (2000) “Averaging”	S&P 500 futures options	In 4/1/88–31/12/91 Out: 2/1/92–31/12/92	D	Implied*: 99%, VIX HIS _{40 trading days} Implied: VIX > Eq4 Implied: WLS > vega > Eq32 > elasticity (ranked)
30. Ederington and Guan (2002) “Efficient Predictor”	S&P 500 futures options	1/1/83–14/9/95	D	Implied _{Black 4NTM} GARCH, HIS _{40 days} (ranked)
31. Edey and Elliot (1992)	Futures options on A\$ 90d-Bill, 10yr bond, Stock index A\$/US\$ options	Futures options: inception to 12/88 A\$/US\$ option: 12/84–12/87	W W	Implied _{BK NTM,call} Implied _{BK NTM,put} (No rank, 1 call and 1 put, selected based on highest trading volume)

Forecasting Horizon	Evaluation & R-square	Comments
1 and 3 months (21 & 63 trading days) with rolling estimation. Actual volatility is calculated as the average absolute return over the forecast horizon.	RMSE, MAE, MAPE, Theil-U, CDC (Correct Directional Change index)	No single model dominates though SV is consistently worst, and implied always improves forecast accuracy. Recommend equal weight combined forecast excluding SV.
Overlapping 10 to 35 days matching maturity of nearest to expiry option. Use SD of daily returns to proxy "actual vol".	Panel R^2 19% and individual R^2 ranges 6–17% (calls) and 15–36% (puts). Implied is biased & inefficient, $\alpha_{\text{implied}} > 0$ and $\beta_{\text{implied}} < 1$ with robust SE.	Information content of implied across strikes exhibit a frown shape with options that are NTM and have moderately high strikes possess largest information content. HIS typically adds 2–3% to the R^2 and nonlinear implied terms add another 2–3%. Implied is unbiased & efficient when measurement error is controlled using Implied $_{t-1}$ and HIS $_{t-1}$.
$n=10, 20, 40, 80$ & 120 days ahead estimated from a 1260-day rolling window; parameters re-estimated every 40 days. Use daily squared deviation to proxy "actual" vol.	RMSE, MAE	GW: geometric weight, MAD: mean absolute deviation, STD: standard deviation. Volatility aggregated over a longer period produces a better forecast. Absolute returns models generally perform better than square returns models (except GARCH > AGARCH). As horizon lengthens, no procedure dominates. GARCH & EGARCH estimations were unstable at times.
Overlapping 10 to 35 days matching maturity of nearest to expiry option. Use SD of daily returns to proxy "actual vol".	RMSE, MAE, MAPE "*" indicates individual implieds were corrected for biasness first before averaging using in-sample regr on realised.	VIX: 2calls + 2puts, NTM weighted to get ATM. Eq4(32): calls + puts equally weighted. WLS, vega and elasticity are other weighting scheme. 99% means 1% of regr error used in weighting all implieds. Once the biasness has been corrected using regr, little is to be gained by any averaging in such a highly liquid S&P 500 futures market.
Overlapping option maturity 7–90, 91–180, 181–365 and 7–365 days ahead. Use sample SD over forecast horizon to proxy "actual vol".	R^2 ranges 22–12% from short to long horizon. Post 87's crash R^2 nearly doubled. Implied is efficient but biased; $\alpha_{\text{implied}} > 0$ and $\beta_{\text{implied}} < 1$ with robust SE.	GARCH parameters were estimated using whole sample. GARCH and HIS add little to 7–90 day R^2 . When 87's crash was excluded HIS add sig. explanatory power to 181–365 day forecast. When measurement errors were controlled using implied $_{t-5}$ and implied $_{t+5}$ as instrument variables implied becomes unbiased for the whole period but remains biased when crash period was excluded.
Option maturity up to 3M. Use sum of (return square plus Implied $_{t+1}$) as "actual vol".	Regression (see comment). In most cases $\alpha_{\text{implied}} > 0$ and $\beta_{\text{implied}} < 1$ with robust SE. For stock index option $\beta_{\text{implied}} = 1$ cannot be rejected using robust SE.	R^2 cannot be compared with other studies because of the way "actual" is derived and lagged squares returns were added to the RHS.
Constant 1M. Use sum of weekly squared returns to proxy "actual vol".		

(Continued)

Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
32. Engle, Ng and Rothschild (1990)	1 to 12 months T-Bill returns, VW index of NYSE & AMSE stocks	Aug. 64–Nov. 85	M	1-Factor ARCH Univariate ARCH-M (ranked)
33. Feinstein (1989)	S&P 500 futures options (CME)	June 83–Dec. 88	Option expiry cycle	Implied: Alanta > average > vega > elasticity Just-OTM _{Call > P+C > Put} HIS _{20 days} (ranked, note pre-crash rank is very different and erratic)
34. Ferreira (1999)	French & German interbank 1M mid rate	In: Jan. 81–Dec. 89 Out: Jan. 90–Dec. 97 (ERM crises: Sep. 92-Sep. 93)	W	ES, HIS _{26, 52, all} GARCH(-L) (E)GJR(-L) (rank varies between French & German rates, sampling method and error statistics.)
35. Figlewski (1997)	S&P 500 3M US T-Bill 20Y T-Bond DM/\$	1/47–12/95 1/47–12/95 1/50–7/93 1/71–11/95	M	HIS _{6, 12, 24, 36, 48, 60m} GARCH(1,1) for S&P and bond yield. (ranked)
	S&P 500 3M US T-Bill 20Y T-Bond DM/\$	2/7/62–29/12/95 2/1/62–29/12/95 2/1/62–29/12/95 4/1/71–30/11/95	D	GARCH(1,1) HIS _{1, 3, 6, 12, 24, 60 months} (S&P's rank, reverse for the others)
36. Figlewski & Green (1999)	S&P 500 US LIBOR 10 yr T-Bond yield	1/4/71–12/31/96 Out: From Feb. 96	D	HIS _{3, 12, 60 months} ES (rank varies)
	DM/\$	1/4/71–12/31/96 Out: From Jan. 92	M	HIS _{26, 60, all months} ES (ranked)
37. Fleming (1998)	S&P 100 (OEX)	10/85–4/92 (All observations that overlap with 87's crash were removed.)	D	Implied _{FW ATM calls} Implied _{FW ATM puts} (Both implieds are WLS using all ATM options in the last 10 minutes before market close) ARCH/GARCH HIS _{H-L, 28 days} (ranked)
38. Fleming, Kirby and Ostdiek (2000)	S&P 500, T-Bond and gold futures	3/1/83–31/12/97	D	Exponentially weighted var-cov matrix

Forecasting Horizon	Evaluation & R-square	Comments
1 month ahead volatility and risk premium of 2 to 12 months T-Bills	Model fit	Equally weighted bill portfolio is effective in predicting (i.e. in an expectation model) volatility and risk premia of individual maturities.
23 non-overlapping forecasts of 57, 38 and 19 days ahead. Use sample SD of daily returns over the option maturity to proxy “actual vol.”	MSE, MAE, ME. T-test indicates all ME > 0 (except HIS) in the post crash period which means implied was upwardly biased.	Alanta: 5-day average of Just-OTM call implied using exponential weights. In general Just-OTM Implied _{call} is the best.
1 week ahead. Use daily squared rate changes to proxy weekly volatility.	Regression, MPE, MAPE, RMSPE. R^2 is 41% for France and 3% for Germany.	L: interest rate level, E: exponential. French rate was very volatile during ERM crises. German rate was extremely stable in contrast. Although there are lots of differences between the two rates, best models are non-parametric; ES (French) and simple level effect (German). Suggest a different approach is needed for forecasting interest rate volatility.
6, 12, 24, 36, 48, 60 months. Use daily returns to compute “actual vol.”	RMSE	Forecast of volatility of the longest horizon is the most accurate. HIS uses the longest estimation period is the best except for short rate.
1, 3, 6, 12, 24 months	RMSE	GARCH is best for S&P but gave worst performance in all the other markets. In general, as out of sample horizon increases, the in sample length should also increase.
1, 3, 12 months for daily data.	RMSE	ES works best for S&P (1–3 month) and short rate (all three horizons). HIS works best for bond yield, exchange rate and long horizon S&P forecast. The longer the forecast horizon, the longer the estimation period.
24 & 60 months for monthly data.		For S&P, bond yield and DM/\$, it is best to use all available “monthly” data. 5 years worth of data works best for short rate.
Option maturity (shortest but > 15 days, average 30 calendar days), 1 and 28 days ahead. Use daily square return deviations to proxy “actual vol”.	R^2 is 29% for monthly forecast and 6% for daily forecast. All $\alpha_{\text{implied}} = 0$, $\beta_{\text{implied}} < 1$ with robust SE for the last two fixed horizon forecasts.	Implied dominates. All other variables related to volatility such as stock returns, interest rate and parameters of GARCH do not possess information incremental to that contained in implied.
Daily rebalanced portfolio	Sharpe ratio (portfolio return over risk)	Efficient frontier of volatility timing strategy plotted above that of fixed weight portfolio.

(Continued)

Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
39. Fleming, Ostdiek & Whaley (1995)	S&P 100 (VIX)	Jan. 86–Dec. 92	D, W	Implied _{VIX} HIS _{20 days} (ranked)
40. Franses & Glijsels (1999)	Dutch, German, Spanish and Italian stock market returns	1983–94	W	AO-GARCH (GARCH adjusted for additive outliers using the ‘less-one’ method) GARCH GARCH- <i>t</i> (ranked)
41. Franses and Van Dijk (1996)	Stock indices (Germany, Netherlands, Spain, Italy, Sweden)	1986–94	W	QGARCH RW GARCH GJR (ranked)
42. Frennberg and Hansson (1996)	VW Swedish stock market returns Index option (European style)	In: 1919–1976 Out: 1977–82, 1983–90 Jan. 87–Dec. 90	M	AR ₁₂ (ABS)-S RW, Implied _{BS ATM Call} (option maturity closest to 1 month) GARCH-S, ARCH-S (ranked) Models that are not adj for seasonality did not perform as well.
43. Fung, Lie and Moreno (1990)	£/\$, C/\$, FFr/\$, DM/\$, ¥/\$ & SrFr/\$ options on PHLX	1/84–2/87 (Pre crash)	D	Implied _{OTM>ATM} Implied _{vega, elasticity} Implied _{equal weight} HIS _{40 days} , Implied _{ITM} (ranked, all implied are from calls.)
44. Fung and Hsieh (1991)	S&P 500, DM/\$ US T-bond Futures and futures options	3/83–7/89 (DM/\$ futures from 26 Feb 85)	D (15min)	RV-AR(n) Implied _{BAW NTM Call/ Put} RV, RW(C-t-C) HL (ranked, some of the differences are small)
45. Gemmill (1986)	13 UK stocks LTOM options. Stock price	May 78–July 83 Jan. 78–Nov. 83	M D	Implied _{ITM} Implied _{ATM, vega WLS} Implied _{equal, OTM, elasticity} HIS _{20 weeks} (ranked, all implied are from calls.)

Forecasting Horizon	Evaluation & R-square	Comments
28 calendar (or 20 trading) days. Use sample SD of daily returns to proxy “actual vol”.	R^2 increased from 15% to 45% when crash is excluded. $\alpha_{VIX} = 0$, $\beta_{VIX} < 1$ with robust SE.	VIX dominates HIS, but is biased upward up to 580 basis point. Orthogonality test rejects HIS when VIX is included. Adjust VIX forecasts with average forecast errors of the last 253 days helps to correct for biasness while retaining implied’s explanatory power.
1 week ahead estimated from previous 4 years. Use weekly squared deviations to proxy “actual vol”.	MSE & MedSE	Forecasting performance significantly improved when parameter estimates are not influenced by ‘outliers’. Performance of GARCH- t is consistently much worse. Same results for all four stock markets.
1 week ahead estimated from rolling 4 years. Use weekly squared deviations to proxy “actual vol”.	MedSE	QGARCH is best if data has no extremes. RW is best when 87’s crash is included. GJR cannot be recommended. Results are likely to be influenced by MedSE that penalize nonsymmetry.
1 month ahead estimated from recursively re-estimated expanding sample. Use daily ret to compile monthly vol, adjusted for autocorrelation.	MAPE, R^2 is 2–7% in first period and 11–24% in second, more volatile period. H_0 : $\alpha_{\text{implied}} = 0$ and $\beta_{\text{implied}} = 1$ cannot be rejected with robust SE.	S: seasonality adjusted. RW model seems to perform remarkably well in such a small stock market where returns exhibit strong seasonality. Option was introduced in 86 and covered 87’s crash; outperformed by RW. ARCH/GARCH did not perform as well in the more volatile second period.
Option maturity; overlapping periods. Use sample SD of daily returns over option maturity to proxy “actual vol”.	RMSE, MAE of overlapping forecasts.	Each day, 5 options were studied; 1 ATM, 2 just in and 2 just out. Define ATM as $S = X$, OTM marginally outperformed ATM. Mixed together implied of different contract months.
1 day ahead. Use 15-min data to construct “actual vol”.	RMSE and MAE of log σ	RV: Realised vol from 15-min returns. AR(n): autoregressive lags of order n . RW(C-t-C): random walk forecast based on close to close returns. HL: Parkinson’s daily high-low method. Impact of 1987 crash does not appear to be drastic possibly due to taking log. In general, high frequency data improves forecasting power greatly.
13-21 non-overlapping option maturity (each average 19 weeks). Use sample SD of weekly returns over option maturity to proxy “actual vol”.	ME, RMSE, MAE aggregated across stocks and time. R^2 are 6–12% (pooled) and 40% (panel with firm specific intercepts). All $\alpha > 0$, $\beta < 1$.	Adding HIS increases R^2 from 12% to 15%. But <i>ex ante</i> combined forecast from HIS and Implied _{ITM} turned out to be worst then individual forecasts. Suffered small sample and nonsynchronicity problems and omitted dividends.

(Continued)

Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
46. Gray (1996)	US 1M T-Bill	1/70–4/94	W	RSGARCH with time varying probability GARCH Constant Variance (ranked)
47. Guo (1996a)	PHLX US\$/¥ options	Jan. 91–Mar. 93	D	Implied _{Heston} Implied _{HW} Implied _{BS} GARCH HIS ₆₀ (ranked)
48. Guo (1996b)	PHLX US\$/¥, US\$/DM options Spot rate	Jan. 86–Feb. 93	Tick D	Implied _{HW} (WLS, $0.8 < S/X < 1.2$, $20 < T < 60$ days) GARCH(1,1) HIS _{60 days} (ranked)
49. Hamid (1998)	S&P 500 futures options	3/83–6/93	D	13 schemes (including HIS, Implied cross strike average and intertemporal averages) (ranked, see comment)
50. Hamilton and Lin (1996)	Excess stock returns (S&P 500 minus T-bill) & Ind Production	1/65–6/93	M	Bivariate RSARCH Univariate RSARCH GARCH+L ARCH+L AR(1) (ranked)
51. Hamilton and Susmel (1994)	NYSE VW stock index	3/7/62–29/12/87	W	RSARCH+L GARCH+L ARCH+L (ranked)
52. Harvey and Whaley (1992)	S&P 100 (OEX)	Oct. 85–July 89	D	Implied _{ATM calls+puts} (American binomial, shortest maturity > 15 days) (Predict changes in implied.)
53. Heynen and Kat (1994)	7 stock indices and 5 exchange rates	1/1/80–31/12/92 In: 80–87 Out: 88–92 (87's crash included in in-sample)	D	SV(? See comment.) EGARCH GARCH RW (ranked)

Forecasting Horizon	Evaluation & R-square	Comments
1 week ahead (model not re-estimated). Use weekly squared deviation to proxy volatility.	R^2 calculated without constant term, is 4 to 8% for RSGARCH, negative for some CV and GARCH. Comparable RMSE and MAE between GARCH & RSGARCH.	Volatility follows GARCH and CIR square root process. Interest rate rise increases probability of switching into high volatility regime. Low volatility persistence and strong rate level mean reversion at high volatility state. At low volatility state, rate appears random walk and volatility is highly persistence.
Information not available.	Regression with robust SE. No information on R^2 and forecast biasness.	Use mid of bid-ask option price to limit 'bounce' effect. Eliminate 'nonsynchronicity' by using simultaneous exchange rate and option price. HIS and GARCH contain no incremental information. Implied _{Heston} and Implied _{HW} are comparable and are marginally better than Implied _{BS} . <i>Only have access to abstract.</i>
60 days ahead. Use sample variance of daily returns to proxy actual volatility.	US\$/DM R^2 is 4, 3, 1% for the three methods. (9, 4, 1% for US\$/¥:) All forecasts are biased $\alpha > 0$, $\beta < 1$ with robust SE.	Conclusion same as Guo(1996a). Use Barone-Adesi/Whaley approximation for American options. No risk premium for volatility variance risk. GARCH has no incremental information. Visual inspection of figures suggests implied forecasts lagged actual.
Non-overlapping 15, 35 and 55 days ahead.	RMSE, MAE	Implied is better than historical and cross strike averaging is better than intertemporal averaging (except during very turbulent periods).
1 month ahead. Use squared monthly residual returns to proxy volatility.	MAE	Found economic recessions drive fluctuations in stock returns volatility. "L" denotes leverage effect. RS model outperformed ARCH/GARCH+L.
1, 4 and 8 weeks ahead. Use squared weekly residual returns to proxy volatility.	MSE, MAE, MSLE, MALE. Errors calculated from variance and log variance.	Allowing up to 4 regimes with t distribution. RSARCH with leverage (L) provides best forecast. Student- t is preferred to GED and Gaussian.
1 day ahead implied for use in pricing next day option.	R^2 is 15% for calls and 4% for puts (excluding 1987 crash.)	Implied volatility changes are statistically predictable, but market was efficient, as simulated transactions (NTM call and put and delta hedged using futures) did not produce profit.
Non-overlapping 5, 10, 15, 20, 25, 50, 75, 100 days horizon with constant update of parameters estimates. Use sample standard deviations of daily returns to proxy "actual vol".	MedSE	SV appears to dominate in index but produces errors that are 10 times larger than (E)GARCH in exchange rate. The impact of 87's crash is unclear. Conclude that volatility model forecasting performance depends on the asset class.

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Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
54. Hol and Koopman (2002)	S&P 100 (VIX)	2/1/86–29/6/2001 Out: Jan. 97–June 01	D	SIV SVX ⁺ SV (ranked)
55. Hwang and Satchell (1998)	LIFFE stock options	23/3/92–7/10/96 240 daily out-of-sample forecasts.	D	Log-ARFIMA-RV Scaled truncated Detrended unscaled truncated MA _{opt n=20} -IV Adj MA _{opt n=20} -RV GARCH-RV (ranked, forecast implied)
56. Jorion (1995)	DM/\$, ¥/\$, SrFr/\$ futures options on CME	1/85–2/92 7/86–2/92 3/85–2/92	D	Implied _{ATM BS call+put} GARCH (1,1), MA ₂₀ (ranked)
57. Jorion (1996)	DM/\$ futures options on CME	Jan. 85–Feb. 92	D	Implied _{Black, ATM} GARCH(1,1) (ranked)
58. Karolyi (1993)	74 stock options	13/1/84–11/12/85	M	Bayesian Implied _{Call} Implied _{Call} HIS _{20,60} (Predict option price not “actual vol”.)
59. Klaassen (2002)	US\$/£, US\$/DM and US\$/¥	3/1/78–23/7/97 Out: 20/10/87–23/7/97	D	RSGARCH RSARCH GARCH(1,1) (ranked)
60. Kroner, Kneafsey and Claessens (1995)	Futures options on Cocoa, cotton, corn, gold, silver, sugar, wheat	Jan. 87–Dec. 90 (Kept last 40 observations for out of sample forecast)	D	GR>COMB Implied _{BAW Call} (WLS>AVG>ATM) HIS _{7 weeks} > GARCH (ranked)
61. Lamoureux and Lastrapes (1993)	Futures prices Stock options for 10 non-dividend paying stocks (CBOE)	Jan. 87–July 91 19/4/82–31/3/84	D	Implied _{Hull-White NTM Call} (intermediate term to maturity, WLS) HIS _{updated expanding estimate} GARCH (ranked, based on regression result)

Forecasting Horizon	Evaluation & R-square	Comments
1,2, 5, 10, 15 and 20 days ahead. Use 10-min. returns to construct “actual vol.”	R^2 ranges between 17 to 33%, MSE, MedSE, MAE. α and β not reported. All forecasts underestimate actuals.	SVX is SV with implied vol_{VIX} as an exogenous variable while SVX^+ is SVX with persistence adjustment. SIV is stochastic implied with persistence parameter set equal to zero.
1, 5, 10, 20, ..., 90, 100, 120 days ahead IV estimated from a rolling sample of 778 daily observations. Different estimation intervals were tested for robustness.	MAE, MFE.	Forecast implied ATM_{BS} of shortest maturity option (with at 15 trading days to maturity). Build MA in IV and ARIMA on log (IV). Error statistics for all forecasts are close except those for GARCH forecasts. The scaling in Log-ARFIMA-RV is to adjust for Jensen inequality.
1 day ahead & option maturity. Use squared returns and aggregate of square returns to proxy actual volatility.	R^2 is 5% (1-day) or 10–15% (option maturity). With robust SE, $\alpha_{\text{implied}} > 0$ and $\beta_{\text{implied}} < 1$ for long horizon and is unbiased for 1-day forecasts.	Implied is superior to the historical methods and least biased. MA and GARCH provide only marginal incremental information.
1 day ahead, use daily squared to proxy actual volatility.	R^2 about 5%. H_0 : $\alpha_{\text{implied}} = 0$, $\beta_{\text{implied}} = 1$ cannot be rejected with robust SE.	R^2 increases from 5% to 19% when unexpected trading volume is included. Implied volatility subsumed information in GARCH forecast, expected futures trading volume and bid-ask spread.
20 days ahead volatility.	MSE	Bayesian adjustment to implied to incorporate cross sectional information such as firm size, leverage and trading volume useful in predicting next period option price.
1 and 10 days ahead. Use mean adjusted 1- and 10-day return squares to proxy actual volatility.	MSE of variance, regression though R^2 is not reported.	GARCH(1,1) forecasts are more variable than RS models. RS provides statistically significant improvement in forecasting volatility for US\$/DM but not the other exchange rates.
225 calendar day (160 working days) ahead, which is longer than average.	MSE, ME	GR: Granger and Ramanathan (1984)'s regression weighted combined forecast, COMB: lag implied in GARCH conditional variance equation. Combined method is best suggests option market inefficiency.
90 to 180 days matching option maturity estimated using rolling 300 observations and expanding sample. Use sample variance of daily returns to proxy “actual vol”.	ME, MAE, RMSE. Average implied is lower than actual for all stocks. R^2 on variance varies between 3–84% across stocks and models.	Implied volatility is best but biased. HIS provides incremental info to implied and has the lowest RMSE. When all three forecasts are included; $\alpha > 0$, $1 > \beta_{\text{implied}} > 0$, $\text{GARCH} = 0$, $\beta_{\text{HIS}} < 0$ with robust SE. Plausible explanations include option traders overreact to recent volatility shocks, and volatility risk premium is non-zero and time varying.

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Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
62. Latane and Rendleman (1976)	24 stock options from CBOE	5/10/73–28/6/74	W	Implied _{vega weighted} HIS _{4 years} (ranked)
63. Lee (1991)	\$/DM, \$/£, \$/¥, \$/FFr, \$/C\$ (Fed Res Bulletin)	7/3/73–4/10/89 Out: 21 Oct. 81– 11 Oct. 89.	W (Wed, 12pm)	Kernel (Gaussian, Truncated) Index (combining ARMA and GARCH) EGARCH (1,1) GARCH (1,1) IGARCH with trend (rank changes see comment for general assessment)
64. Li (2002)	\$/DM, \$/£, \$/¥ OTC ATM options \$/£, \$/¥ \$/DM	3/12/86–30/12/99 In: 12/8/86–11/5/95 19/6/94–13/6/99 19/6/94–30/12/98	Tick (5 min) D D	Implied _{GK OTC ATM} ARFIMA _{realised} (Implied better at shorter horizon and ARFIMA better at long horizon.)
65. Lopez (2001)	C\$/US\$, DM/US\$, ¥/US\$, US\$/£	1980–1995 In: 1980–1993 Out: 1994–1995	D	SV-AR(1)-normal GARCH-gev EWMA-normal GARCH-normal, -t EWMA-t AR(10)-Sq, -Abs Constant (approx rank, see comments)
66. Loudon, Watt and Yadav (2000)	FT All Share	Jan. 71–Oct. 97 Sub-periods: Jan. 71–Dec. 80 Jan. 81–Dec. 90 Jan. 91–Oct. 97	D	EGARCH, GJR, TS-GARCH, TGARCH NGARCH, VGARCH, GARCH, MGARCH (No clear rank, forecast GARCH vol)
67. Martens and Zein (2002)	S&P 500 futures, ¥/US\$ futures, Crude oil futures	Jan. 94–Dec. 2000 Jan. 96–Dec. 2000 June 93–Dec. 2000	Tick	Implied _{BAW VIX style} Log-ARFIMA GARCH (ranked, see comment also)
68. McKenzie (1999)	21 A\$ bilateral exchange rates	Various length from 1/1/86 or 4/11/92 to 31/10/95	D	Square vs. power transformation (ARCH models with various lags. See comment for rank.)

Forecasting Horizon	Evaluation & R-square	Comments
In-sample forecast and forecast that extend partially into the future. Use weekly and monthly returns to calculate actual volatility of various horizons.	Cross-section correlation between volatility estimates for 38 weeks and a 2-year period.	Used European model on American options and omitted dividends. "Actual" is more correlated (0.686) with "Implied" than HIS volatility (0.463). Highest correlation is that between implied and actual standard deviations which were calculated partially into the future.
1 week ahead (451 observations in sample and 414 observations out-of-sample)	RMSE, MAE. It is not clear how actual volatility was estimated.	Nonlinear models are, in general, better than linear GARCH. Kernel method is best with MAE. But most of the RMSE and MAE are very close. Over 30 kernel models were fitted, but only those with smallest RMSE and MAE were reported. It is not clear how the non-linear equivalence was constructed. Multi-step forecast results were mentioned but not shown.
1, 2, 3 and 6 months ahead. Parameters not re-estimated. Use 5-min returns to construct "actual vol".	MAE. R^2 ranges 0.3–51% (Implied), 7.3–47% (LM), 16–53% (emcompass). For both models, H_0 : $\alpha = 0$, $\beta = 1$ are rejected and typically $\beta < 1$ with robust SE.	Both forecasts have incremental information especially at long horizon. Forcing: $\alpha = 0$, $\beta = 1$ produce low/negative R^2 (especially for long horizon). Model realised standard deviation as ARFIMA without log transformation and with no constant, which is awkward as a theoretical model for volatility.
1 day ahead and probability forecasts for four "economic events", viz. cdf of specific regions. Use daily squared residuals to proxy volatility. Use empirical distribution to derive cdf.	MSE, MAE, LL, HMSE, GMLE and QPS (quadratic probability scores).	LL is the logarithmic loss function from Pagan and Schwert (1990), HMSE is the heteroskedasticity-adj MSE from Bollerslev and Ghysels (1996) and GMLE is the Gaussian quasi-ML function from Bollerslev, Engle and Nelson (1994). Forecasts from all models are indistinguishable. QPS favours SV-n, GARCH-g and EWMA-n.
Parameters estimated in period 1 (or 2) used to produce conditional variances in period 2 (or 3). Use GARCH squared residuals as "actual" volatility.	RMSE, regression on log volatility and a list of diagnostics. R^2 is about 4% in period 2 and 5% in period 3.	TS-GARCH is an absolute return version of GARCH. All GARCH specifications have comparable performance though non-linear, asymmetric versions seem to fare better. Multiplicative GARCH appears worst, followed by NGARCH and VGARCH (Engle and Ng 1993).
Non-overlapping 1, 5, 10, 20, 30 and 40 days ahead. 500 daily observations in in-sample which expands on each iteration.	Heteroskedasticity adjusted RMSE. R^2 ranges 25–52% (implied), 15–48% (LM) across assets and horizons. Both models provide incremental info to encompassing regr.	Scaled down one large oil price. Log-ARFIMA truncated at lag 100. Based on R^2 , Implied outperforms GARCH in every case, and beats Log-ARFIMA in ¥/US\$ and Crude oil. Implied has larger HRMSE than Log-ARFIMA in most cases. <i>Difficult to comment on implied's biasness from information presented.</i>
1 day ahead absolute returns.	RMS, ME, MAE. Regressions suggest all ARCH forecasts are biased. No R^2 was reported.	The optimal power is closer to 1 suggesting squared return is not the best specification in ARCH type model for forecasting purpose.

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Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
69. McMillan, Speight and Gwilym (2000)	FTSE100 FT All Share	Jan. 84–July 96 Jan. 69–July 96 Out: 1996–1996 for both series.	D, W, M	RW, MA, ES, EWMA GARCH, TGARCH, EGARCH, CGARCH HIS, Regression, (ranked)
70. Noh, Engle and Kane (1994)	S&P 500 index options	Oct. 85–Feb. 92	D	GARCH adj for weekend & hols Implied _{BS} weighted by trading volume (ranked, predict option price not “actual vol”)
71. Pagan and Schwert (1990)	US stock market	1834–1937 Out: 1900–1925 (low volatility), 1926–37 (high volatility)	M	EGARCH(1,2) GARCH(1,2) 2-step conditional variance RS-AR(m) Kernel (1 lag) Fourier (1 or 2 lags) (ranked)
72. Pong, Shackleton, Taylor and Xu (2002)	US\$/£	In: July 87–Dec. 93 Out: Jan. 94–Dec. 98	5-, 30-minute	Implied _{ATM, OTC quote} (bias adj using rolling regr on last 5 years monthly data) Log-ARMA(2,1) Log-ARFIMA(1,d,1) GARCH(1,1) (ranked)
73. Poteshman (2000)	S&P 500 (SPX) & futures	1 Jun. 88–29 Aug. 97 Heston estimation: 1 June 93–29 Aug. 97	D futures Tick	Implied _{Heston} Implied _{BS} (both implieds are from WLS of all options < 7 months but > 6 calendar days) HIS _{1, 2, 3, 6 months} (ranked)
	S&P 500	7 June 62–May 93	M	
74. Randolph and Najand (1991)	S&P 500 futures options	2/1/1986–31/12/88 (Crash included)	Daily opening Tick	MRM _{ATM HIS} MRM _{ATM implied} GARCH(1,1)
	ATM Calls only	In: First 80 observations		HIS _{20 day} Implied _{Black} (ranked though the error statistics are close)
75. Schmalensee and Trippi (1978)	6 CBOE stock options	29/4/74–23/5/75 56 weekly observations	W	Implied _{BS call} (simple average of all strikes and all maturities.) (Forecast implied not actual volatility.)

Forecasting Horizon	Evaluation & R-square	Comments
$j = 1$ day, 1 week and 1 month ahead based on the three data frequencies. Use j period squared returns to proxy actual volatility.	ME, MAE, RMSE for symmetry loss function. MME(U) & MME(O), mean mixed error that penalize under/over predictions.	CGARCH is the component GARCH model. Actual volatility is proxied by mean adjusted squared returns, which is likely to be extremely noisy. Evaluation conducted on variance, hence forecast error statistics are very close for most models. RW, MA, ES dominate at low frequency and when crash is included. Performances of GARCH models are similar though not as good.
Option maturity. Based on 1,000 days rolling period estimation	Equate forecastability with profitability under the assumption of an inefficient option market	Regression with call+put implieds, daily dummies and previous day returns to predict next day implied and option prices. Straddle strategy is not vega neutral even though it might be delta neutral assuming market is complete. It is possible that profit is due to now well-documented post 87's crash higher option premium.
One month ahead. Use squared residual monthly returns to proxy actual volatility.	R^2 is 7–11% for 1900–25 and 8% for 1926–37. Compared with R^2 for variance, R^2 for log variance is smaller in 1900–25 and larger in 1926–37.	The nonparametric models fared worse than the parametric models. EGARCH came out best because of the ability to capture volatility asymmetry. Some prediction bias was documented.
1 month and 3 month ahead at 1-month interval	ME, MSE, regression. R^2 ranges between 22–39% (1-month) and 6–21% (3-month)	Implied, ARMA and ARFIMA have similar performance. GARCH(1,1) clearly inferior. Best combination is Implied+ARMA(2,1). Log-AR(FI)MA forecasts adjusted for Jensen inequality. <i>Difficult to comment on implied's biasness from information presented.</i>
Option maturity (about 3.5 to 4 weeks, non-overlapping). Use 5-min futures inferred index return to proxy “actual vol”.	BS R^2 is over 50%. Heston implied produced similar R^2 but very close to being unbiased.	F test for $H_0: \alpha_{BS} = 0, \beta_{BS} = 1$ are rejected though t -test supports H_0 on individual coefficients. Show biasness is not caused by bid-ask spread. Using $\ln\sigma$, high frequency realised vol, and Heston model, all help to reduce implied biasness.
Non-overlapping 20 days ahead, re-estimated using expanding sample.	ME, RMSE, MAE, MAPE	Mean reversion model (MRM) sets drift rate of volatility to follow a mean reverting process taking implied _{ATM} (or HIS) as the previous day vol. Argue that GARCH did not work as well because it tends to provide a persistent forecast, which is valid only in period when changes in vol are small.
1 week ahead. “Actual” proxied by weekly range and average price deviation.	Statistical tests reject the hypothesis that IV responds positively to current volatility.	Find implied rises when stock price falls, negative serial correlation in changes of IV and a tendency for IV of different stocks to move together. Argue that IV might correspond better with <i>future</i> volatility.

(Continued)

Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
76. Scott and Tucker (1989)	DM/\$, £/\$, C\$/¥, ¥/\$ & SrFr/\$ American options on PHLX	14/3/83–13/3/87 (Pre crash)	Daily closing tick	Implied _{GK} (vega, Inferred ATM, NTM) Implied _{CEV} (similar rank)
77. Sill (1993)	S&P 500	1959–92	M	HIS with exo variables HIS (See comment)
78. Szakmary, Ors, Kim and Davidson (2002)	Futures options on S&P 500, 9 interest rates, 5 currency, 4 energy, 3 metals, 10 agriculture 3 livestock	Various dates between Jan. 83–May 01	D	Implied _{BK, NTM 2Calls+2Puts equal weight} HIS ₃₀ , GARCH (ranked)
79. Taylor JW (2001)	DAX, S&P 500, Hang Seng, FTSE100, Amsterdam EOE, Nikkei, Singapore All Share	6/1/88–30/8/95 (equally split between in- and out-)	W	STES (E, AE, EAE) GJR (+Smoothed variations) GARCH MA _{20 weeks} , Riskmetrics (ranked)
80. Taylor SJ (1986)	15 US stocks FT30 6 metal £/\$ 5 agricultural futures 4 interest rate futures	Jan. 66–Dec. 76 July 75–Aug. 82 Various length Nov. 74–Sep. 82 Various length Various length	D	EWMA Log-AR(1) ARMACH-Abs ARMACH-Sq HIS (ranked) ARMACH-Sq is similar to GARCH
81. Taylor SJ (1987)	DM/\$ futures	1977–83	D	High, low and closing prices (see comment)
82. Taylor SJ and Xu (1997)	DM/\$ DM/\$ options on PHLX	1/10/92–30/9/93 In: 9 months Out: 3 months	Quote D	Implied + ARCH combined Implied, ARCH HIS _{9 months} HIS _{last hour realised vol} (ranked) See comment for details on implied & ARCH
83. Tse (1991)	Topix Nikkei Stock Average	In: 1986–87 Out: 1988–89	D	EWMA HIS ARCH, GARCH (ranked)

Forecasting Horizon	Evaluation & R-square	Comments
Non-overlapping option maturity: 3, 6 and 9 months. Use sample SD of daily returns to proxy “actual vol”.	MSE, R^2 ranges from 42 to 49%. In all cases, $\alpha > 0$, $\beta < 1$. HIS has no incremental info content.	Simple B-S forecasts just as well as sophisticated CEV model. Claimed omission of early exercise is not important. Weighting scheme does not matter. Forecasts for different currencies were mixed together.
1 month ahead	R^2 increase from 1% to 10% when additional variables were added.	Volatility is higher in recessions than in expansions, and the spread between commercial-paper and T-Bill rates predict stock market volatility.
Overlapping option maturity, shortest but > 10 days. Use sample SD of daily returns over forecast horizon to proxy “actual vol”.	R^2 smaller for financial (23–28%), higher for metal & agricult (30–37%), highest for livestock & energy (47, 58%)	HIS ₃₀ and GARCH have little or no incremental information content. $\alpha_{\text{implied}} > 0$ for 24 cases (or 69%), all 35 cases $\beta_{\text{implied}} < 1$ with robust SE.
1 week ahead using a moving window of 200 weekly returns. Use daily squared residual returns to construct weekly “actual” volatility.	ME, MAE, RMSE, R^2 (about 30% for HK and Japan and 6% for US)	Models estimated based on minimizing in-sample forecast errors instead of ML. STES-EAE (smooth transition exponential smoothing with return and absolute return as transition variables) produced consistently better performance for 1-step ahead forecasts.
1 and 10 days ahead absolute returns. 2/3 of sample used in estimation. Use daily absolute returns deviation as “actual vol”.	Relative MSE	Represent one of the earliest studies in ARCH class forecasts. The issue of volatility stationarity is not important when forecast over short horizon. Non-stationary series (e.g. EWMA) has the advantage of having fewer parameter estimates and forecasts respond to variance change fairly quickly.
1, 5, 10 & 20 days ahead. Estimation period, 5 years.	RMSE	Best model is a weighted average of present and past high, low and closing prices with adjustments for weekend and holiday effects.
1 hour ahead estimated from 9 months in-sample period. Use 5-min returns to proxy “actual vol”.	MAE and MSE on std deviation & variance Friday macro news seasonal factors have no impact on forecast accuracy.	5-min return has information incremental to daily implied when forecasting hourly volatility. ARCH model includes with hourly and 5-min returns in the last hr plus 120 hour/day/week seasonal factors. Implied derived from NTM shortest maturity (> 9 calendar days) Call + Put using BAW.
25 days ahead estimated from rolling 300 observations	ME, RMSE, MAE, MAPE of variance of 21 non-overlapping 25-day periods.	Use dummies in mean equation to control for 1987 crash. Non-normality provides a better fit but a poorer forecast. ARCH/GARCH models are slow to react to abrupt change in volatility. EWMA adjust to changes very quickly.

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Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
84. Tse and Tung (1992)	Singapore, 5 VW market & industry indices	19/3/75–25/10/88	D	EWMA HIS GARCH (ranked)
85. Vasilellis and Meade (1996)	Stock options 12 UK stocks (LIFFE)	In: 28/3/86–27/6/86 In2 (for combine forecast): 28/6/86–25/3/88 Out: 6/7/88–21/9/91	W	Combine(Implied + GARCH) Implied (various, see comment) GARCH EWMA HIS _{3 months} (ranked, results not sensitive to basis use to combine)
86. Vilasuso (2002)	C\$/\$, FFr/\$, DM/\$, ¥/\$, £/\$	In: 13/3/79–31/12/97 Out: 1/1/98–31/12/99	D	FIGARCH GARCH, IGARCH (ranked, GARCH marginally better than IGARCH)
87. Walsh and Tsou (1998)	Australian indices: VW20, VW50 & VW300	1 Jan. 93–31 Dec. 95 In: 1 year Out: 2 years	5-min to form H, D & W returns	EWMA GARCH (not for weekly returns) HIS, IEV (Improved extreme-value method) (ranked)
88. Wei and Frankel (1991)	SrFr/\$, DM/\$, ¥/\$, £/\$ options (PHLX) Spot rates	2/83–1/90	M D	Implied _{GK} ATM call (shortest maturity)
89. West and Cho (1995)	C\$/\$, FFr/\$, DM/\$, ¥/\$, £/\$	14/3/73–20/9/89 In: 14/3/73–17/6/81 Out: 24/6/81–12/4/89	W	GARCH(1,1) IGARCH(1,1) AR(12) in absolute AR(12) in squares Homoskedastic Gaussian kernel (No clear rank)
90. Wiggins (1992)	S&P 500 futures	4/82–12/89	D	ARMA model with 2 types of estimators: 1. Parkinson/Garmen-Klass extreme value estimators 2. Close-to-close estimator (ranked)
91. Xu and Taylor (1995)	£/\$, DM/\$, ¥/\$ & SrFr/\$ PHLX options Corresponding futures rates	In: Jan. 85–Oct. 89 Out: 18 Oct. 89–4 Feb. 92	D	Implied _{BAW} NTM TS or short GARCH _{Normal} or GED HIS _{4 weeks} (ranked)

Forecasting Horizon	Evaluation & R-square	Comments
25 days ahead estimated from rolling 425 observations	RMSE, MAE	EWMA is superior, GARCH worst. Absolute returns > 7% are truncated. Sign of non-stationarity. Some GARCH non-convergence
3 months ahead. Use sample SD of daily returns to proxy "actual vol".	RMSE	Implied: 5-day average dominates 1-day implied vol. Weighting scheme: max vega > vega weighted > elasticity weighted > max elasticity with '>' indicates better forecasting performance. Adjustment for div & early exercise: Rubinstein > Roll > constant yield. Crash period might have disadvantaged time series methods.
1, 5 and 10 days ahead. Used daily squared returns to proxy actual volatility.	MSE, MAE, & Diebold-Mariano's test for sig. difference.	Significantly better forecasting performance from FIGARCH. Built FIARMA (with a constant term) on conditional variance without taking log. Truncated at lag 250.
1 hour, 1 day and 1 week ahead estimated from a 1-year rolling sample. Use square of price changes (non cumulative) as "actual vol".	MSE, RMSE, MAE, MAPE	Index with larger number of stock is easier to forecast due to diversification, but gets harder as sampling interval becomes shorter due to problem of non-synchronous trading. None of the GARCH estimations converged for the weekly series, probably too few observations.
Non-overlapping 1 month ahead. Use sample SD of daily exchange rate return to proxy "actual vol"	R^2 30% (£), 17% (DM), 3% (SrFr), 0% (¥). $\alpha > 0$, $\beta < 1$ (except that for £/\$, $\alpha > 0$, $\beta = 1$) with heteroske consistent SE.	Use European formula for American style option. Also suffers from non-synchronicity problem. Other tests reveal that Implied tends to over-predict high vol and under-predict low vol. Forecast/implied could be made more accurate by placing more weight on long run average.
$j = 1, 12, 24$ weeks estimated from rolling 432 weeks. Use j period squared returns to proxy actual volatility.	RMSE and regression test on variance, R^2 varies from 0.1% to 4.5%.	Some GARCH forecasts mean revert to unconditional variance in 12- to 24-weeks. It is difficult to choose between models. Nonparametric method came out worst though statistical tests for do not reject null of no significant difference in most cases.
1 week ahead and 1 month ahead. Compute actual volatility from daily observations.	Bias test, efficiency test, regression	Modified Parkinson approach is least biased. C-t-C estimator is three times less efficient than EV estimators. Parkinson estimator is also better than C-t-C at forecasting. 87's crash period excluded from analysis.
Non-overlapping 4 weeks ahead, estimated from a rolling sample of 250 weeks daily data. Use cumulative daily squared returns to proxy "actual vol".	ME, MAE, RMSE. When α_{implied} is set equal to 0, $\beta_{\text{implied}} = 1$ cannot be rejected.	Implied works best and is unbiased. Other forecasts have no incremental information. GARCH forecast performance not sensitive to distributional assumption about returns. The choice of implied predictor (term structure, TS, or short maturity) does not affect results.

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Author(s)	Asset(s)	Data Period	Data Freq	Forecasting Methods & Rank
92. Yu (2002)	NZSE40	Jan. 80-Dec. 98 In: 1980–93 Out: 1994–98	D	SV (of log variance) GARCH(3,2), GARCH(1,1) HIS, MA _{5yr} or 10 _{yr} ES and EWMA (monthly revision) Regression _{lag-1} ARCH(9), RW, (ranked)
93. Zumbach (2002)	USD/CHF, USD/JPY	1/1/89–1/7/2000	H	LM-ARCH F-GARCH GARCH And their integrated counterparts (ranked)

Ranked: models appear in the order of forecasting performance; best performing model at the top. If two weighting schemes or two forecasting models appear at both sides of “>”, it means the LHS is better than the RHS in terms of forecasting performance. **SE:** Standard error.

ATM: At the money, **NTM:** Near the money, **OTM:** Out of the money, **WLS:** an implied volatility weighting scheme used in Whaley (1982) designed to minimize the pricing errors of a collection of options. In some cases the pricing errors are multiplied by trading volume or vega to give ATM implied a greater weight.

REFERENCES

- Aggarwal, Reena; Carla Inclan and Ricardo Leal. 1999. “Volatility in Emerging Stock Markets,” *J. Finan. Quant. Anal.* 34:1, pp. 33–55.
- Akgiray, Vedat. 1989. “Conditional Heteroskedasticity in Time Series of Stock Returns: Evidence and Forecasts,” *J. Bus.* 62, pp. 55–80.
- Alford, Andrew W. and James R. Boatsman. 1995. “Predicting Long-Term Stock Return Volatility: Implications for Accounting and Valuation of Equity Derivatives,” *Acc. Rev.* 70:4, pp. 599–618.
- Amin, Kaushik and Victor Ng. 1997. “Inferring Future Volatility from the Information in Implied Volatility in Eurodollar Options: A New Approach,” *Rev. Finan. Stud.* 10, pp. 333–67.
- Andersen, Torben G. 1996. “Return Volatility and Trading Volume: An Information Flow Interpretation of Stochastic Volatility,” *J. Finance* 51:1, pp. 169–204.
- Andersen, Torben and Tim Bollerslev. 1998. “Answering the Skeptics: Yes, Standard Volatility Models Do Provide Accurate Forecasts,” *Int. Econ. Rev.* 39:4, pp. 885–905.
- Andersen, Torben G.; Tim Bollerslev and Steve Lange. 1999. “Forecasting Financial Market Volatility: Sample Frequency vis-à-vis Forecast Horizon,” *J. Empirical Finance* 6:5, pp. 457–77.
- Andersen, Torben G.; Tim Bollerslev, Francis X. Diebold and Heiko Ebens. 2001. “The Distribution of Realized Stock Return Volatility,” *J. Finan. Econ.* 61:1, pp. 43–76.
- Andersen, Torben G.; Tim Bollerslev, Francis X. Diebold and Paul Labys. 2001. “The Distribution of Realized Exchange Rate Volatility,” *J. Amer. Statist. Assoc.* 96:453, pp. 42–57.
- . 2002. “Modeling and Forecasting Realized Volatility,” work. paper, North Western U., Duke U., and U. Pennsylvania.
- Andersen, Torben G. and Bent E. Sorensen. 1997. “GMM and QML Asymptotic Standard Deviations in Stochastic Volatility Models,” *J. Econometrics* 76, pp. 397–403.
- Aydemir, Abdurrahman B. 1998. “Volatility Modelling in Finance,” in *Forecasting Volatility in the Financial Markets*. John Knight and Stephen Satchell, eds. Butterworth Heinemann, ch. 1, pp. 1–46.
- Baillie, Richard T. and Tim Bollerslev. 1989. “The Message in Daily Exchange Rates: A Conditional-Variance Tale,” *J. Bus. Econ. Statist.* 7:3, pp. 297–305.
- Baillie, Richard T.; Tim Bollerslev and Hans O. Mikkelsen. 1996. “Fractionally Integrated Generalized Autoregressive Conditional Heteroskedasticity,” *J. Econometrics* 74:1, pp. 3–30.
- Bali, Turan G. 2000. “Testing The Empirical Performance of Stochastic Volatility Models of the Short-Term Interest Rate,” *J. Finan. Quant. Anal.* 35:2, pp. 191–215.
- Ball, Clifford A. and Walter N. Torous. 1984. “The Maximum Likelihood Estimation of Security Price Volatility: Theory, Evidence and Application to Option Pricing,” *J. Bus.* 57:1, pp. 97–113.
- Bates, David S. 1996. “Testing Option Pricing Models,” in *Handbook of Statistics, v.14: Statistical Methods in Finance*. G.S. Maddala and C.R. Rao, eds. Amsterdam: Elsevier, North-Holland, pp. 567–611.

Forecasting Horizon	Evaluation & R-square	Comments
1 month ahead estimated from previous 180 to 228 months of daily data. Use aggregate of daily squared returns to construct actual monthly volatility.	RMSE, MAE, Theil-U and LINEX on variance.	Range of the evaluation measures for most models is very narrow. Within this narrow range, SV ranked first, performance of GARCH was sensitive to evaluation measure; regression and EWMA methods did not perform well. Worst performance from ARCH(9) and RW. Volatile periods (Oct. 87 and Oct. 97) included in in- and out-of samples.
1 day ahead estimated from previous 5.5 years	RMSE. Realized volatility measured using hourly returns.	LM-ARCH, aggregates high frequency squared returns with a set of power law weights, is the best though difference is small. All integrated versions are more stable across time.

HIS: Historical volatility constructed based on past variance/standard deviation. **VIX:** Chicago Board of Option Exchange's volatility index derived from S&P 100 options. **RS:** Regime Switching,

BS: Black-Scholes, **BK:** Black model for pricing futures option. **BAW:** Barone-Adesi and Whaley American option pricing formula, **HW:** Hull and White option pricing model with stochastic volatility, **FW:** Fleming and Whaley (1994) modified binomial method that takes into account wildcard option, **GK:** Garman and Kohlhagan model for pricing European currency option. **HJM:** Heath, Jarrow and Morton (1992) forward rate model for interest rates.

- Beckers, Stan. 1981. "Standard Deviations Implied in Option Prices as Predictors of Future Stock Price Variability," *J. Banking Finance* 5, pp. 363–82.
- Benzoni, Luca. 2001. "Pricing Options under Stochastic Volatility: An Empirical Investigation," work. paper, Carlson School of Management, Minneapolis.
- Bera, Anil K. and Matthew L. Higgins. 1993. "ARCH Models: Properties, Estimation and Testing," *J. Econ. Surveys* 7, 4, pp. 305–65.
- Bera, Anil and Matthew Higgins. 1997. "ARCH and Bilinearity as Competing Models for Nonlinear Dependence," *J. Bus. Econ. Statist.* 15:1, pp. 43–50.
- Black, Fischer and Myron Scholes. 1973. "The Pricing of Options and Corporate Liabilities," *J. Polit. Econ.* 81:May/June, pp. 637–54.
- Blair, Bevan; Ser-Huang Poon and Stephen J. Taylor. 2001. "Forecasting S&P 100 Volatility: The Incremental Information Content of Implied Volatilities and High Frequency Index Returns," *J. Econometrics* 105, pp. 5–26.
- Bluhm, Hagen H.W. and Jun Yu. 2000. "Forecasting Volatility: Evidence from the German Stock Market," work. paper, U. Auckland.
- Bollerslev, Tim 1986. "Generalized Autoregressive Conditional Heteroskedasticity," *J. Econometrics* 31, pp. 307–28.
- . 1987. "A Conditionally Heteroskedastic Time Series Model for Speculative Prices and Rates of Return," *Rev. Econ. Statist.* 69:3, pp. 542–47.
- Bollerslev, Tim; Ray Y. Chou and Kenneth P. Kroner. 1992. "ARCH Modeling in Finance: A Review of the Theory and Empirical Evidence," *J. Econometrics* 52, pp. 5–59.
- Bollerslev, Tim; Robert F. Engle and Daniel B. Nelson. 1994. "ARCH models," in *Handbook of Econometrics*, Vol. IV. Robert F. Engle and D.L. McFadden, eds. Amsterdam: North-Holland, pp. 2959–3038.
- Bollerslev, Tim and Eric Ghysels. 1996. "Periodic Autoregressive Conditional Heteroskedasticity," *J. Bus. Econ. Statist.* 14:2, pp. 139–51.
- Bollerslev, Tim and Hans O. Mikkelsen. 1996. "Modeling and Pricing Long Memory in Stock Market Volatility," *J. Econometrics* 73:1, pp. 151–84.
- Boudoukh, Jacob; Matthew Richardson and Robert F. Whitelaw. 1997. "Investigation of a Class of Volatility Estimators," *J. Derivatives* 4:3, pp. 63–71.
- Brace, Alan and Allan Hodgson. 1991. "Index Futures Options in Australia—An Empirical Focus on Volatility," *Acc. Finance* 31:2, pp. 13–31.
- Brailsford, Timothy J. and Robert W. Faff. 1996. "An Evaluation of Volatility Forecasting Techniques," *J. Banking Finance*, 20:3, pp. 419–38.
- Brooks, Chris. 1998. "Predicting Stock Market Volatility: Can Market Volume Help?" *J. Forecasting* 17:1, pp. 59–80.
- Brown, Stephen. 1990. "Estimating Volatility," in *Financial Options: From Theory to Practice*. Stephen Figlewski, William L. Silber and Martin G. Subrahmanyam, eds. Chicago: Irwin.
- Canina, Linda and Stephen Figlewski. 1993. "The Informational Content of Implied Volatility," *Rev. Finan. Stud.* 6:3, pp. 659–81.
- Cao, Charles Q. and Ruey S. Tsay. 1992. "Nonlinear Time-Series Analysis of Stock Volatilities," *J. Applied Econometrics* 1:supp. pp. 165–85.
- Chernov, Mikhail. 2001. "Implied Volatilities as Forecasts of Future Volatility, the Market Risk Premia, and Returns Variability," work. paper, Columbia Business School.

- Chernov, Mikhail and Eric Ghysels. 2000. "A Study towards a Unified Approach to the Joint Estimation of Objective and Risk Neutral Measures for the Purposes of Options Valuation," *J. Finan. Econ.* 56:3, pp. 407–58.
- Chiras, Donald and Steven Manaster. 1978. "The Information Content of Option Prices and a Test of Market Efficiency," *J. Finan. Econ.* 6, pp. 213–34.
- Cho, D. Chinyung and Edward W. Frees. 1988. "Estimating the Volatility of Discrete Stock Prices," *J. Finance* 43:2, pp. 451–66.
- Chong, Yock Y. and David F. Hendry. 1986. "Econometric Evaluation of Linear Macro-Economic Models," *Rev. Econ. Stud.* 53, pp. 671–90.
- Christensen, Bent J. and Nagpurnanand R. Prabhala. 1998. "The Relation between Implied and Realized Volatility," *J. Finan. Econ.* 50:2, pp. 125–50.
- Christodoulakis, George A. and Stephen E. Satchell. 1998. "Hashing GARCH: A Reassessment of Volatility Forecast and Performance," in *Forecasting Volatility in the Financial Markets*. John Knight and Stephen Satchell, eds. Butterworth Heinemann, ch. 6, pp. 168–92.
- Christoffersen, Peter F. and Francis X. Diebold. 2000. "How Relevant Is Volatility Forecasting for Risk Management?" *Rev. Econ. Statist.* 82:1, pp. 12–22.
- Clark, Peter. 1973. "A Subordinated Stochastic Process Model with Finite Variance for Speculative Prices," *Econometrica* 41, pp. 135–56.
- Cox, John C. and Mark Rubinstein. 1985. *Options Markets*. Englewood Cliffs, NJ: Prentice Hall.
- Cumby, Robert; Stephen Figlewski and Joel Hasbrouck. 1993. "Forecasting Volatilities and Correlations with EGARCH Models," *J. Derivatives* 1, pp. 51–63.
- Danielsson, Jon. 1994. "Stochastic Volatility in Asset Prices: Estimation with Simulated Maximum Likelihood," *J. Econometrics* 64, pp. 375–400.
- Davidian, Marie and Raymond J. Carroll. 1987. "Variance Function Estimation," *J. Amer. Statist. Assoc.* 82, pp. 1079–91.
- Day, Ted E. and Craig M. Lewis. 1992. "Stock Market Volatility and the Information Content of Stock Index Options," *J. Econometrics* 52, pp. 267–87.
- . 1993. "Forecasting Futures Market Volatility," *J. Derivatives* 1, pp. 33–50.
- Diebold, Francis X. 1988. *Empirical Modeling of Exchange Rate Dynamics*. NY: Springer Verlag.
- Diebold, Francis X.; Andrew Hickman, Atsushi Inoue and Til Schuermann. 1998. "Scale Models," *RISK Magazine* 11, pp. 104–107.
- Diebold, Francis X. and Atsushi Inoue. 2001. "Long Memory and Regime Switching," *J. Econometrics* 105:1, pp. 131–59.
- Diebold, Francis X. and Jose A. Lopez. 1995. "Modeling Volatility Dynamics," in *Macroeconomics: Developments, Tensions and Prospects*. Kevin D. Hoover, ed. Boston, Dordrecht, and London: Kluwer Academic, pp. 427–66.
- Diebold, Francis X. and Roberto S. Mariano. 1995. "Comparing Predictive Accuracy," *J. Bus. Econ. Statist.* 13, pp. 253–63.
- Diebold, Francis X. and Peter Pauly. 1987. "Structural Change and the Combination of Forecasts," *J. Forecast.* 6, 1, pp. 21–40.
- Dimson, Elroy and Paul Marsh. 1990. "Volatility Forecasting without Data-Snooping," *J. Banking Finance*, 14:2–3, pp. 399–421.
- Ding, Zhuangxin; Clive W.J. Granger and Robert F. Engle. 1993. "A Long Memory Property Of Stock Market Returns And A New Model," *J. Empirical Finance*, 1, pp. 83–106.
- Doiidge, Craig and Jason Z. Wei. 1998. "Volatility Forecasting and the Efficiency of the Toronto 35 Index Options Market," *Canadian J. Admin. Science* 15:1, pp. 28–38.
- Drost, Feike C. and Theo E. Nijman. 1993. "Temporal Aggregation of GARCH Process," *Econometrica* 61:4, pp. 909–27.
- Duffie, Durrell and Kenneth J. Singleton. 1993. "Simulated Moments Estimation of Markov Models of Asset Prices," *Econometrica* 61, pp. 929–52.
- Dunis, Chris L.; Jason Laws and Stephane Chauvin. 2000. "The Use of Market Data and Model Combination to Improve Forecast Accuracy," work. paper, Liverpool Business School.
- Durbin, J. and Siem Jan Koopman. 2000. "Time Series Analysis of Non-Gaussian Observations Based on State Space Models from Both Classical and Bayesian Perspectives," *J. Royal Statist. Society Series* 62:1, pp. 3–56.
- Ederington, Louis H. and Wei Guan. 1999. "The Information Frown in Option Prices," work. paper, U. Oklahoma.
- . 2000a. "Forecasting Volatility," work. paper, U. Oklahoma.
- . 2000b. "Measuring Implied Volatility: Is an Average Better?" work. paper, U. Oklahoma.
- . 2002. "Is Implied Volatility an Informationally Efficient and Effective Predictor of Future Volatility?" *J. Risk* 4, 3.
- Ederington, Louis H. and Jae Ha Lee. 2001. "Intraday Volatility in Interest Rate and Foreign Exchange Markets: ARCH, Announcement, and Seasonality Effects," *J. Futures Markets* 21:6, pp. 517–52.
- Edey, Malcolm and Graham Elliot. 1992. "Some Evidence on Option Prices as Predictors of Volatility," *Oxford Bull. Econ. Statist.* 54:4, pp. 567–78.
- Engle, Robert F. 1982. "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation," *Econometrica* 50:4, pp. 987–1007.
- . 1993. "Statistical Models for Financial Volatility," *Finan. Analysts J.* 49:1, pp. 72–78.
- Engle, Robert F. and Tim Bollerslev. 1986. "Modelling the Persistence of Conditional Variances," *Econometric Rev.* 5, pp. 1–50.
- Engle, Robert F.; Victor Ng and Michael Rothschild. 1990. "Asset Pricing with a Factor-ARCH Covariance Structure: Empirical Estimates for Treasury Bills," *J. Econometrics* 45, pp. 213–39.
- Engle, Robert F. and Jeff R. Russell. 1998. "Autoregressive Conditional Duration: A New Model for Irregularly Spaced Transaction Data," *Econometrica* 66:5, pp. 1127–62.
- Fair, Ray C. and Robert J. Shiller. 1989. "The Informational Content of Ex Ante Forecasts," *Rev. Econ. Statist.* 71:2, pp. 325–32.
- . 1990. "Comparing Information in Forecasts from Econometric Models," *Amer. Econ. Rev.* 80:3, pp. 375–80.

- Feinstein, Steven P. 1989a. "The Black-Scholes Formula Is Nearly Linear in Sigma for At-The-Money Options; Therefore Implied Volatilities from At-The-Money Options Are Virtually Unbiased," work. paper, Fed. Reserve Bank Atlanta.
- . 1989b. "Forecasting Stock Market Volatility Using Options on Index Futures," *Econ. Rev.*, Fed. Reserve Bank Atlanta, 74:3, pp. 12–30.
- Ferreira, Miguel A. 1999. "Forecasting Interest Rate Volatility from the Information in Historical Data," work. paper, dept. finance, U. Wisconsin-Madison.
- Figlewski, Stephen. 1997. "Forecasting Volatility," in *Financial Markets, Inst. Instruments*. NYU, Salomon Center, 6:1, pp. 1–88.
- Figlewski, Stephen and T. Clifton Green. 1999. "Market Risk and Model Risk for a Financial Institution Writing Options," *J. Finance* 54:4, pp. 1465–999.
- Fleming, Jeff. 1998. "The quality of market volatility forecasts implied by S&P 100 index option prices," *J. Empirical Finance*, 5, 4, pp. 317–345.
- Fleming, Jeff; Chris Kirby and Barbara Ostdiek. 2000. "The Economic Value of Volatility Timing," *J. Finance* 56:1.
- . 2002. "The Economic Value of Volatility Timing Using Realized Volatility," *J. Finan. Econ.* Forthcoming.
- Fleming, Jeff; Barbara Ostdiek and Robert E. Whaley. 1995. "Predicting Stock Market Volatility: A New Measure," *J. Futures Markets* 15:3, pp. 265–302.
- Franke, Guenter; Richard C. Stapleton and Martin G. Subrahmanyam. 1998. "Who Buys and Who Sells Options: The Role of Options in an Economy with Background Risk," *J. Econ. Theory* 82:1, pp. 89–109.
- Franses, Philip H. and Dick Van Dijk. 1996. "Forecasting Stock Market Volatility Using (Non-Linear) GARCH Models," *J. Forecast.* 15, 3, pp. 229–35.
- . 2000. *Non-Linear Time Series Models in Empirical Finance*. Cambridge U. Press.
- Franses, Philip H. and Hendrik Ghysels. 1999. "Additive Outliers, GARCH and Forecasting Volatility," *Int. J. Forecast.* 15, pp. 1–9.
- French, Kenneth R.; Schwert G. William and Robert F. Stambaugh. 1987. "Expected Stock Returns and Volatility," *J. Finan. Econ.* 19:1, pp. 3–30.
- Frennberg, Per and Bjorn Hansson. 1996. "An Evaluation of Alternative Models for Predicting Stock Volatility," *J. Int. Finan. Markets, Inst. Money* 5, pp. 117–34.
- Fridman, Moshe and Lawrence Harris. 1998. "A Maximum Likelihood Approach for Non Gaussian Stochastic Volatility Models," *J. Bus. Econ. Statist.* 16, pp. 284–91.
- Friedman, Benjamin M. and David I. Laibson. 1989. "Economic Implications of Extraordinary Movements in Stock Prices," *Brooking Pap. Econ. Act.* 2, pp. 137–89.
- Fung, Hung-Gay; Chin-Jen Lie and Abel Moreno. 1990. "The Forecasting Performance of the Implied Standard Deviation in Currency Options," *Managerial Finance* 16:3, pp. 24–29.
- Fung, William K.H. and David A. Hsieh. 1991. "Empirical Analysis of Implied Volatility: Stocks, Bonds and Currencies," work. paper, dept. finance, Fuqua School of Business.
- Gallant, A. Ronald; Peter E. Rossi and George Tauchen. 1993. "Nonlinear Dynamic Structures," *Econometrica* 61:4, pp. 871–907.
- Garman, Mark B. and Michael J. Klass. 1980. "On the Estimation of Security Price Volatilities from Historical Data," *J. Bus.* 53:1, pp. 67–78.
- Gemmell, Gordon. 1986. "The Forecasting Performance of Stock Options on the London Traded Options Markets," *J. Bus. Finance Acc.* 13:4, pp. 535–46.
- Ghysels, Eric; Andrew Harvey and Eric Renault. 1996. "Stochastic Volatility," in *Handbook of Statistics: Statistical Methods in Finance*, Vol.14. G.S. Maddala and C.R. Rao, eds. Amsterdam: Elsevier Science, pp. 119–91.
- Glosten, Lawrence R.; Ravi Jagannathan and David E. Runkle. 1993. "On the Relation Between the Expected Value and the Volatility of the Nominal Excess Return on Stocks," *J. Finance* 48, pp. 1779–801.
- Granger, Clive W.J. 1999. *Empirical Modeling in Economics: Specification and Evaluation*. Cambridge U. Press.
- . 2001. "Long Memory Processes—An Economist's Viewpoint," work. paper, UC-San Diego.
- Granger, Clive W.J. and Namwon Hyung. 2000. "Occasional Structural Breaks and Long Memory," work. paper 99–14, UC-San Diego.
- Granger, Clive W.J.; Zhuaxin Ding and Scott Spear. 2000. "Stylized Facts on the Temporal and Distributional Properties of Absolute Returns: An Update," work. paper, UC-San Diego.
- Gray, Stephen F. 1996. "Modeling the Conditional Distribution of Interest Rates as a Regime-Switching Process," *J. Finan. Econ.* 42:1, pp. 27–62.
- Guo, Dajiang. 1996. "The Predictive Power of Implied Stochastic Variance from Currency Options," *J. Futures Markets* 16:8, pp. 915–42.
- . 1996. "The Information Content of Implied Stochastic Volatility from Currency Options," *Canadian J. Econ.* 29:5, pp. 559–61.
- Hamid, Shaikh. 1998. "Efficient Consolidation of Implied Volatilities and a Test of Intertemporal Averaging," *Derivatives Quart.* 4:3, pp. 35–49.
- Hamilton, James D. 1989. "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle," *Econometrica* 57, pp. 357–84.
- Hamilton, James D. and Gang Lin. 1996. "Stock Market Volatility and the Business Cycle," *J. Applied Econometrics* 11:5, pp. 573–93.
- Hamilton, James D. and Rauli Susmel. 1994. "Autoregressive Conditional Heteroskedasticity and Changes in Regime," *J. Econometrics* 64:1–2, pp. 307–33.
- Hansen, Lars P. and Robert J. Hodrick. 1980. "Forward Exchange Rates as Optimal Predictors Of Future Spot Rates: An Econometric Analysis," *J. Polit. Econ.* 88, pp. 829–53.
- Harvey, Andrew C. 1998. "Long Memory in Stochastic Volatility," in *Forecasting Volatility in the Financial Markets*. John Knight and Stephen Satchell, eds. Ch. 12, pp. 307–20. Butterworth Heinemann.
- Harvey, Andrew C.; Esther Ruiz and Neil Shephard. 1994. "Multivariate Stochastic Variance Models," *Rev. Econ. Stud.* 61, pp. 247–64.

- Harvey, Campbell R. and Robert E. Whaley. 1992. "Market Volatility Prediction and the Efficiency of the S&P100 Index Option Market," *J. Finan. Econ.* 31:1, pp. 43-74.
- Hentschel L. 2001. "Errors in Implied Volatility Estimation," work. paper, U. Rochester.
- Heston, Steven L. 1993. "A Closed Solution for Options with Stochastic Volatility, with Application to Bond and Currency Options," *Rev. Finan. Stud.* 6:2, pp. 327-43.
- Heynen, Ronald C. 1995. *Essays on Derivatives Pricing Theory*. Amsterdam: Thesis Publishers.
- Heynen, Ronald C. and Harry M. Kat. 1994. "Volatility Prediction: A Comparison of Stochastic Volatility, GARCH(1,1) and EGARCH(1,1) Models," *J. Derivatives* pp. 50-65.
- Hol, Eugenie and Siem Jan Koopman. 2001. "Forecasting the Variability of Stock Index Returns with Stochastic Volatility Models and Implied Volatility," work. paper, Free U. Amsterdam.
- Hsieh, David A. 1989. "Modeling Heteroscedasticity in Daily Foreign Exchange Rates," *J. Bus. Econ. Statist.* 7:3, pp. 307-17.
- Hu, Michael Y. and Christos Tsoukalas. 1999. "Combining Conditional Volatility Forecasts Using Neural Networks: An Application to the EMS Exchange Rates," *J. Int. Finan. Markets, Inst. Money* 9, pp. 407-22.
- Hull, John C. 2002. *Fundamentals of Futures and Options Markets*. 4th ed., Prentice Hall.
- Hull, John and Alan White. 1987. "The Pricing of Options on Assets with Stochastic Volatilities," *J. Finance* 42:2, pp. 281-300.
- . 1988. "An Analysis Of The Bias In Option Pricing Caused By A Stochastic Volatility," *Advanc. Futures Options Res.* 3, pp. 27-61.
- Hwang, Soosung and Stephen Satchell. 1998. "Implied Volatility Forecasting: A Comparison of Different Procedures Including Fractionally Integrated Models with Applications to U.K. Equity Options," in *Forecasting Volatility in the Financial Markets*. John Knight and Stephen Satchell, eds. Butterworth Heinemann, ch. 7, pp. 193-225.
- Inclan, Carla and George C. Tiao. 1994. "Use of cumulative sum of squares for retrospective detection of change of variance," *J. Amer. Statist. Assoc.* 89, pp. 913-23.
- Jacquier, Eric; Nicholas G. Polson and Peter E. Rossi. 1994. "Bayesian Analysis of Stochastic Volatility Models: Reply," *J. Bus. Econ. Statist.* 12:4, pp. 413-17.
- Jones, Chris S. 2001. "The Dynamics of Stochastic Volatility: Evidence from Underlying and Options Markets," work. paper, Simon School Business, U. Rochester.
- Jones, Charles M.; Owen Lamont and Robin L. Lumsdaine. 1998. "Macroeconomic News and Bond Market Volatility," *J. Finan. Econ.* 47, pp. 315-37.
- Jorion, Phillip. 1995. "Predicting Volatility in the Foreign Exchange Market," *J. Finance* 50:2, pp. 507-28.
- . 1996. "Risk and Turnover in the Foreign Exchange Market," in *The Microstructure of Foreign Exchange Markets*. J.A. Frankel, G. Galli, and A. Giovannini, eds. Chicago: U. Chicago Press.
- Karatzas, Ioannis and Stephen E. Shreve. 1988. *Brownian Motion and Stochastic Calculus*. NY: Springer Verlag.
- Karolyi, G. Andrew. 1993. "A Bayesian Approach to Modeling Stock Return Volatility and Option Valuation," *J. Finan. Quant. Anal.* 28:4, pp. 579-95.
- Karpoff, Jonathan M. 1987. "The Relation Between Price Changes and Trading Volume: A Survey," *J. Finan. Quant. Anal.* 22:1, pp. 109-26.
- Kearns, Phillip and Adrian R. Pagan. 1993. "Australian Stock Market Volatility, 1875-1987," *Econ. Record* 69, pp. 163-78.
- Kim, Sangjoon, Neil Shephard and Siddhartha Chib. 1998. "Stochastic Volatility: Likelihood Inference and Comparison with ARCH Models," *Rev. Econ. Stud.* 65:224, pp. 361-93.
- Klaassen, Franc. 2002. "Improving GARCH Volatility Forecasts," *Empir. Econ.* 27:2, pp. 363-94.
- Knight, John and Stephen Satchell. 1998. ed. *Forecasting Volatility in the Financial Markets*. Butterworth Heinemann.
- Kroner, Kenneth F. 1996. "Creating and Using Volatility Forecasts," *Derivatives Quart.* 3:2, pp. 39-53.
- Kroner, Kenneth; Kevin Kneafsey and Stijn Claessens. 1995. "Forecasting Volatility in Commodity Markets," *J. Forecast.* 14, pp. 77-95.
- Lamoureux, Christopher G. and William D. Lastrapes. 1990. "Persistence in Variance, Structural Change and the GARCH Model," *J. Bus. Econ. Statist.* 8:2, pp. 225-34.
- Lamoureux, Christopher and William Lastrapes. 1993. "Forecasting Stock-Return Variance: toward an Understanding of Stochastic Implied Volatilities," *Rev. Finan. Stud.* 6:2, pp. 293-326.
- Latane, Henry and Richard J. Rendleman. 1976. "Standard Deviations of Stock Price Ratios Implied in Option Prices," *J. Finance* 31:2, pp. 369-81.
- Lee, Keun Yeong. 1991. "Are the GARCH Models Best in Out-Of-Sample Performance?" *Econ. Letters* 37:3, pp. 305-308.
- Li, Kai. 2002. "Long-Memory versus Option-Implied Volatility Prediction," *J. Derivatives* 9:3, pp. 9-25.
- Lopez, Jose A. 2001. "Evaluating the Predictive Accuracy of Volatility Models," *J. Forecast.* 20:2, pp. 87-109.
- Loudon, Geoffrey F.; Wing H. Watt and Pradeep K. Yadav. 2000. "An Empirical Analysis of Alternative Parametric ARCH Models," *J. Appl. Econometrics* 15, pp. 117-36.
- Markowitz, Harry M. 1991. *Portfolio Selection*. Blackwell Publishers.
- Martens, Martens and Jason Zein. 2002. "Predicting Financial Volatility: High-Frequency Time-Series Forecasts vis-à-vis Implied Volatility," work. paper, Erasmus U.
- Mayhew, Stewart. 1995. "Implied Volatility," *Finan. Analyst J.* 51, pp. 8-20.
- McCurdy, Thomas H. and Ieuan G. Morgan. 1987. "Tests of the Martingale Hypothesis for Foreign Currency Futures with Time Varying Volatility," *Int. J. Forecast.* 3, pp. 131-48.
- McKenzie, Michael D. 1999. "Power Transformation and Forecasting the Magnitude of Exchange Rate Changes," *Int. J. Forecast.* 15, pp. 49-55.
- McMillan, David; Alan H. Speight and Owain ap Gwilym. 2000. "Forecasting U.K. Stock Market Volatility," *J. Applied Econ.* 10, pp. 435-48.

- Milhoj, Anders. 1987. "A Conditional Variance Model for Daily Observations of an Exchange Rate," *J. Bus. Econ. Statist.* 5, pp. 99–103.
- Nasar, Sylvia. 1992. "For Fed, a New Set of Tea Leaves," *New York Times*.
- Nelson, Daniel B. 1991. "Conditional Heteroskedasticity in Asset Returns: A New Approach," *Econometrica* 59:2, pp. 347–70.
- . 1992. "Filtering and Forecasting with Misspecified ARCH Models I: Getting the Right Variance with the Wrong Model," *J. Econometrics* 52, pp. 61–90.
- Nelson, Daniel B. and Dean P. Foster. 1995. "Filtering and Forecasting with Misspecified ARCH Models II: Making the Right Forecast with the Wrong Model," *J. Econometrics* 67:2, pp. 303–35.
- Noh, Jaesun, Robert F. Engle and Alex Kane. 1994. "Forecasting Volatility and Option Prices of the S&P 500 Index," *J. Derivatives* 2, pp. 17–30.
- Pagan, Adrian R. and G. William Schwert. 1990. "Alternative Models for Conditional Stock Volatility," *J. Econometrics* 45:1–2, pp. 267–90.
- Parkinson, Michael. 1980. "The Extreme Value Method for Estimating the Variance of the Rate of Return," *J. Bus.* 53, pp. 61–65.
- Pitt, Michael J. and Neil Shephard. 1997. "Likelihood Analysis of Non-Gaussian Measurement Time Series," *Biometrika* 84, pp. 653–67.
- Perry, Philip R. 1982. "The Time-Variance Relationship of Security Returns: Implications for the Return-Generating Stochastic Process," *J. Finance* 37, pp. 857–70.
- Pong, Shiu-Yan; Mark B. Shackleton, Stephen J. Taylor and Xinzhong Xu. 2002. "Forecasting Sterling/Dollar Volatility: A Comparison of Implied Volatilities and AR(FI)MA Models," work. paper, Lancaster U.
- Poon, Ser-Huang and Stephen J. Taylor. 1992. "Stock Returns and Stock Market Volatilities," *J. Banking Finance*, 16, pp. 37–59.
- Poteshman, Allen M. 2000. "Forecasting Future Volatility from Option Prices," work. paper, U. Illinois Urbana-Champaign.
- Randolph, William L. and Mohammad Najand. 1991. "A Test of Two Models in Forecasting Stock Index Futures Price Volatility," *J. Futures Markets* 11:2, pp. 179–90.
- Schmalensee, Richard and Robert R. Trippi. 1978. "Common Stock Volatility Expectations Implied by Option Premia," *J. Finance* 33:1, pp. 129–47.
- Scott, Elton and Alan L. Tucker. 1989. "Predicting currency return volatility," *J. Banking Finance* 13:6, pp. 839–51.
- Sill, D. Keith. 1993. "Predicting Stock-Market Volatility," *Business Rev.*, Jan/Feb, Fed. Reserve Bank Philadelphia, pp. 15–27.
- Singleton, Kenneth. 2001. "Estimation of Affine Asset Pricing Models Using the Empirical Characteristic Function," *J. Econometrics* 102, pp. 111–41.
- Szakmary, Andrew; Evren Ors, Jin K. Kim and Wallace D. Davidson III. 2002. "The Predictive Power of Implied Volatility: Evidence from 35 Futures Markets," work. paper, Southern Illinois U.
- Tauchen, George and Mark Pitts. 1983. "The Price Variability-Volume Relationship on Speculative Markets," *Econometrica* 51, pp. 485–505.
- Taylor, James W. 2001. "Volatility Forecasting with Smooth Transition Exponential Smoothing," work. paper, Oxford U.
- Taylor, Stephen J. 1986. *Modelling Financial Time Series*. Wiley.
- . 1987. "Forecasting of the Volatility of Currency Exchange Rates," *Int. J. Forecast.* 3, pp. 159–70.
- Taylor, Stephen J. and Gary Xinzhong Xu. 1997. "The Incremental Volatility Information in One Million Foreign Exchange Quotations," *J. Empirical Finance* 4:4, pp. 317–40.
- Theil, Henri. 1966. *Applied Economic Forecasting*. Amsterdam: North-Holland.
- Tse, Yiu Kuen. 1991. "Stock return volatility in the Tokyo Stock Exchange," *Japan World Econ.* 3, pp. 285–98.
- Tse, Yiu Kuen and Siew H. Tung. 1992. "Forecasting Volatility in the Singapore Stock Market," *Asia Pacific J. Manage.* 9:1, pp. 1–13.
- Vasilellis, George A. and Nigel Meade. 1996. "Forecasting Volatility for Portfolio Selection," *J. Bus. Finance Acc.* 23, 1, pp. 125–143.
- Vlasuso, Jon. 2002. "Forecasting Exchange Rate Volatility," *Econ. Letters* 76, pp. 59–64.
- Walsh, David M. and Glenn Yu-Gen Tsou. 1998. "Forecasting Index Volatility: Sampling Integral and Non-Trading Effects," *Appl. Finan. Econ.* 8, 5, pp. 477–85.
- Wei, Shang Jin and Jaffrey A. Frankel. 1991. "Are Option-Implied Forecasts of Exchange Rate Volatility Excessively Variable?" NBER work. paper 3910.
- West, Kenneth D. 1996. "Asymptotic Inference about Predictive Ability," *Econometrica* 64, pp. 1067–84.
- West, Kenneth D. and Dongchul Cho. 1995. "The Predictive Ability of Several Models of Exchange Rate Volatility," *J. Econometrics* 69:2, pp. 367–91.
- West, Kenneth D.; Hali J. Edison and Dongchul Cho. 1993. "A Utility Based Comparison of Some Methods of Exchange Rate Volatility," *J. Int. Econ.* 35:1–2, pp. 23–45.
- West, Kenneth D. and M. McCracken. 1998. "Regression Based Tests of Predictive Ability," *Int. Econ. Rev.* 39, pp. 817–40.
- Whaley, Robert E. 1982. "Valuation of American Call Options on Dividend-Paying Stocks," *J. Finan. Econ.* 10, pp. 29–58, Vanderbilt U.
- . 1993. "Derivatives on Market Volatility: Hedging Tools Long Overdue," *J. Derivatives* (fall), pp. 71–84.
- Wiggins, James B. 1987. "Option Values under Stochastic Volatility: Theory and Empirical Estimates," *J. Finan. Econ.* 19, pp. 351–72.
- . 1992. "Estimating the Volatility of S&P 500 Futures Prices Using the Extreme-Value Method," *J. Futures Markets* 12:3, pp. 256–73.
- Xu, Xinzhong and Stephen J. Taylor. 1995. "Conditional Volatility and the Informational Efficiency of the PHLX Currency Options Market," *J. Banking Finance* 19:5, pp. 803–21.
- Yu, Jun. 2002. "Forecasting Volatility in the New Zealand Stock Market," *Applied Finan. Econ.* 12, pp. 193–202.
- Zumbach, Gilles. 2002. "Volatility Processes and Volatility Forecast with Long Memory," work. paper, Olsen Associates.
- Zumbach, Gilles; Olivier V. Pictet and Oliver Masutti. 2001. "Genetic Programming with Syntactic Restrictions Applied to Financial Volatility Forecasting," work. paper, Olsen Associates.

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1. B. Karali, G. J. Power. 2013. Short- and Long-Run Determinants of Commodity Price Volatility. *American Journal of Agricultural Economics* **95**:3, 724-738. [[CrossRef](#)]
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3. Ihsan Ullah Badshah. 2013. Quantile Regression Analysis of the Asymmetric Return-Volatility Relation. *Journal of Futures Markets* **33**:3, 235-265. [[CrossRef](#)]
4. Christian Thomann. 2013. The Impact of Catastrophes on Insurer Stock Volatility. *Journal of Risk and Insurance* **80**:1, 65-94. [[CrossRef](#)]
5. Yaw-Huei Wang. 2013. Volatility Information in the Trading Activity of Stocks, Options, and Volatility Options. *Journal of Futures Markets* n/a-n/a. [[CrossRef](#)]
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7. Vasyl Golosnoy, Yarema Okhrin. 2013. Using information quality for volatility model combinations. *Quantitative Finance* 1-19. [[CrossRef](#)]
8. Jung-Bin Su. 2013. How to mitigate the impact of inappropriate distributional settings when the parametric value-at-risk approach is used. *Quantitative Finance* 1-21. [[CrossRef](#)]
9. Chaiyuth Padungsaksawasdi, Robert T. Daigler. 2013. The Return-Implied Volatility Relation for Commodity ETFs. *Journal of Futures Markets* n/a-n/a. [[CrossRef](#)]
10. Baohua Wang, Hejiao Huang, Xiaolong Wang. 2013. A support vector machine based MSM model for financial short-term volatility forecasting. *Neural Computing and Applications* **22**:1, 21-28. [[CrossRef](#)]
11. David G. McMillan, Raquel Quiroga García. 2013. Does Information Help Intra-Day Volatility Forecasts?. *Journal of Forecasting* **32**:1, 1-9. [[CrossRef](#)]
12. Costas Siriopoulos, Athanasios Fassas. 2012. Dynamic relations of uncertainty expectations: a conditional assessment of implied volatility indices. *Review of Derivatives Research* . [[CrossRef](#)]
13. Guillermo Benavides, Carlos Capistrán. 2012. Forecasting exchange rate volatility: The superior performance of conditional combinations of time series and option implied forecasts. *Journal of Empirical Finance* **19**:5, 627-639. [[CrossRef](#)]
14. CHIA-LIN CHANG, MICHAEL McALEER, ROENGCHAI TANSUCHAT. 2012. MODELLING LONG MEMORY VOLATILITY IN AGRICULTURAL COMMODITY FUTURES RETURNS. *Annals of Financial Economics* **07**:02, 1250010. [[CrossRef](#)]
15. Rick N. Francis, Patricia Eason. 2012. Accruals and the naïve out-of-sample prediction of operating cash flow. *Advances in Accounting* **28**:2, 226-234. [[CrossRef](#)]
16. Cathy W.S. Chen, Richard Gerlach, Edward M.H. Lin, W. C. W. Lee. 2012. Bayesian Forecasting for Financial Risk Management, Pre and Post the Global Financial Crisis. *Journal of Forecasting* **31**:8, 661-687. [[CrossRef](#)]
17. Robert T. Daigler, Ann Marie Hibbert, Ivelina Pavlova. 2012. Examining the Return-Volatility Relation for Foreign Exchange: Evidence from the Euro VIX. *Journal of Futures Markets* n/a-n/a. [[CrossRef](#)]

18. John F. Garvey, Liam A. Gallagher. 2012. The Realised-Implied Volatility Relationship: Recent Empirical Evidence from FTSE-100 Stocks. *Journal of Forecasting* **31**:7, 639-660. [[CrossRef](#)]
19. C. Emre Alper, Salih Fendoglu, Burak Saltoglu. 2012. MIDAS volatility forecast performance under market stress: Evidence from emerging stock markets. *Economics Letters* **117**:2, 528-532. [[CrossRef](#)]
20. Wolfgang Drobetz, Dieter Kaiser, Jasper Zimbehl. 2012. Efficient hedge fund style allocations: A rules-based model. *Journal of Derivatives & Hedge Funds* **18**:4, 274-300. [[CrossRef](#)]
21. Qian Chen, Richard Gerlach, Zudi Lu. 2012. Bayesian Value-at-Risk and expected shortfall forecasting via the asymmetric Laplace distribution. *Computational Statistics & Data Analysis* **56**:11, 3498-3516. [[CrossRef](#)]
22. Silvia Muzzioli. 2012. The Forecasting Performance of Corridor Implied Volatility in the Italian Market. *Computational Economics* . [[CrossRef](#)]
23. Anton Sorin Gabriel. 2012. Evaluating the Forecasting Performance of GARCH Models. Evidence from Romania. *Procedia - Social and Behavioral Sciences* **62**, 1006-1010. [[CrossRef](#)]
24. A. Buss, G. Vilkov. 2012. Measuring Equity Risk with Option-implied Correlations. *Review of Financial Studies* **25**:10, 3113-3140. [[CrossRef](#)]
25. Dimitrios P. Louzis, Spyros Xanthopoulos-Sisinis, Apostolos P. Refenes. 2012. Stock index realized volatility forecasting in the presence of heterogeneous leverage effects and long range dependence in the volatility of realized volatility. *Applied Economics* **44**:27, 3533-3550. [[CrossRef](#)]
26. Barry Harrison, Winston Moore. 2012. Forecasting Stock Market Volatility in Central and Eastern European Countries. *Journal of Forecasting* **31**:6, 490-503. [[CrossRef](#)]
27. Nikolay Gospodinov, Ibrahim Jamali. 2012. The effects of Federal funds rate surprises on S&P 500 volatility and volatility risk premium. *Journal of Empirical Finance* **19**:4, 497-510. [[CrossRef](#)]
28. Dilip Kumar, S. Maheswaran. 2012. Modelling asymmetry and persistence under the impact of sudden changes in the volatility of the Indian stock market. *IIMB Management Review* **24**:3, 123-136. [[CrossRef](#)]
29. JIANHUA GANG, NAN YE, CHENGSI ZHANG. 2012. FINANCIAL CRISIS, RISK PERCEPTION AND THE IMPLIED VOLATILITY TRANSMISSION: A CROSS-REGION STUDY*. *The Manchester School* **80**, 92-120. [[CrossRef](#)]
30. Wing Lon Ng. Improving Long-Term Financial Risk Forecasts using High-Frequency Data and Scaling Laws 255-278. [[CrossRef](#)]
31. Gordon Sirr, John Garvey, Liam Gallagher. 2012. A quantitative approach to guiding the promotional efforts of IPAs in emerging markets. *International Business Review* **21**:4, 618-630. [[CrossRef](#)]
32. Doojin Ryu. 2012. Implied Volatility Index of KOSPI200: Information Contents and Properties. *Emerging Markets Finance and Trade* **48**:0, 24-39. [[CrossRef](#)]
33. Hubert Dichtl, Wolfgang Drobetz. 2012. Zur Rendite-Risiko-Beziehung am deutschen Aktienmarkt – Eine empirische Analyse der Beziehung zwischen dem Deutschen Aktienindex DAX und dem Volatilitätsindex VDAX. *Kredit und Kapital* **45**:3, 373-406. [[CrossRef](#)]
34. S.T. Boris Choy, Cathy W.S. Chen, Edward M.H. Lin. 2012. Bivariate asymmetric GARCH models with heavy tails and dynamic conditional correlations. *Quantitative Finance* 1-17. [[CrossRef](#)]

35. Yuanyuan Zhang, Stephen J. Taylor, Lili Wang. 2012. Investigating the Information Content of the Model-Free Volatility Expectation by Monte Carlo Methods. *Journal of Futures Markets* n/a-n/a. [[CrossRef](#)]
36. Dong Wan Shin, Sangun Park. 2012. Efficient realized variance, regression coefficient, and correlation coefficient under different sampling frequencies. *Economics Letters* **115**:3, 334-337. [[CrossRef](#)]
37. George J. Jiang, Eirini Konstantinidi, George Skiadopoulos. 2012. Volatility spillovers and the effect of news announcements. *Journal of Banking & Finance* . [[CrossRef](#)]
38. Lloyd P. Blenman, Guan Jun Wang. 2012. New Insights on the Implied and Realized Volatility Relation. *Review of Pacific Basin Financial Markets and Policies* **15**:01, 1250001. [[CrossRef](#)]
39. Teng-Hao Huang, Yaw-Huei Wang. 2012. The Volatility and Density Prediction Performance of Alternative GARCH Models. *Journal of Forecasting* **31**:2, 157-171. [[CrossRef](#)]
40. Rosy Oh, Dong-Wan Shin. 2012. Market Microstructure Noise and Optimal Sampling Frequencies for the Realized Variances of Stock Prices of Four Leading Korean Companies. *Korean Journal of Applied Statistics* **25**:1, 15-27. [[CrossRef](#)]
41. Liang Peng, Rainer Schulz. 2012. Does the Diversification Potential of Securitized Real Estate Vary Over Time and Should Investors Care?. *The Journal of Real Estate Finance and Economics* . [[CrossRef](#)]
42. XIN CHENG, JOSEPH K.W. FUNG. 2012. The Information Content of Model-Free Implied Volatility. *Journal of Futures Markets* n/a-n/a. [[CrossRef](#)]
43. Jae-Won Jung, Kyungsik Kim. 2012. Return volatilities of the Korea treasury bond in financial markets. *Journal of the Korean Physical Society* **60**:4, 637-640. [[CrossRef](#)]
44. Ritab Al-Khouri, Abdulkhader Abdallah. 2012. Market liberalization and volatility of returns in emerging markets: The case of Qatar Exchange (QSC). *International Journal of Islamic and Middle Eastern Finance and Management* **5**:2, 106-115. [[CrossRef](#)]
45. S.S.S. Kumar. 2012. A first look at the properties of India's volatility index. *International Journal of Emerging Markets* **7**:2, 160-176. [[CrossRef](#)]
46. Jian Zhou. 2012. Extreme risk measures for REITs: a comparison among alternative methods. *Applied Financial Economics* **22**:2, 113-126. [[CrossRef](#)]
47. Costas Siriopoulos, Athanasios Fassas. 2012. An investor sentiment barometer — Greek Implied Volatility Index (GRIV). *Global Finance Journal* **23**:2, 77-93. [[CrossRef](#)]
48. Matthias Hartmann, Helmut Herwartz. 2011. Causal relations between inflation and inflation uncertainty—Cross sectional evidence in favour of the Friedman-Ball hypothesis. *Economics Letters* . [[CrossRef](#)]
49. Bing Xu, Jamal Ouenniche. 2011. A data envelopment analysis-based framework for the relative performance evaluation of competing crude oil prices' volatility forecasting models. *Energy Economics* . [[CrossRef](#)]
50. Jianhua Gang, Xiang Li. 2011. RISK PERCEPTION AND EQUITY RETURNS: EVIDENCE FROM THE SPX AND VIX. *Bulletin of Economic Research* no-no. [[CrossRef](#)]
51. Georgios Chortareas, Ying Jiang, John. C. Nankervis. 2011. Forecasting exchange rate volatility using high-frequency data: Is the euro different?. *International Journal of Forecasting* **27**:4, 1089-1107. [[CrossRef](#)]

52. B. Dupoyet, H.R. Fiebig, D.P. Musgrove. 2011. Replicating financial market dynamics with a simple self-organized critical lattice model. *Physica A: Statistical Mechanics and its Applications* **390**:18-19, 3120-3135. [[CrossRef](#)]
53. David G. McMillan, Mark E. Wohar. 2011. Structural breaks in volatility: the case of UK sector returns. *Applied Financial Economics* **21**:15, 1079-1093. [[CrossRef](#)]
54. References 180-189. [[CrossRef](#)]
55. Rossitsa Yalamova, Bill McKelvey. 2011. Explaining What Leads Up to Stock Market Crashes: A Phase Transition Model and Scalability Dynamics. *Journal of Behavioral Finance* **12**:3, 169-182. [[CrossRef](#)]
56. Erhan Deniz, James T. Luxhøj. 2011. A Scenario Generation Method with Heteroskedasticity and Moment Matching. *The Engineering Economist* **56**:3, 231-253. [[CrossRef](#)]
57. Yue Peng, Wing Lon Ng. 2011. Analysing financial contagion and asymmetric market dependence with volatility indices via copulas. *Annals of Finance* . [[CrossRef](#)]
58. Michal Dzielinski, Marc Oliver Rieger, Tönn Talpsepp Volatility asymmetry, news, and private investors 255-270. [[CrossRef](#)]
59. Tanya Tang, Michael Firth. 2011. Can book-tax differences capture earnings management and tax Management? Empirical evidence from China. *The International Journal of Accounting* . [[CrossRef](#)]
60. W.L. Tung, C. Quek. 2011. Financial volatility trading using a self-organising neural-fuzzy semantic network and option straddle-based approach. *Expert Systems with Applications* **38**:5, 4668-4688. [[CrossRef](#)]
61. Brice Dupoyet, Robert T. Daigler, Zhiyao Chen. 2011. A simplified pricing model for volatility futures. *Journal of Futures Markets* **31**:4, 307-339. [[CrossRef](#)]
62. Andrew J. Patton. 2011. Data-based ranking of realised volatility estimators. *Journal of Econometrics* **161**:2, 284-303. [[CrossRef](#)]
63. David G. Mcmillan, Alan E. H. Speight. 2011. Daily FX Volatility Forecasts: Can the GARCH(1,1) Model be Beaten using High-Frequency Data?. *Journal of Forecasting* n/a-n/a. [[CrossRef](#)]
64. Heping Liu, Ergin Erdem, Jing Shi. 2011. Comprehensive evaluation of ARMA-GARCH(-M) approaches for modeling the mean and volatility of wind speed. *Applied Energy* **88**:3, 724-732. [[CrossRef](#)]
65. Her-Jiun Sheu, Yu-Chen Wei. 2011. Options Trading Based on the Forecasting of Volatility Direction with the Incorporation of Investor Sentiment. *Emerging Markets Finance and Trade* **47**:2, 31-47. [[CrossRef](#)]
66. Pasquale Della Corte, Lucio Sarno, Ilias Tsiakas. 2011. Spot and forward volatility in foreign exchange#. *Journal of Financial Economics* . [[CrossRef](#)]
67. Alex YiHou Huang. 2011. Volatility forecasting by quantile regression. *Applied Economics* 1-11. [[CrossRef](#)]
68. Lester Hadsell. 2011. Inefficiency in deregulated wholesale electricity markets: the case of the New England ISO. *Applied Economics* **43**:5, 515-525. [[CrossRef](#)]
69. Sheri Markose, Amadeo Alentorn. 2011. The Generalized Extreme Value Distribution, Implied Tail Index, and Option Pricing. *The Journal of Derivatives* **18**:3, 35-60. [[CrossRef](#)]

70. Erik Haugom, Sjur Westgaard, Per Bjarte Solibakke, Gudbrand Lien. 2011. Realized volatility and the influence of market measures on predictability: Analysis of Nord Pool forward electricity data. *Energy Economics* . [[CrossRef](#)]
71. Alex YiHou Huang, Sheng-Pen Peng, Fangjhy Li, Ching-Jie Ke. 2011. Volatility forecasting of exchange rate by quantile regression#. *International Review of Economics & Finance* . [[CrossRef](#)]
72. Carlin C.F. Chu, K.P. Lam. 2011. Modeling intraday volatility: A new consideration. *Journal of International Financial Markets, Institutions and Money* . [[CrossRef](#)]
73. David McMillan, Pako Thupayagale. 2011. Measuring volatility persistence and long memory in the presence of structural breaks: Evidence from African stock markets. *Managerial Finance* **37**:3, 219-241. [[CrossRef](#)]
74. Her-Jiun Sheu, Yu-Chen Wei. 2011. Effective options trading strategies based on volatility forecasting recruiting investor sentiment. *Expert Systems with Applications* **38**:1, 585-596. [[CrossRef](#)]
75. Bartosz Gębka. 2011. THE DYNAMIC RELATION BETWEEN RETURNS, TRADING VOLUME, AND VOLATILITY: LESSONS FROM SPILLOVERS BETWEEN ASIA AND THE UNITED STATES. *Bulletin of Economic Research* no-no. [[CrossRef](#)]
76. Christian Tallau. 2011. Volatilitätsprognosen auf Basis der DAX-Volatilitätsindizes. *Kredit und Kapital* **44**:1, 47-74. [[CrossRef](#)]
77. Andrew J. Patton. 2011. Volatility forecast comparison using imperfect volatility proxies#. *Journal of Econometrics* **160**:1, 246-256. [[CrossRef](#)]
78. San-Lin Chung, Wei-Che Tsai, Yaw-Huei Wang, Pei-Shih Weng. 2011. The information content of the S&P 500 index and VIX options on the dynamics of the S&P 500 index. *Journal of Futures Markets* n/a-n/a. [[CrossRef](#)]
79. Ahmed A. A. Khalifa, Hong Miao, Sanjay Ramchander. 2011. Return distributions and volatility forecasting in metal futures markets: Evidence from gold, silver, and copper. *Journal of Futures Markets* **31**:1, 55-80. [[CrossRef](#)]
80. Neda Todorova, Sven Husmann. 2011. A comparative study of range-based stock return volatility estimators for the German market. *Journal of Futures Markets* n/a-n/a. [[CrossRef](#)]
81. Timo Terasvirta, Zhenfang Zhao. 2011. Stylized facts of return series, robust estimates and three popular models of volatility. *Applied Financial Economics* **21**:1, 67-94. [[CrossRef](#)]
82. Turgut Kısınbay. 2010. Predictive ability of asymmetric volatility models at medium-term horizons. *Applied Economics* **42**:30, 3813-3829. [[CrossRef](#)]
83. Baruch Lev, Siyi Li, Theodore Sougiannis. 2010. The usefulness of accounting estimates for predicting cash flows and earnings. *Review of Accounting Studies* **15**:4, 779-807. [[CrossRef](#)]
84. Yaw-Huei Wang, Yu-Jen Hsiao. 2010. The Impact of Non-trading Periods on the Measurement of Volatility. *Review of Pacific Basin Financial Markets and Policies* **13**:04, 607-620. [[CrossRef](#)]
85. Hongquan Li, Yongmiao Hong. 2010. Financial volatility forecasting with range-based autoregressive volatility model. *Finance Research Letters* . [[CrossRef](#)]
86. Wing Hong Chan, Xin Cheng, Joseph K.W. Fung. 2010. Forecasting volatility: Roles of sampling frequency and forecasting horizon. *Journal of Futures Markets* **30**:12, 1167-1191. [[CrossRef](#)]
87. Tripathy. 2010. Suitability of Volatility Models for Forecasting Stock Market Returns: A Study on the Indian National Stock Exchange. *American Journal of Applied Sciences* **7**:11, 1487-1494. [[CrossRef](#)]

88. Maria T. Gonzalez-Perez, Alfonso Novales. 2010. The information content in a volatility index for Spain. *SERIEs* . [[CrossRef](#)]
89. Francesco Audrino, Dominik Colangelo. 2010. Semi-parametric forecasts of the implied volatility surface using regression trees. *Statistics and Computing* **20**:4, 421-434. [[CrossRef](#)]
90. Risk Measurement and Volatility 63-108. [[CrossRef](#)]
91. Joon Young Kim, Jungryol Kim, Sang Bong Kim. 2010. Transmission of Stock Prices and Volatility from the Influential Major Markets on the Emerging Market: A Case Study of the Korean Stock Market. *Global Economic Review* **39**:3, 247-268. [[CrossRef](#)]
92. Camilo Serrano, Martin Hoesli. 2010. Are Securitized Real Estate Returns more Predictable than Stock Returns?. *The Journal of Real Estate Finance and Economics* **41**:2, 170-192. [[CrossRef](#)]
93. Peter Brous, Ufuk Ince, Ivilina Popova. 2010. Volatility forecasting and liquidity: Evidence from individual stocks. *Journal of Derivatives & Hedge Funds* **16**:2, 144-159. [[CrossRef](#)]
94. Louis H. Ederington, Wei Guan. 2010. Longer-Term Time-Series Volatility Forecasts. *Journal of Financial and Quantitative Analysis* **45**:04, 1055-1076. [[CrossRef](#)]
95. Theodore Tsekeris, Antony Stathopoulos. 2010. Short-Term Prediction of Urban Traffic Variability: Stochastic Volatility Modeling Approach. *Journal of Transportation Engineering* **136**:7, 606-613. [[CrossRef](#)]
96. Gyu-Hyen Moon, Wei-Choun Yu. 2010. Volatility Spillovers between the US and China Stock Markets: Structural Break Test with Symmetric and Asymmetric GARCH Approaches. *Global Economic Review* **39**:2, 129-149. [[CrossRef](#)]
97. Leonidas S. Rompolis, Elias Tzavalis. 2010. RISK PREMIUM EFFECTS ON IMPLIED VOLATILITY REGRESSIONS. *Journal of Financial Research* **33**:2, 125-151. [[CrossRef](#)]
98. Jie Ding, Nigel Meade. 2010. Forecasting accuracy of stochastic volatility, GARCH and EWMA models under different volatility scenarios. *Applied Financial Economics* **20**:10, 771-783. [[CrossRef](#)]
99. Md. Sabiruzzaman, Md. Monimul Huq, Rabiul Alam Beg, Sajid Anwar. 2010. Modeling and forecasting trading volume index: GARCH versus TGARCH approach. *The Quarterly Review of Economics and Finance* **50**:2, 141-145. [[CrossRef](#)]
100. Chaker Aloui, Samir Mabrouk. 2010. Value-at-risk estimations of energy commodities via long-memory, asymmetry and fat-tailed GARCH models. *Energy Policy* **38**:5, 2326-2339. [[CrossRef](#)]
101. Jian Zhou, Randy I. Anderson. 2010. Extreme Risk Measures for International REIT Markets. *The Journal of Real Estate Finance and Economics* . [[CrossRef](#)]
102. Gilles Zumbach. 2010. Volatility conditional on price trends. *Quantitative Finance* **10**:4, 431-442. [[CrossRef](#)]
103. Stephen J. Taylor, Pradeep K. Yadav, Yuanyuan Zhang. 2010. The information content of implied volatilities and model-free volatility expectations: Evidence from options written on individual stocks. *Journal of Banking & Finance* **34**:4, 871-881. [[CrossRef](#)]
104. George J. Jiang, Yisong S. Tian. 2010. Forecasting Volatility Using Long Memory and Comovements: An Application to Option Valuation under SFAS 123R. *Journal of Financial and Quantitative Analysis* **45**:02, 503-533. [[CrossRef](#)]
105. Amélie Charles. 2010. The day-of-the-week effects on the volatility: The role of the asymmetry. *European Journal of Operational Research* **202**:1, 143-152. [[CrossRef](#)]

106. Edward J. Lusk, Michael Halperin, Atanas Tetikov, Niya Stefanova. 2010. Forecasting Financial Market Annual Performance Measures: Further Evidence. *American Journal of Economics and Business Administration* 2:3, 300-306. [[CrossRef](#)]
107. Brianna Cain, Ralf Zurbrugg. 2010. Can switching between risk measures lead to better portfolio optimization?. *Journal of Asset Management* 10:6, 358-369. [[CrossRef](#)]
108. Myoung Shik Choi. 2010. Currency risks hedging for major and minor currencies: constant hedging versus speculative hedging. *Applied Economics Letters* 17:3, 305-311. [[CrossRef](#)]
109. Esben Høg, Leonidas Tsiaras. 2010. Density forecasts of crude-oil prices using option-implied and ARCH-type models. *Journal of Futures Markets* n/a-n/a. [[CrossRef](#)]
110. Anastassios A. Drakos, Georgios P. Kouretas, Leonidas P. Zarangas. 2010. Forecasting financial volatility of the Athens stock exchange daily returns: an application of the asymmetric normal mixture GARCH model. *International Journal of Finance & Economics* n/a-n/a. [[CrossRef](#)]
111. S. Muzzioli. 2010. Option-based forecasts of volatility: an empirical study in the DAX-index options market. *The European Journal of Finance* 16:6, 561-586. [[CrossRef](#)]
112. Wayne W. Yu, Evans C.K. Lui, Jacqueline W. Wang. 2010. The predictive power of the implied volatility of options traded OTC and on exchanges. *Journal of Banking & Finance* 34:1, 1-11. [[CrossRef](#)]
113. Hachmi Ben AmeerChapter 9 GARCH Models with CPPI Application 20, 187-205. [[CrossRef](#)]
114. Chuang-Chang Chang, Pei-Fang Hsieh, Yaw-Huei Wang. 2010. Information content of options trading volume for future volatility: Evidence from the Taiwan options market. *Journal of Banking & Finance* 34:1, 174-183. [[CrossRef](#)]
115. Mark C. Hutchinson, Liam A. Gallagher. 2010. Convertible Bond Arbitrage: Risk and Return. *Journal of Business Finance & Accounting* 37:1-2, 206-241. [[CrossRef](#)]
116. Balaji Raman, Vladimir Pozdnyakov. 2010. On Gaussian HJM framework for Eurodollar Futures. *Applied Stochastic Models in Business and Industry* n/a-n/a. [[CrossRef](#)]
117. Hung-Chun Liu, Jui-Cheng Hung. 2010. Forecasting volatility and capturing downside risk of the Taiwanese futures markets under the financial tsunami. *Managerial Finance* 36:10, 860-875. [[CrossRef](#)]
118. Erik Lindström. 2010. Implications of Parameter Uncertainty on Option Prices. *Advances in Decision Sciences* 2010, 1-15. [[CrossRef](#)]
119. Guan Jun Wang, Pierre Yourougou, Yue Dong Wang. 2009. Which implied volatility provides the best measure of future volatility?. *Journal of Economics and Finance* . [[CrossRef](#)]
120. andrew stuart duncan, guangling“dave” liu. 2009. MODELLING SOUTH AFRICAN CURRENCY CRISES AS STRUCTURAL CHANGES IN THE VOLATILITY OF THE RAND. *South African Journal of Economics* 77:3, 363-379. [[CrossRef](#)]
121. Yingfu Xie, Jun Yu, Bo Ranneby. 2009. Forecasting using locally stationary wavelet processes. *Journal of Statistical Computation and Simulation* 79:9, 1067-1082. [[CrossRef](#)]
122. Jian Zhou, Zhixin Kang. 2009. A Comparison of Alternative Forecast Models of REIT Volatility. *The Journal of Real Estate Finance and Economics* . [[CrossRef](#)]
123. Xiaoquan Liu, Mark Shackleton, Stephen Taylor, Xinzhong Xu. 2009. Empirical pricing kernels obtained from the UK index options market. *Applied Economics Letters* 16:10, 989-993. [[CrossRef](#)]

124. António Câmara, San-Lin Chung, Yaw-Huei Wang. 2009. Option implied cost of equity and its properties. *Journal of Futures Markets* **29**:7, 599-629. [[CrossRef](#)]
125. Hojin Lee. 2009. Out-of-sample Forecasting Performance of Won/Dollar Exchange Rate Return Volatility Model. *Journal of East Asian Economic Integration* **13**:1, 57-88. [[CrossRef](#)]
126. Sang Hoon Kang, Seong-Min Yoon. 2009. Modeling and Forecasting the Volatility of Eastern European Emerging Markets. *Journal of East Asian Economic Integration* **13**:1, 113-132. [[CrossRef](#)]
127. Roberto Ferulano. 2009. A Mixed Historical Formula to forecast volatility. *Journal of Asset Management* **10**:2, 124-136. [[CrossRef](#)]
128. Simone Bianco, Roberto Reno. 2009. Unexpected volatility and intraday serial correlation. *Quantitative Finance* **9**:4, 465-475. [[CrossRef](#)]
129. Ernst Konrad. 2009. The impact of monetary policy surprises on asset return volatility: the case of Germany. *Financial Markets and Portfolio Management* **23**:2, 111-135. [[CrossRef](#)]
130. David McMillan, Raquel Quiroga Garcia. 2009. Intra-day volatility forecasts. *Applied Financial Economics* **19**:8, 611-623. [[CrossRef](#)]
131. P AGNOLUCCI. 2009. Volatility in crude oil futures: A comparison of the predictive ability of GARCH and implied volatility models. *Energy Economics* **31**:2, 316-321. [[CrossRef](#)]
132. Charlie Charoenwong, Nattawut Jenwittayaroje, Buen Sin Low. 2009. Who knows more about future currency volatility?. *Journal of Futures Markets* **29**:3, 270-295. [[CrossRef](#)]
133. José Dias Curto, José Castro Pinto, Gonçalo Nuno Tavares. 2009. Modeling stock markets' volatility using GARCH models with Normal, Student's t and stable Paretian distributions. *Statistical Papers* **50**:2, 311-321. [[CrossRef](#)]
134. Kent Wang. 2009. Volatility linkages of the equity, bond and money markets: an implied volatility approach. *Accounting & Finance* **49**:1, 207-219. [[CrossRef](#)]
135. Wei Liu, Bruce Morley. 2009. Volatility Forecasting in the Hang Seng Index using the GARCH Approach. *Asia-Pacific Financial Markets* **16**:1, 51-63. [[CrossRef](#)]
136. Yan-Leung Cheung, Yin-Wong Cheung, Alan T. K. Wan. 2009. A high-low model of daily stock price ranges. *Journal of Forecasting* **28**:2, 103-119. [[CrossRef](#)]
137. P LIN, P KO. 2009. Portfolio value-at-risk forecasting with GA-based extreme value theory. *Expert Systems with Applications* **36**:2, 2503-2512. [[CrossRef](#)]
138. Kun Zhang, Laiwan Chan. 2009. Efficient factor GARCH models and factor-DCC models. *Quantitative Finance* **9**:1, 71-91. [[CrossRef](#)]
139. C NEELY. 2009. Forecasting foreign exchange volatility: Why is implied volatility biased and inefficient? And does it matter?. *Journal of International Financial Markets, Institutions and Money* **19**:1, 188-205. [[CrossRef](#)]
140. Shiyi Chen, Wolfgang K. Härdle, Kiho Jeong. 2009. Forecasting volatility with support vector machine-based GARCH model. *Journal of Forecasting* n/a-n/a. [[CrossRef](#)]
141. Malay Bhattacharyya, Dileep Kumar M, Ramesh Kumar. 2009. Optimal sampling frequency for volatility forecast models for the Indian stock markets. *Journal of Forecasting* **28**:1, 38-54. [[CrossRef](#)]
142. Louis H. Ederington, Wei Guan. 2009. The bias in time series volatility forecasts. *Journal of Futures Markets* n/a-n/a. [[CrossRef](#)]
143. References and Bibliography 327-332. [[CrossRef](#)]

144. Stavros Degiannakis, Alexandra Livada, Epaminondas Panas. 2008. Rolling-sampled parameters of ARCH and Levy-stable models. *Applied Economics* **40**:23, 3051-3067. [[CrossRef](#)]
145. M BHATTACHARYYA, A CHAUDHARY, G YADAV. 2008. Conditional VaR estimation using Pearson's type IV distribution#. *European Journal of Operational Research* **191**:2, 386-397. [[CrossRef](#)]
146. Nicholas Taylor. 2008. The predictive value of temporally disaggregated volatility: evidence from index futures markets. *Journal of Forecasting* **27**:8, 721-742. [[CrossRef](#)]
147. M MEDEIROS, M MCALEER, D SLOTTJE, V RAMOS, J REYMAQUEIRA. 2008. An alternative approach to estimating demand: Neural network regression with conditional volatility for high frequency air passenger arrivals. *Journal of Econometrics* **147**:2, 372-383. [[CrossRef](#)]
148. Trino-Manuel Níguez. 2008. Volatility and VaR forecasting in the Madrid Stock Exchange. *Spanish Economic Review* **10**:3, 169-196. [[CrossRef](#)]
149. R Fildes, K Nikolopoulos, S F Crone, A A Syntetos. 2008. Forecasting and operational research: a review. *Journal of the Operational Research Society* **59**:9, 1150-1172. [[CrossRef](#)]
150. Dima Alberg, Haim Shalit, Rami Yosef. 2008. Estimating stock market volatility using asymmetric GARCH models. *Applied Financial Economics* **18**:15, 1201-1208. [[CrossRef](#)]
151. Mark Hutchinson, Liam Gallagher. 2008. Simulating convertible bond arbitrage portfolios. *Applied Financial Economics* **18**:15, 1247-1262. [[CrossRef](#)]
152. Ramón Yepes Rodríguez. 2008. Real option valuation of free destination in long-term liquefied natural gas supplies. *Energy Economics* **30**:4, 1909-1932. [[CrossRef](#)]
153. Ikhlaas Gurrib. 2008. Do large hedgers and speculators react to events? A stability and events analysis. *Applied Financial Economics Letters* **4**:4, 259-267. [[CrossRef](#)]
154. David G. McMillan, Raquel Quiroga Garcia. 2008. Efficiency of the IBEX spot-futures basis: The impact of the mini-futures. *Journal of Futures Markets* **28**:4, 398-415. [[CrossRef](#)]
155. Thomas Lux. 2008. The Markov-Switching Multifractal Model of Asset Returns. *Journal of Business and Economic Statistics* **26**:2, 194-210. [[CrossRef](#)]
156. X LI, J MIFFRE, C BROOKS, N OSULLIVAN. 2008. Momentum profits and time-varying unsystematic risk. *Journal of Banking & Finance* **32**:4, 541-558. [[CrossRef](#)]
157. Jun Nagayasu. 2008. Japanese stock movements from 1991 to 2005: evidence from high- and low-frequency data. *Applied Financial Economics* **18**:4, 295-307. [[CrossRef](#)]
158. Amélie Charles, Olivier Darné. 2008. A Note on Unit Root Tests and GARCH Errors: A Simulation Experiment. *Communications in Statistics - Simulation and Computation* **37**:2, 314-319. [[CrossRef](#)]
159. David E. Rapach, Jack K. Strauss, Mark E. WoharChapter 10 Forecasting Stock Return Volatility in the Presence of Structural Breaks **3**, 381-416. [[CrossRef](#)]
160. Dimitris N. Politis, Dimitrios D. ThomakosChapter 11 Financial Time Series and Volatility Prediction using NoVaS Transformations **3**, 417-447. [[CrossRef](#)]
161. Michael McAleer, Marcelo Medeiros. 2008. Realized Volatility: A Review. *Econometric Reviews* **27**:1, 10-45. [[CrossRef](#)]
162. Timotheos Angelidis, Stavros Degiannakis. 2008. Forecasting one-day-ahead VaR and intra-day realized volatility in the Athens Stock Exchange Market. *Managerial Finance* **34**:7, 489-497. [[CrossRef](#)]

163. M BHATTACHARYYA, G RITOLIA. 2008. Conditional VaR using EVT – Towards a planned margin scheme. *International Review of Financial Analysis* 17:2, 382-395. [[CrossRef](#)]
164. Vyacheslav M. Abramov, Fima C. Klebaner. 2007. Estimation and Prediction of a Non-Constant Volatility. *Asia-Pacific Financial Markets* 14:1-2, 1-23. [[CrossRef](#)]
165. S. M. Duarte Queirós. 2007. On a generalised model for time-dependent variance with long-term memory. *Europhysics Letters (EPL)* 80:3, 30005. [[CrossRef](#)]
166. Vipul, Joshy Jacob. 2007. Forecasting performance of extreme-value volatility estimators. *Journal of Futures Markets* 27:11, 1085-1105. [[CrossRef](#)]
167. Andraz Grum. 2007. Measuring Market Risk for Commercial Banks in the Volatile Environment of an Emerging Market Economy. *South East European Journal of Economics and Business* 2:2, 89-94. [[CrossRef](#)]
168. Twm Evans, David McMillan. 2007. Volatility forecasts: the role of asymmetric and long-memory dynamics and regional evidence. *Applied Financial Economics* 17:17, 1421-1430. [[CrossRef](#)]
169. Lance E. Davis, Larry Neal, Eugene White. 2007. The Highest Price Ever: The Great NYSE Seat Sale of 1928–1929 and Capacity Constraints. *The Journal of Economic History* 67:03. . [[CrossRef](#)]
170. Eric Girard, Rita Biswas. 2007. Trading Volume and Market Volatility: Developed versus Emerging Stock Markets. *The Financial Review* 42:3, 429-459. [[CrossRef](#)]
171. R BECKER, A CLEMENTS, S WHITE. 2007. Does implied volatility provide any information beyond that captured in model-based volatility forecasts?. *Journal of Banking & Finance* 31:8, 2535-2549. [[CrossRef](#)]
172. M. Pilar Muñoz, M. Dolores Marquez, Lesly M. Acosta. 2007. Forecasting volatility by means of threshold models. *Journal of Forecasting* 26:5, 343-363. [[CrossRef](#)]
173. José R. Aragonés, Carlos Blanco, Pablo García Estévez. 2007. Neural network volatility forecasts. *Intelligent Systems in Accounting, Finance & Management* 15:3-4, 107-121. [[CrossRef](#)]
174. Elijah Brewer, James M. Carson, Elyas Elyasiani, Iqbal Mansur, William L. Scott. 2007. Interest Rate Risk and Equity Values of Life Insurance Companies: A GARCH?M Model. *Journal of Risk & Insurance* 74:2, 401-423. [[CrossRef](#)]
175. Steven L. Beach, Alexei G. Orlov. 2007. An application of the Black–Litterman model with EGARCH-M-derived views for international portfolio management. *Financial Markets and Portfolio Management* 21:2, 147-166. [[CrossRef](#)]
176. X LIU, M SHACKLETON, S TAYLOR, X XU. 2007. Closed-form transformations from risk-neutral to real-world distributions. *Journal of Banking & Finance* 31:5, 1501-1520. [[CrossRef](#)]
177. Yasemin Ulu. 2007. Sampling properties of criteria for evaluating GARCH volatility forecasts. *Applied Financial Economics* 17:8, 671-681. [[CrossRef](#)]
178. E ELYASIANI, I MANSUR, M PAGANO. 2007. Convergence and risk-return linkages across financial service firms. *Journal of Banking & Finance* 31:4, 1167-1190. [[CrossRef](#)]
179. Yong Bao, Tae-Hwy Lee, Burak Saltoğlu. 2007. Comparing density forecast models. *Journal of Forecasting* 26:3, 203-225. [[CrossRef](#)]
180. J. Duarte. 2007. The Causal Effect of Mortgage Refinancing on Interest Rate Volatility: Empirical Evidence and Theoretical Implications. *Review of Financial Studies* 21:4, 1689-1731. [[CrossRef](#)]

181. Thomas K. Lee, John Zyren. 2007. Volatility Relationship between Crude Oil and Petroleum Products. *Atlantic Economic Journal* 35:1, 97-112. [[CrossRef](#)]
182. Thorsten M. Egelkraut, Philip Garcia, Bruce J. Sherrick. 2007. The Term Structure of Implied Forward Volatility: Recovery and Informational Content in the Corn Options Market. *American Journal of Agricultural Economics* 89:1, 1-11. [[CrossRef](#)]
183. Timotheos Angelidis, Alexandros Benos, Stavros Degiannakis. 2007. A robust VaR model under different time periods and weighting schemes. *Review of Quantitative Finance and Accounting* 28:2, 187-201. [[CrossRef](#)]
184. Antonios Antoniou, Gioia M. Pescetto, Ibrahim Stevens. 2007. Market-wide and sectoral integration: Evidence from the UK, USA and Europe. *Managerial Finance* 33:3, 173-194. [[CrossRef](#)]
185. Thomas A. Silvey An investigation of the relative performance of GARCH models versus simple rules in forecasting volatility 101-129. [[CrossRef](#)]
186. N TAYLOR. 2007. A note on the importance of overnight information in risk management models. *Journal of Banking & Finance* 31:1, 161-180. [[CrossRef](#)]
187. Linlan Xiao, Abdurrahman Aydemir Volatility modelling and forecasting in finance 1-45. [[CrossRef](#)]
188. David G. McMillan, Alan E.H. Speight. 2007. Weekly volatility forecasts with applications to risk management. *The Journal of Risk Finance* 8:3, 214-229. [[CrossRef](#)]
189. George J Jiang, Yisong S Tian. 2007. Extracting Model-Free Volatility from Option Prices. *The Journal of Derivatives* 14:3, 35-60. [[CrossRef](#)]
190. Anders Wilhelmsson. 2006. Garch forecasting performance under different distribution assumptions. *Journal of Forecasting* 25:8, 561-578. [[CrossRef](#)]
191. Cathy W.S. Chen, Richard Gerlach, Mike K.P. So. 2006. Comparison of nonnested asymmetric heteroskedastic models. *Computational Statistics & Data Analysis* 51:4, 2164-2178. [[CrossRef](#)]
192. WO-CHIANG LEE. 2006. FORECASTING HIGH-FREQUENCY FINANCIAL DATA VOLATILITY VIA NONPARAMETRIC ALGORITHMS: EVIDENCE FROM TAIWAN'S FINANCIAL MARKETS. *New Mathematics and Natural Computation* 02:03, 345-359. [[CrossRef](#)]
193. David Ashton, Mark Tippet. 2006. Mean Reversion and the Distribution of United Kingdom Stock Index Returns. *Journal of Business Finance & Accounting* 33:9-10, 1586-1609. [[CrossRef](#)]
194. SHU-HENG CHEN. 2006. PREFACE. *New Mathematics and Natural Computation* 02:03, 255-260. [[CrossRef](#)]
195. J. M. Vanden. 2006. Information Quality and Options. *Review of Financial Studies* 21:6, 2635-2676. [[CrossRef](#)]
196. Nikolay Gospodinov, Athanasia Gavala, Deming Jiang. 2006. Forecasting volatility. *Journal of Forecasting* 25:6, 381-400. [[CrossRef](#)]
197. Lester Hadsell. 2006. A TARCH examination of the return volatility-volume relationship in electricity futures. *Applied Financial Economics* 16:12, 893-901. [[CrossRef](#)]
198. Ercan Balaban, Charalambos Constantinou. 2006. Volatility clustering and event-induced volatility: Evidence from UK mergers and acquisitions. *The European Journal of Finance* 12:5, 449-453. [[CrossRef](#)]
199. Zhiyao Chen, Robert T. Daigler, Ali M. Parhizgari. 2006. Persistence of volatility in futures markets. *Journal of Futures Markets* 26:6, 571-594. [[CrossRef](#)]

200. HUI GUO, ROBERT F. WHITE LAW. 2006. Uncovering the Risk–Return Relation in the Stock Market. *The Journal of Finance* **61**:3, 1433–1463. [[CrossRef](#)]
201. Naren Pappu, S Lakshmivarahan, Duane R Stock. 2006. Models of Conditional Variance for Bond Prices. *The Journal of Fixed Income* **16**:1, 65–70. [[CrossRef](#)]
202. Stefano Sampietro. 2006. Bayesian analysis of mixture of autoregressive components with an application to financial market volatility. *Applied Stochastic Models in Business and Industry* **22**:3, 225–242. [[CrossRef](#)]
203. Alexei G. Orlov. 2006. Capital controls and stock market volatility in frequency domain. *Economics Letters* **91**:2, 222–228. [[CrossRef](#)]
204. Francesco Audrino, Fabio Trojani. 2006. Estimating and predicting multivariate volatility thresholds in global stock markets. *Journal of Applied Econometrics* **21**:3, 345–369. [[CrossRef](#)]
205. B RAUNIG. 2006. The longer-horizon predictability of German stock market volatility#. *International Journal of Forecasting* **22**:2, 363–372. [[CrossRef](#)]
206. Thierry Ané. 2006. Short and long term components of volatility in Hong Kong stock returns. *Applied Financial Economics* **16**:6, 439–460. [[CrossRef](#)]
207. Yong Bao, Tae-Hwy Lee, Burak Saltoglu. 2006. Evaluating predictive performance of value-at-risk models in emerging markets: a reality check. *Journal of Forecasting* **25**:2, 101–128. [[CrossRef](#)]
208. Joseph Persky, Gilbert W. Bassett. 2006. Conceptualizing Inequality and Risk. *Journal of the History of Economic Thought* **28**:01, 81. [[CrossRef](#)]
209. P HANSEN, A LUNDE. 2006. Consistent ranking of volatility models. *Journal of Econometrics* **131**:1–2, 97–121. [[CrossRef](#)]
210. T BOLLERSLEV, H ZHOU. 2006. Volatility puzzles: a simple framework for gauging return–volatility regressions. *Journal of Econometrics* **131**:1–2, 123–150. [[CrossRef](#)]
211. 2006. Comparison of a Class of Nonlinear Time Series models (GARCH, IGARCH, EGARCH). *Korean Journal of Applied Statistics* **19**:1, 33–41. [[CrossRef](#)]
212. Ercan Balaban, Asli Bayar, Robert Faff. 2006. Forecasting stock market volatility: Further international evidence. *The European Journal of Finance* **12**:2, 171–188. [[CrossRef](#)]
213. Timo Teräsvirta Chapter 8 Forecasting economic variables with nonlinear models **1**, 413–457. [[CrossRef](#)]
214. Gilles Zumbach Relevance of Volatility Forecasting in Financial Risk Management 583–603. [[CrossRef](#)]
215. Ahsan Habib. 2006. Disaggregated earnings and prediction of future profitability: evidence from industrial groups in Japan. *Review of Accounting and Finance* **5**:4, 355–369. [[CrossRef](#)]
216. Chiara Pederzoli. 2006. Stochastic Volatility and GARCH: a Comparison Based on UK Stock Data. *The European Journal of Finance* **12**:1, 41–59. [[CrossRef](#)]
217. Peter R. Hansen, Asger Lunde. 2005. A forecast comparison of volatility models: does anything beat a GARCH(1,1)? *Journal of Applied Econometrics* **20**:7, 873–889. [[CrossRef](#)]
218. Yasemin Ulu. 2005. Out-of-sample forecasting performance of the QGARCH model. *Applied Financial Economics Letters* **1**:6, 387–392. [[CrossRef](#)]

219. S. M.D. Queirós, C. Tsallis. 2005. On the connection between financial processes with stochastic volatility and nonextensive statistical mechanics. *The European Physical Journal B* **48**:1, 139-148. [[CrossRef](#)]
220. L. EDERINGTON, W. GUAN. 2005. The information frown in option prices. *Journal of Banking & Finance* **29**:6, 1429-1457. [[CrossRef](#)]
221. James W. Taylor. 2005. Generating Volatility Forecasts from Value at Risk Estimates. *Management Science* **51**:5, 712-725. [[CrossRef](#)]
222. SERGEI FEDOTOV, ABBY TAN. 2005. LONG MEMORY STOCHASTIC VOLATILITY IN OPTION PRICING. *International Journal of Theoretical and Applied Finance* **08**:03, 381-392. [[CrossRef](#)]
223. Louis H. Ederington, Wei Guan. 2005. Forecasting volatility. *Journal of Futures Markets* **25**:5, 465-490. [[CrossRef](#)]
224. S. M. Duarte Queirós, C. Tsallis. 2005. Bridging a paradigmatic financial model and nonextensive entropy. *Europhysics Letters (EPL)* **69**:6, 893-899. [[CrossRef](#)]
225. Ser-Huang Poon, Clive Granger. 2005. Practical Issues in Forecasting Volatility. *Financial Analysts Journal* **61**:1, 45-56. [[CrossRef](#)]
226. Timotheos Angelidis, Stavros Degiannakis. 2005. Modeling risk for long and short trading positions. *The Journal of Risk Finance* **6**:3, 226-238. [[CrossRef](#)]
227. Pierre Giot. 2005. Relationships Between Implied Volatility Indexes and Stock Index Returns. *The Journal of Portfolio Management* **31**:3, 92-100. [[CrossRef](#)]
228. Stavros Degiannakis. 2004. Volatility forecasting: evidence from a fractional integrated asymmetric power ARCH skewed-t model. *Applied Financial Economics* **14**:18, 1333-1342. [[CrossRef](#)]
229. Martin Martens, Jason Zein. 2004. Predicting financial volatility: High-frequency time-series forecasts vis-à-vis implied volatility. *Journal of Futures Markets* **24**:11, 1005-1028. [[CrossRef](#)]
230. S. PONG, M. SHACKLETON, S. TAYLOR, X. XU. 2004. Forecasting currency volatility: A comparison of implied volatilities and AR(FI)MA models. *Journal of Banking & Finance* **28**:10, 2541-2563. [[CrossRef](#)]
231. J. Farmer, Fabrizio Lillo. 2004. On the origin of power-law tails in price fluctuations. *Quantitative Finance* **4**:1, C7-C11. [[CrossRef](#)]
232. Peter Reinhard Hansen, Asger Lunde, James M. Nason. 2003. Choosing the Best Volatility Models: The Model Confidence Set Approach*. *Oxford Bulletin of Economics and Statistics* **65**:s1, 839-861. [[CrossRef](#)]